

Modeling, Simulation, and Design of a Ge-on-Si PIN Waveguide Photodetector for 400 Gb/s Optical Interconnects

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Abstract

This paper presents the numerical modeling, simulation, and design of a high-speed, waveguide-coupled Germanium-on-Silicon (Ge-on-Si) p-i-n photodetector optimized for 400 Gb/s (4×106.25 Gb/s PAM-4) optical interconnects. Using three-dimensional finite-difference time-domain (FDTD) and electrical drift-diffusion solvers, the photodetector geometry is optimized. The nominal $8\text{ }\mu\text{m} \times 5\text{ }\mu\text{m}$ device achieves a transport-limited responsivity of 0.931 A W^{-1} at 1310 nm (external quantum efficiency of 88.08 %) and a low dark current of 0.684 nA at -1.0 V bias. By integrating a 300 pH series peaking inductor, the 3 dB electro-optic bandwidth is boosted from a baseline of 56.1 GHz to 89.7 GHz. System-level transmission simulations confirm open PAM-4 eye diagrams at 106.25 GBd with a bit error rate (BER) of 6.35×10^{-3} at -10 dBm , comfortably satisfying the KP4 forward error correction (FEC) limits for intra-data-center links.

1 Introduction

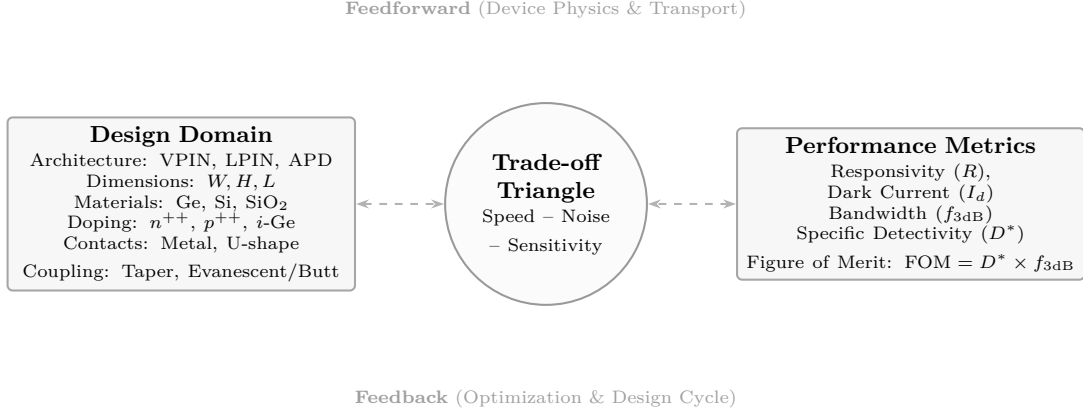


Figure 1: Photodetector design methodology and optimization workflow. The feedforward path maps parameters from the design domain to the performance metrics, while the feedback path drives the design cycle based on multi-objective optimizations. The central trade-off triangle governs the device constraints.

Data-centre interconnect capacity doubles roughly every two years, and the silicon-photonics ecosystem has responded by standardising on four-lane parallel single-mode interfaces in the O-band. IEEE 802.3 Clause 140 defines 400 GBASE-DR4 as one such interface: four lanes at 1310 nm, each carrying $106.25 \text{ Gbit s}^{-1}$ with PAM-4 signalling, over reaches of up to 500 m. A receiver-side photodetector must therefore achieve a -3 dB electrical bandwidth of at least $0.75f_s = 39.8 \text{ GHz}$, a responsivity sufficient to meet the -6.0 dBm minimum OMA sensitivity, and a dark current low enough that shot noise does not materially degrade link margin. this paper studies those requirements at the device physics and simulation level, using an existing high-speed Ge-on-Si photodetector as the working example.

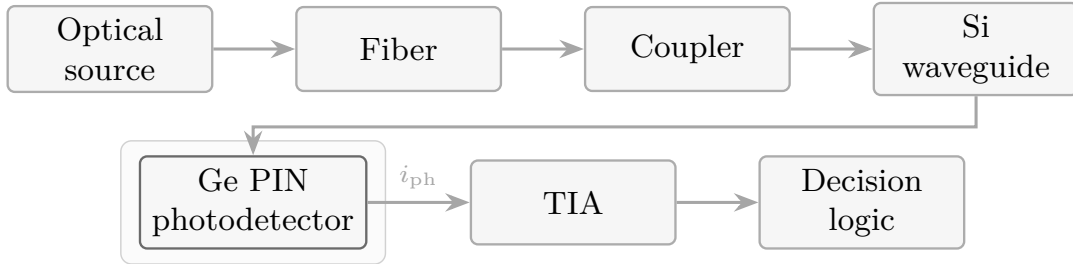


Figure 2: Optical receiver signal chain in a silicon photonics transceiver. The photodetector converts the guided-mode optical signal into a photocurrent i_{ph} , which is amplified by the TIA and processed by the DSP. The grating coupler and Si waveguide precede the Ge photodetector; performance at the coupler-detector interface determines the chip-level responsivity discussed throughout this paper.

The study is built on the U-shaped vertical Ge-on-Si photodetector demonstrated experimentally by Shi *et al.* [1], which holds the bandwidth record of 103 GHz for vertical Ge-on-Si devices. That architecture is taken as-is, layer stack, electrode geometry, and coupling scheme, and simulated at O-band conditions using a coupled multiphysics chain: three-dimensional finite-difference time-domain optical modelling, two-dimensional drift-diffusion electrical simulation, numerical post-processing for compact-model extraction, and a layout editor for geometry validation. No fabrication, RF on-wafer characterisation, or Monte Carlo process analysis is part of this work; all performance figures are simulation-derived and checked against the published record of Shi *et al.* [1].

the paper traces the physical reasoning from electromagnetic absorption and carrier transport through explicit trade-off quantification and into the simulation results, so that each design choice is grounded in a physical argument rather than accepted without explanation. This is an educational simulation study; the U-shaped vertical architecture, the layer stack, and the electrode geometry are taken directly from Shi *et al.* [1] and nothing new is added to the experimental record.

This paper covers the receiver photodetector branch only. Transmitter, modulator, DSP, and link-level analysis belong to companion papers and appear here solely as interface requirements. IEEE 802.3 400 GBASE-DR4 receiver sensitivity is used as a practical design reference throughout.

the paper is organised as follows. Section 2 establishes the physics of photodetection, deriving the Beer-Lambert absorption law from the Helmholtz wave equation, developing the drift-diffusion carrier transport framework through the coupled Poisson and continuity equations, and quantifying the recombination mechanisms that set the dark-current floor. Section 3 defines the evaluation language: responsivity, quantum efficiency, bandwidth (transit-time and RC limits), dark current, and the coupled trade-offs among them that connect every geometry parameter to a measurable performance figure. Section 4 surveys the main photodetector device classes, from the photoconductor through the avalanche photodiode, building the physical argument for choosing the waveguide-integrated PIN junction for high-speed integrated O-band links. Section 5 covers germanium-on-silicon as the absorber material, including its spectral suitability at 1310 nm, the CMOS-compatible epitaxial integration path, and the threading-dislocation challenge that limits the dark-current floor. Section 6 reviews the experimental record of Ge-on-Si waveguide photodetectors, tracing successive bandwidth improvements from early SOI demonstrations to the 103 GHz U-shaped benchmark of Shi *et al.* [1], and explains why that device is used as the study object. Section 7 presents the device architecture taken from Shi *et al.* [1], including the vertical PIN layer stack, the 40 μm adiabatic taper, and the U-shaped electrode topology, with the geometry-to-performance mapping at 1310 nm. Section 8 describes the multiphysics simulation chain: three-dimensional finite-difference time-domain optical modelling, two-dimensional drift-diffusion electrical simulation, numerical post-processing for compact-model extraction, and the validation checks that confirm the solver outputs are physically consistent. Section 9 discusses the simulation outputs, optical generation profiles, absorber-length and intrinsic-layer sweeps, current-voltage characteristics, bandwidth estimates, and sensitivity analysis,

and connects each result back to the physics developed earlier. Section 10 consolidates the key observations and explains the quantitative effect of the U-shaped electrode geometry as seen through the simulation and analytical framework of this paper.

2 Fundamentals of Photodetection

Photodetection starts when optical energy is absorbed by the semiconductor and converted into mobile charge. The useful device metrics in section 3 all trace back to three physical questions: how rapidly photons are absorbed, how the generated carriers move, and how many carriers are lost to recombination before they reach the terminals.

2.1 Optical Absorption in Semiconductors

Before tracing the electromagnetic derivation, it is useful to locate the operating photon energy on the Ge band structure, because the absorption mechanism, direct vs. indirect interband transition, determines both the magnitude of α and its spectral slope.

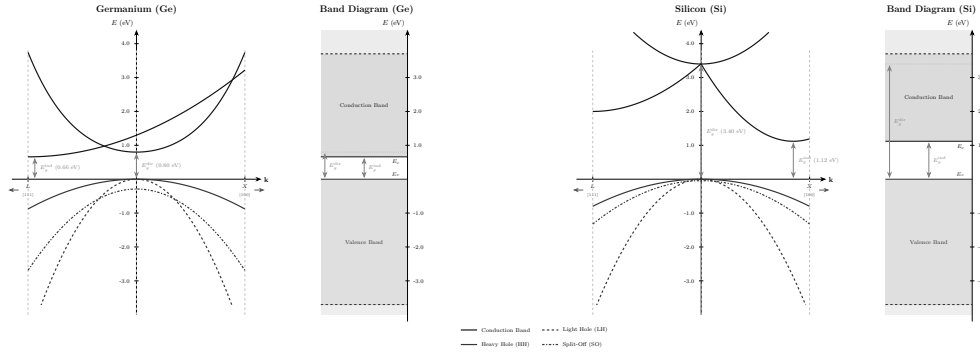


Figure 3: Schematic E - k band structure of bulk Germanium along the Γ - L and Γ - X directions at 300 K, and corresponding band diagram. The direct gap $E_{g,\text{dir}} = 0.80$ eV is smaller than the photon energy at 1310 nm ($\hbar\omega \approx 0.946$ eV), enabling direct interband absorption.

The starting point for all optical analysis in this paper is the macroscopic form of Maxwell's equations in a non-magnetic, source-free medium. In SI units and in the frequency domain, obtained by assuming a time dependence $e^{-i\omega t}$, these read:

$$\nabla \times \tilde{\mathbf{E}}(\mathbf{r}, \omega) = i\omega\mu_0 \tilde{\mathbf{H}}(\mathbf{r}, \omega), \quad (1)$$

$$\nabla \times \tilde{\mathbf{H}}(\mathbf{r}, \omega) = -i\omega\tilde{\epsilon}(\omega) \tilde{\mathbf{E}}(\mathbf{r}, \omega), \quad (2)$$

where μ_0 is the permeability of free space and $\tilde{\epsilon}(\omega)$ is the complex permittivity. Taking the curl of eq. (1) and substituting eq. (2) yields the Helmholtz wave equation:

$$\nabla^2 \tilde{\mathbf{E}}(\mathbf{r}, \omega) + \tilde{k}^2(\omega) \tilde{\mathbf{E}}(\mathbf{r}, \omega) = \mathbf{0}, \quad (3)$$

where $\tilde{k}(\omega) = \omega \sqrt{\mu_0 \tilde{\varepsilon}(\omega)}$. Defining the *complex refractive index*:

$$\tilde{n}(\omega) \equiv \frac{c \tilde{k}(\omega)}{\omega} = n(\omega) + i\kappa(\omega), \quad (4)$$

the plane-wave solution propagating along $+\hat{z}$ is:

$$\tilde{\mathbf{E}}(z, \omega) = \mathbf{E}_0 \exp\left[i n(\omega) \frac{\omega z}{c}\right] \exp\left[-\kappa(\omega) \frac{\omega z}{c}\right]. \quad (5)$$

The time-averaged Poynting intensity $I(z) \propto |\tilde{\mathbf{E}}|^2$ decays as $\exp[-2\kappa\omega z/c]$, giving the Beer-Lambert absorption coefficient:

$$\alpha(\omega) = \frac{2\omega\kappa(\omega)}{c} = \frac{4\pi\kappa(\lambda)}{\lambda}, \quad (6)$$

so that the Beer-Lambert law:

$$I(z) = I_0 \exp[-\alpha(\lambda) z] \quad (7)$$

emerges from eq. (3) rather than from an independent empirical assumption. The absorbed power density follows directly:

$$\mathcal{P}_{\text{abs}}(\mathbf{r}) = \alpha(\lambda) I(\mathbf{r}). \quad (8)$$

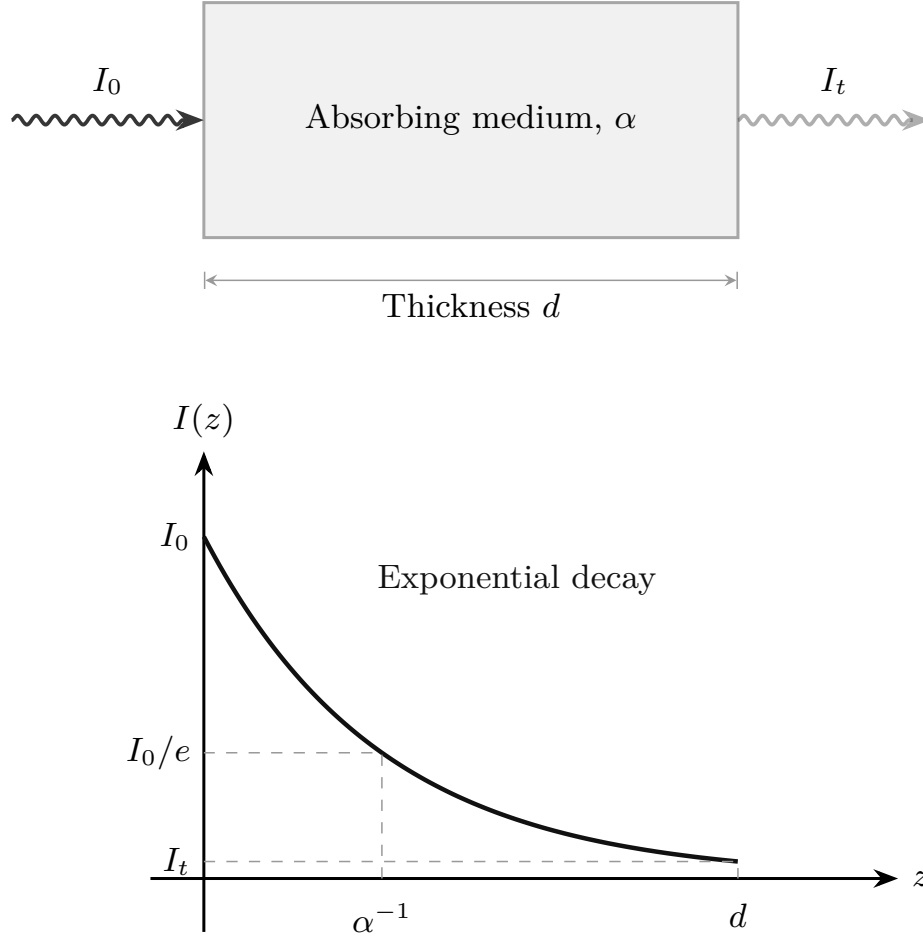


Figure 4: Beer-Lambert absorption profile. The intensity $I(z)$ decays exponentially with characteristic depth α^{-1} (top), and the volumetric generation rate $G_{\text{opt}}(z)$ mirrors this profile (bottom). The characteristic Ge absorption dimensions at 1310 nm are indicated.

Equation (6) shows that α is proportional to the extinction coefficient κ , a purely material quantity. Its quantitative wavelength dependence is characterised in the solver spectral outputs rather than duplicated as a standalone conceptual plot.

Poynting-vector formulation of the generation rate.

The optical generation rate $G_{\text{opt}}(\mathbf{r})$ is more generally obtained from the time-averaged divergence of the Poynting vector. The divergence theorem applied to the time-averaged Poynting flux $\langle \mathbf{S} \rangle = \frac{1}{2} \text{Re}[\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*]$ over a closed surface $\partial\Omega$ enclosing a volume Ω of absorbing material gives:

$$\int_{\Omega} G_{\text{opt}}(\mathbf{r}) \hbar\omega \, dV = - \oint_{\partial\Omega} \langle \mathbf{S} \rangle \cdot \hat{n} \, dA, \quad (9)$$

so that the local volumetric generation rate is the Poynting-flux sink:

$$G_{\text{opt}}(\mathbf{r}) = -\frac{\nabla \cdot \langle \mathbf{S}(\mathbf{r}) \rangle}{\hbar\omega} = \frac{\omega\varepsilon_0 \text{Im}[\tilde{\varepsilon}_r(\mathbf{r})]|\tilde{\mathbf{E}}(\mathbf{r})|^2}{2\hbar\omega}. \quad (10)$$

This expression reduces to eq. (15) under the plane-wave approximation, but it is the appropriate definition when the three-dimensional field contains interference terms from modal coupling, metal reflections, or end-facet diffraction, all of which occur in the adiabatic taper region studied here. The FDTD solver employed in section 8.2 evaluates G_{opt} from eq. (10); the scalar Beer-Lambert law of eq. (7) serves only as the analytical validation baseline.

Square-law photodetection and the domain of linear operation.

For a quasi-monochromatic field at carrier frequency ω_0 with slowly varying complex envelope $\tilde{s}(t)$:

$$\tilde{\mathbf{E}}(t) = \frac{1}{2}\tilde{s}(t)e^{-i\omega_0 t}\hat{\mathbf{e}} + \text{c.c.}, \quad (11)$$

the time-averaged Poynting intensity at the detector facet is $I(t) = |\tilde{s}(t)|^2/(2\eta_0)$, where $\eta_0 = \sqrt{\mu_0/\varepsilon_0}$. The photocurrent produced by the device is therefore:

$$i_{\text{ph}}(t) = \mathcal{R} \cdot P(t) = \mathcal{R} \cdot \frac{|\tilde{s}(t)|^2}{2\eta_0}, \quad (12)$$

which is *linear in optical power* $P(t)$ but *quadratic in optical field amplitude* $|\tilde{s}(t)|$. When the field contains both a data-bearing modulated component $\tilde{u}(t)$ and a residual DC carrier c , $\tilde{s}(t) = \tilde{u}(t) + c$, the photocurrent expands as:

$$i(t) = \mathcal{R} \left[\underbrace{c^2}_{\text{DC bias}} + \underbrace{2c\text{Re}\{\tilde{u}(t)\}}_{\text{linear signal term}} + \underbrace{|\tilde{u}(t)|^2}_{\text{SSBI}} \right], \quad (13)$$

where SSBI denotes signal-signal beating interference. The linear approximation $i_{\text{ph}} \approx \mathcal{R} \cdot P_{\text{opt}}$ is valid when the carrier-to-signal power ratio $\text{CSPR} \equiv c^2/\langle |\tilde{u}|^2 \rangle \gg 1$, so that SSBI is negligible relative to the cross-beat term.

Integrating eq. (8) through an absorber of length L gives the absorbed fraction:

$$\mathcal{A}(L) = 1 - \exp[-\alpha(\lambda)L]. \quad (14)$$

The absorption efficiency of the implemented device across the O-band, extracted from the FDTD solver, is presented alongside the material-selection argument in fig. 20.

2.2 Carrier Photogeneration and Transport

An absorbed photon creates an electron-hole pair. Whether that pair contributes to photocurrent or recombines inside the absorber depends on the electric field, carrier mobility, and defect density of the semiconductor.

The volumetric optical generation rate $G_{\text{opt}}(\mathbf{r})$ quantifies the number of electron-hole pairs created per unit volume per unit time:

$$G_{\text{opt}}(\mathbf{r}) = \frac{\mathcal{P}_{\text{abs}}(\mathbf{r})}{\hbar\omega} = \frac{\alpha(\lambda) I(\mathbf{r})}{\hbar\omega} \quad [\text{cm}^{-3} \text{s}^{-1}]. \quad (15)$$

Because $I(\mathbf{r})$ decays exponentially along the propagation direction via eq. (7), G_{opt} is spatially non-uniform: it is concentrated near the input facet of the absorber and falls off with an effective length set by the Ge material absorption, the optical confinement factor, and the simulated taper-to-Ge mode overlap.

Drift-diffusion transport.

The self-consistent carrier distribution and electrostatic field are governed by the coupled Poisson and drift-diffusion equations [2], [3]. Poisson's equation is:

$$\nabla \cdot [\varepsilon(\mathbf{r}) \nabla \phi(\mathbf{r}, t)] = -q[p - n + N_D^+ - N_A^-], \quad (16)$$

and the electron and hole current densities in the drift-diffusion approximation are:

$$\mathbf{J}_n = q\mu_n n \mathbf{E} + qD_n \nabla n, \quad (17)$$

$$\mathbf{J}_p = q\mu_p p \mathbf{E} - qD_p \nabla p, \quad (18)$$

where the carrier diffusivities follow the Einstein relation [2]:

$$D_{n,p} = \mu_{n,p} \frac{k_B T}{q}. \quad (19)$$

The general, time-dependent carrier continuity equations are:

$$\frac{\partial n}{\partial t} = \frac{1}{q} \nabla \cdot \mathbf{J}_n - U_{\text{net}} + G_{\text{opt}}, \quad (20)$$

$$\frac{\partial p}{\partial t} = -\frac{1}{q} \nabla \cdot \mathbf{J}_p - U_{\text{net}} + G_{\text{opt}}, \quad (21)$$

where $U_{\text{net}} = U_{\text{SRH}} + U_{\text{Auger}} + U_{\text{rad}}$. Under *steady-state* CW illumination, $\partial/\partial t \equiv 0$:

$$\frac{1}{q} \nabla \cdot \mathbf{J}_n = U_{\text{net}} - G_{\text{opt}}, \quad (22)$$

$$\frac{1}{q} \nabla \cdot \mathbf{J}_p = -(U_{\text{net}} - G_{\text{opt}}). \quad (23)$$

The complete self-consistent system, from eq. (16) through eq. (23), is the foundational PDE set solved numerically by the commercial finite-element drift-diffusion solver and reduced numerically for the geometry-derived bandwidth and noise models.

The drift-diffusion picture separates fast field-driven transport from slower concentration-gradient-driven motion. A sufficiently large reverse-bias field sweeps carriers across the

intrinsic region before recombination, whereas diffusion dominates in quasi-neutral regions outside the depletion zone and produces slow response tails when depletion is incomplete. A reverse-biased PIN junction suppresses that tail by confining useful photogeneration to the depleted region, making drift the dominant transport mechanism and carrier transit time the key speed parameter.

Figure 5 contrasts the two components of carrier current (eqs. (17) and (18)) spatially across a reverse-biased PIN junction, showing why the drift term dominates once the intrinsic layer is fully depleted.

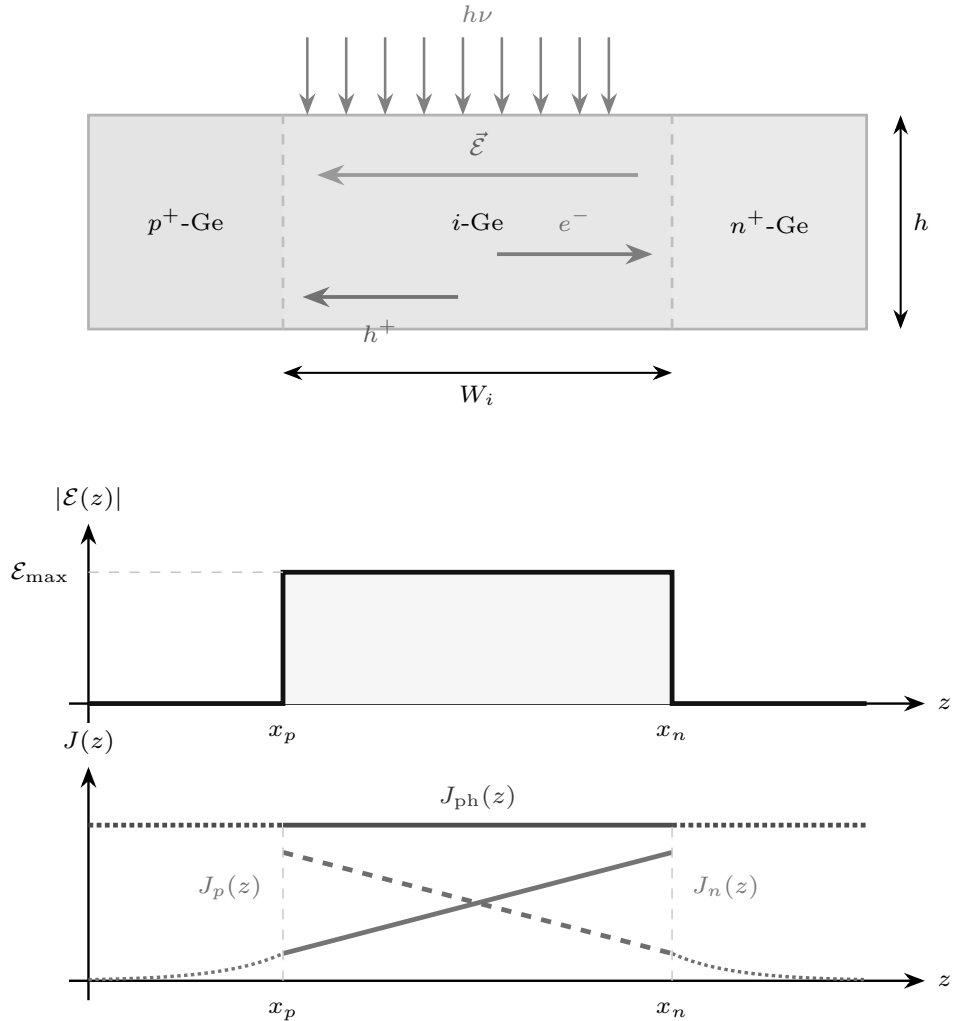


Figure 5: Spatial separation of drift-dominated and diffusion-dominated carrier transport in a reverse-biased PIN junction. Within the depleted intrinsic layer, the electric field is high and drift dominates, providing fast carrier sweep-out. In the quasi-neutral contact regions, transport is slow and diffusion-dominated.

Velocity saturation and field-dependent mobility.

An important refinement to the linear drift picture arises at the field magnitudes present in a reverse-biased Ge PIN junction. The carrier drift velocity does not increase indefinitely with electric field but saturates at $v_{d,\text{sat}}$ when the field exceeds a critical value $\mathcal{E}_c$. A widely adopted empirical model for the field-dependent drift velocity is the Caughey-Thomas expression [2]:

$$v_{n,p}(\mathcal{E}) = \frac{\mu_{n,p}^{(0)} \mathcal{E}}{\left[1 + \left(\mu_{n,p}^{(0)} \mathcal{E} / v_{d,\text{sat},n/p}\right)^{\beta_{n,p}}\right]^{1/\beta_{n,p}}}, \quad (24)$$

where $\mu_{n,p}^{(0)}$ is the low-field mobility, $v_{d,\text{sat},n/p}$ is the saturation velocity, and $\beta_{n,p}$ is an empirical shape parameter. For germanium at 300 K, representative values are $\mu_n^{(0)} \approx 3900 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, $\mu_p^{(0)} \approx 1900 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, $v_{d,\text{sat},e} \approx 6.0 \times 10^6 \text{ cm s}^{-1}$, and $v_{d,\text{sat},h} \approx 4.0 \times 10^6 \text{ cm s}^{-1}$, with $\beta \approx 2$ for electrons and $\beta \approx 1$ for holes [2]. At a reverse bias of 2 V across a 1 μm intrinsic layer, the mean field is 20 kV cm^{-1} , which substantially exceeds the critical field $\mathcal{E}_c = v_{d,\text{sat}}/\mu^{(0)} \approx 1 \text{ kV cm}^{-1}$ to 3 kV cm^{-1} and places the device firmly in the velocity-saturated regime. The constant velocity assumption adopted in eq. (55) and the analytical transfer function of eq. (25) is therefore well justified at standard operating bias, and the transit time is accurately estimated as $\tau_{\text{tr}} = W_i/v_{d,\text{sat}}$ rather than the low-field value $W_i^2/(\mu V)$.

Analytical photocurrent transfer function.

In the frequency domain, neglecting diffusion and assuming a uniform electric field at saturation value throughout $z \in [0, W_i]$, the Fourier amplitude of the terminal current density admits a closed-form expression:

$$\begin{aligned} J_t(\omega) = & qG_0(\omega)W_i \\ & \times \left\{ \frac{e^{-\alpha W_i} - 1}{\alpha W_i(\alpha W_i - j\omega\tau_{\text{tr},h})} + e^{-\alpha W_i} \frac{e^{j\omega\tau_{\text{tr},h}} - 1}{j\omega\tau_{\text{tr},h}(\alpha W_i - j\omega\tau_{\text{tr},h})} \right. \\ & \left. + \frac{1 - e^{-\alpha W_i}}{\alpha W_i(j\omega\tau_{\text{tr},e} + \alpha W_i)} + \frac{1 - e^{j\omega\tau_{\text{tr},e}}}{j\omega\tau_{\text{tr},e}(j\omega\tau_{\text{tr},e} + \alpha W_i)} \right\}, \end{aligned} \quad (25)$$

where $G_0(\omega) = \frac{\alpha\eta(1-R_{\text{refl}})\lambda P_{\text{opt}}(\omega)}{hcA_j}$ is the carrier generation rate amplitude at the illuminated front interface, and $\tau_{\text{tr},e} = W_i/v_{d,\text{sat},e}$ and $\tau_{\text{tr},h} = W_i/v_{d,\text{sat},h}$ are the electron and hole transit times. In the limits $\omega\tau_{\text{tr},h} \ll 1$ and $\alpha W_i \gg 1$, eq. (25) reduces to the rectangular-pulse sinc-function roll-off yielding the bandwidth formula of eq. (55).

The frequency response evaluated from these model parameters at the design point is presented with the electrode design discussion in fig. 24, after the compact-circuit parameters have been introduced.

Derivation of the factorised small-signal transfer function.

Consider an electron-hole pair generated at depth $z \in [0, W_i]$ inside the intrinsic layer. Under the uniform-field, velocity-saturated approximation, the electron and hole transit times are:

$$\tau_e(z) = \frac{z}{v_{d,\text{sat},e}}, \quad \tau_h(z) = \frac{W_i - z}{v_{d,\text{sat},h}}. \quad (26)$$

By the Ramo-Shockley theorem, the terminal current induced by each carrier is non-zero only while the carrier is moving, giving the local impulse response:

$$h_z(t) = \frac{qv_{d,\text{sat},e}}{W_i} u(t) u[\tau_e(z) - t] + \frac{qv_{d,\text{sat},h}}{W_i} u(t) u[\tau_h(z) - t], \quad (27)$$

where $u(t)$ is the Heaviside function.

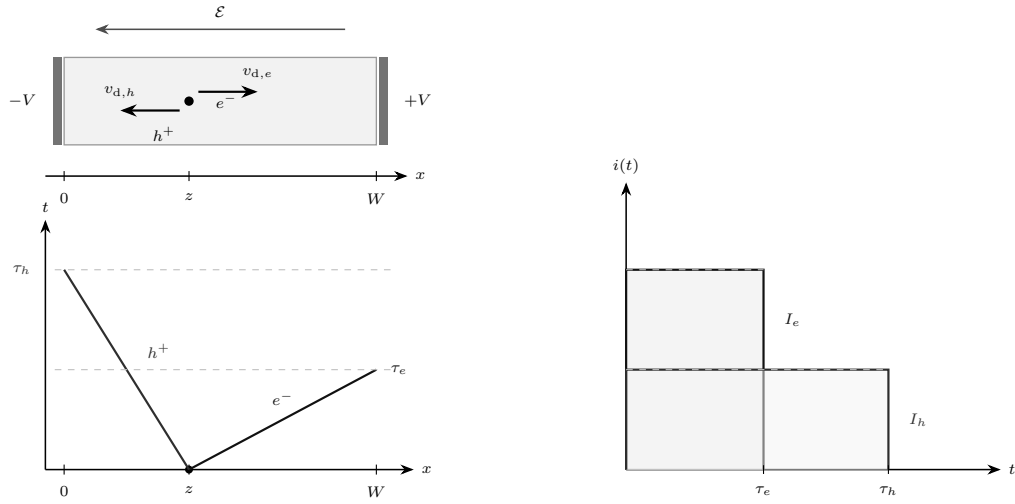


Figure 6: Ramo-Shockley induced terminal current. Drift trajectories of an electron-hole pair generated at depth z are shown (top), along with the corresponding induced current pulses (bottom). The slower hole transit governs the tail of the total impulse response.

Fourier transformation yields:

$$H_z(\omega) = \frac{q}{W_i} \left[\frac{1 - e^{-j\omega\tau_e(z)}}{j\omega} + \frac{1 - e^{-j\omega\tau_h(z)}}{j\omega} \right]. \quad (28)$$

If the optical generation is approximately uniform across the depletion depth, the transit response follows from averaging eq. (28) over z :

$$H_{\text{tt}}(\omega) \propto \int_0^{W_i} H_z(\omega) \frac{dz}{W_i}. \quad (29)$$

The integral evaluates to a weighted sum of electron and hole sinc terms. In the practically relevant limit where the slower hole branch dominates the 3 dB roll-off, the envelope reduces

to the single-sinc form used in eq. (62). The electrical loading is then appended through the junction capacitance and total series resistance. For a current source driving C_j into $R_{\text{tot}} = R_s + R_L$:

$$H_{\text{RC}}(\omega) = \frac{1}{1 + j\omega R_{\text{tot}} C_j} = \frac{1}{1 + j\omega/\omega_{\text{RC}}}, \quad (30)$$

with $\omega_{\text{RC}} = (R_{\text{tot}} C_j)^{-1}$. The total response is the product:

$$H_{\text{PD}}(\omega) = H_{\text{tt}}(\omega) H_{\text{RC}}(\omega), \quad (31)$$

which is the physical content hidden inside the compact form of eq. (62): transport determines the sinc-like envelope, while the external electrical boundary condition determines the low-pass pole.

Scharfetter-Gummel discretisation and numerical stability.

The finite-element discretisation of eqs. (17) and (18) on a non-uniform mesh faces a well-known numerical instability when the local Péclet number $\text{Pe} = q|\mathbf{E}|\Delta x/(2kT)$ exceeds unity. The Scharfetter-Gummel scheme approximates the current between adjacent mesh nodes i and $i + 1$ as:

$$J_{n,i+1/2} = \frac{qD_n}{\Delta x} \left[B\left(\frac{q\Delta\phi}{kT}\right) n_{i+1} - B\left(-\frac{q\Delta\phi}{kT}\right) n_i \right], \quad (32)$$

where $B(u) = u/(e^u - 1)$ is the Bernoulli function and $\Delta\phi = \phi_{i+1} - \phi_i$. This scheme recovers centred differences for $|u| \ll 1$ (low-field) and upstream weighting for $|u| \gg 1$ (high-field), providing second-order accuracy and unconditional positivity across the 30 kV cm^{-1} field regime of the present device.

2.3 The PIN Junction: Electrostatics and Operating Regime

The vertical PIN architecture, a lightly doped or nominally intrinsic germanium absorber sandwiched between heavily doped p^{++} and n^{++} contact layers, is the canonical structure for high-speed photodetection. Its central merit is the ability to sustain a large, spatially uniform electric field across the entire absorber volume under modest reverse bias, simultaneously achieving full depletion, velocity-saturated carrier transport, and minimal free-carrier absorption.

2.3.1 Depletion Approximation and Built-in Potential

In the absence of applied bias, the Fermi levels of the p^{++} and n^{++} regions equilibrate by charge redistribution, establishing a built-in potential:

$$V_{\text{bi}} = \frac{\Phi_{\text{Si}} - \Phi_{\text{Ge}}}{q}, \quad (33)$$

where Φ_{Si} and Φ_{Ge} are the work functions of the p^{++} -Si platform and n^{++} -Ge cathode, respectively. At 300 K, the intrinsic carrier concentration $n_i^{\text{Ge}} \approx 2.4 \times 10^{13} \text{ cm}^{-3}$ is approximately three orders of magnitude larger than in silicon, which has important consequences for dark current as discussed in section 2.4. For typical contact doping levels of $N_{A,D} \approx 10^{19} \text{ cm}^{-3}$, and incorporating the band offsets at the Si/Ge interface, the heterojunction built-in potential is $V_{\text{bi}} \approx 0.68 \text{ V}$ (compared to $\approx 0.51 \text{ V}$ for a Ge-only homojunction).

Under the depletion approximation, the space charge in the p^{++} and n^{++} layers is assumed to be that of fully ionised impurities with sharp depletion edges. In the ideal case where the intrinsic layer is fully depleted at zero bias by the built-in field, the charge neutrality condition reads:

$$N_A x_p = N_D x_n, \quad (34)$$

where x_p and x_n are the partial penetrations of the depletion region into the p^{++} and n^{++} layers, respectively. Since $N_{A,D} \gg n_i$, the depletion extension into the heavily doped contact layers is negligible ($x_p, x_n \ll W_i$), and the electric field profile is approximately uniform across the intrinsic layer with peak value:

$$\mathcal{E}_{\text{max}} = \frac{q N_A x_p}{\varepsilon_0 \varepsilon_{\text{Ge}}} \approx \frac{V_{\text{bi}} + |V_{\text{bias}}|}{W_i}, \quad (35)$$

where the second equality applies in the fully depleted regime. For $V_{\text{bias}} = -2 \text{ V}$ and $W_i = 1 \mu\text{m}$, eq. (35) gives $\mathcal{E}_{\text{max}} \approx 25 \text{ kV cm}^{-1}$, confirming velocity saturation. The flatness of the field profile is a key advantage of the PIN geometry over a simple pn junction: it ensures that every photogenerated carrier, regardless of its generation depth within $[0, W_i]$, experiences the same drift force and therefore the same transit time, making eq. (26) well-conditioned for all z .

To appreciate the advantage of the PIN topology, it is instructive to contrast its electrostatics with those of a simple PN junction of the same total doping (fig. 7).

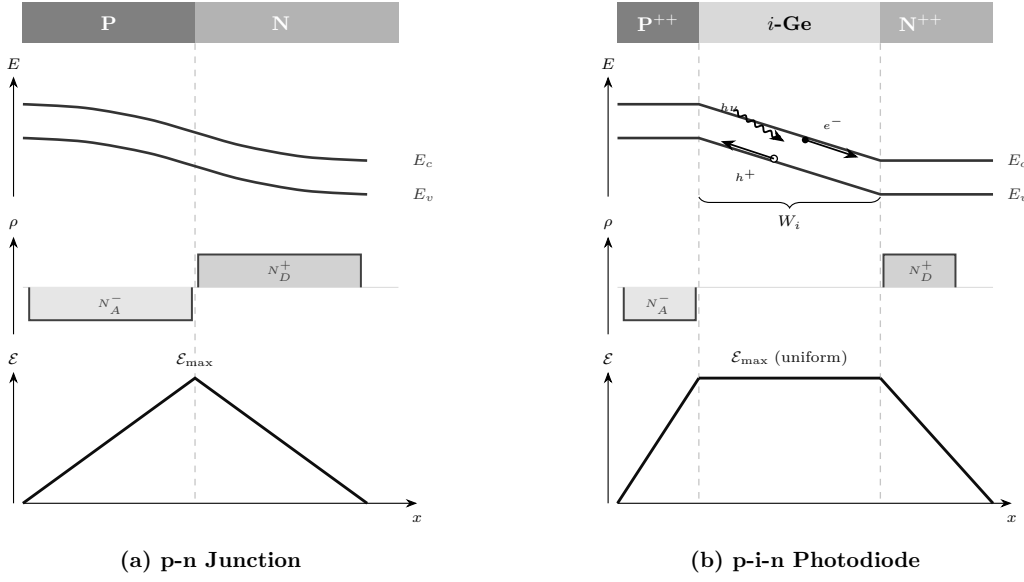


Figure 7: Side-by-side comparison of depletion-region electrostatics for a PN junction (left column) and a PIN junction of equal doping (right column), showing doping profiles, space-charge density, electric field, and electrostatic potential. The PIN junction provides a uniform electric field across the intrinsic layer, eliminating transit-time dispersion.

2.3.2 Dark Current Mechanisms and Decomposition

The dark current I_d under reverse bias is the primary figure of merit for detector noise and static power dissipation. In a Ge-on-Si PIN device, it receives contributions from both the junction bulk volume and the mesa perimeter, and it is important to distinguish these because they scale differently with device geometry:

$$I_d = J_{\text{bulk}} A_j + J_{\text{surf}} P_j, \quad (36)$$

where J_{bulk} is the bulk dark-current density, J_{surf} is an effective surface dark-current line density, and P_j is the mesa perimeter [2]. The bulk component in the fully depleted intrinsic region is dominated by SRH thermal generation (as established in section 2.4) and is given by:

$$J_{\text{bulk}} = q U_{\text{SRH,dep}} W_i = \frac{q n_i W_i}{2\tau_0}, \quad (37)$$

which scales linearly with W_i and inversely with the effective SRH lifetime τ_0 . The surface component arises from interface states at the Ge/SiO₂ sidewall and Ge/Si heterointerface and is written as:

$$J_{\text{surf}} = q v_s n_i, \quad (38)$$

where v_s is the effective surface recombination velocity. Because n_i of germanium is orders of magnitude larger than that of silicon, both J_{bulk} and J_{surf} in Ge-on-Si are intrinsically larger than in silicon-only detectors by the same factor, placing a fundamental material

constraint on achievable dark current density. As device dimensions are scaled to improve bandwidth, reducing A_j and P_j together but with $A_j/P_j \propto W_j$ decreasing, the surface term becomes proportionally more important, and surface passivation through dielectric encapsulants such as Al_2O_3 or GeO_x becomes critical. The decomposition of eq. (36) is exploited in the calibration procedure of section 8.3, where measurements on devices of varying junction width W_j are used to extract J_{bulk} and J_{surf} independently.

2.4 Recombination and Loss Mechanisms

Three recombination mechanisms are quantitatively relevant in the germanium absorber [2], [4].

Shockley-Read-Hall (SRH) recombination, mediated by deep-level trap states, is [2], [5]:

$$U_{\text{SRH}} = \frac{np - n_i^2}{\tau_p(n + n_1) + \tau_n(p + p_1)}, \quad (39)$$

where the effective trap-state carrier densities are:

$$n_1 = n_i \exp\left[\frac{E_t - E_i}{k_B T}\right], \quad p_1 = n_i \exp\left[\frac{E_i - E_t}{k_B T}\right]. \quad (40)$$

Fossum doping-dependent SRH lifetime.

In the presence of a background doping $N_A + N_D$, the SRH carrier lifetime is not constant but falls with increasing impurity concentration. The Fossum model [5] captures this dependence as:

$$\tau_{\text{SRH},n/p} = \frac{\tau_{\text{SRH},n/p}^{(0)}}{a_1 + a_2 \mathcal{N}_{n/p} + a_3 \mathcal{N}_{n/p}^{a_4}}, \quad \mathcal{N}_{n/p} = \frac{N_A + N_D}{N_{n/p}^{(\text{ref})}}, \quad (41)$$

where $\tau_{\text{SRH},n/p}^{(0)}$ is the low-concentration lifetime and $a_1, \dots, a_4$ are fitting parameters. In the limit $a_2 = 0$, $a_1 = a_3 = a_4 = 1$, eq. (41) reduces to the standard Shockley-Read-Hall result $\tau = \tau^{(0)}/(1 + N/N^{(\text{ref})})$. For the heavily doped n^{++} -Ge cathode at $N_D = 10^{19} \text{ cm}^{-3}$, eq. (41) gives a lifetime reduction of approximately one order of magnitude relative to $\tau^{(0)}$, confirming that carrier recombination is concentrated in the contact region rather than the intrinsic absorber. Within the fully depleted intrinsic region at the midgap trap condition $E_t = E_i$, eq. (39) reduces to:

$$U_{\text{SRH,dep}} \approx \frac{n_i}{\tau_n + \tau_p} \equiv \frac{n_i}{2\tau_0}, \quad (42)$$

where $\tau_0 = (\tau_n + \tau_p)/2$. SRH is the dominant mechanism in Ge-on-Si because the threading dislocations formed during heteroepitaxial growth act as efficient midgap trap centres.

Auger recombination is [2], [5]:

$$U_{\text{Auger}} = (C_n n + C_p p)(np - n_i^2), \quad (43)$$

where C_n and C_p are the electron and hole Auger capture coefficients. The combined effective carrier lifetime accounting for both SRH and Auger mechanisms is:

$$\frac{1}{\tau_{n/p}} = \frac{1}{\tau_{\text{Auger},n/p}} + \frac{1}{\tau_{\text{SRH},n/p}}, \quad (44)$$

where $\tau_{\text{Auger},n/p}$ is derived from eq. (43) evaluated at the local carrier densities [2], [5]. Equation (44) defines the effective recombination time used in the drift-diffusion simulation of section 8.3.

The three mechanisms operate through physically distinct band-to-band paths; fig. 8 places all three on a common energy-band canvas so their relative rates (eqs. (39), (43) and (45)) can be interpreted intuitively.

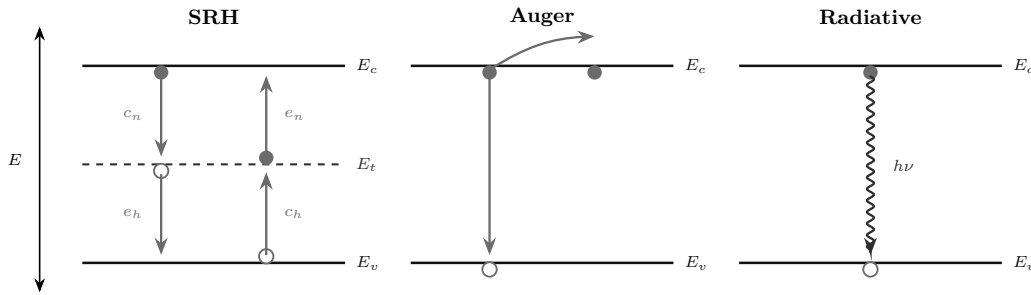


Figure 8: The three recombination mechanisms in the Ge absorber, shown on a simplified E - x band diagram. Shockley-Read-Hall (SRH) recombination dominates in the depleted region, while Auger and radiative processes are negligible due to the indirect band gap of Germanium and low carrier concentrations.

Radiative recombination is modelled as:

$$U_{\text{rad}} = r_{\text{opt}}(np - n_i^2), \quad (45)$$

where $r_{\text{opt}}^{\text{Ge}} \approx 6.4 \times 10^{-14} \text{ cm}^3 \text{ s}^{-1}$ at 300 K, approximately four orders of magnitude smaller than for a direct-gap III-V of comparable bandgap. In the fully depleted intrinsic region, where $np \approx n_i^2$, the radiative rate vanishes identically and eq. (42) constitutes the sole generation mechanism. Simulated generation and recombination rates confirm that SRH exceeds Auger by more than two orders of magnitude and radiative recombination by more than five orders of magnitude across $0 \leq |V_{\text{bias}}| \leq 3 \text{ V}$. This justifies the single-parameter calibration strategy of section 8.3, in which only τ_0 is adjusted to reproduce I_d .

Surface recombination at the Ge/SiO₂ and Ge/Si interfaces constitutes a separate loss channel characterised by the surface recombination velocity v_s . Its contribution to the

dark current is proportional to the mesa perimeter P_j rather than the junction area A_j , as formalised in eqs. (36) and (38). As device dimensions shrink to improve bandwidth, the area-to-perimeter ratio A_j/P_j decreases and surface recombination becomes increasingly significant, setting a practical lower bound on the achievable dark current density that surface passivation, for example through Ge_xO_y or Al_2O_3 , must address.

2.5 Design Synthesis: Linking Physics to Device Targets

The physical models above can now be condensed into the parameter dependencies that matter for device design before the numerical models are introduced.

The absorber-length/bandwidth Pareto front.

For a waveguide-integrated detector, optical absorption and bandwidth are in direct tension through the absorber length L . Responsivity benefits from long L via $\mathcal{A}(L) = 1 - e^{-\alpha L}$, while bandwidth degrades primarily through the junction capacitance $C_j \propto A_j = W_j \cdot L$, which enters the RC bandwidth eq. (58). Differentiating $f_{3\text{dB}}$ with respect to L at fixed W_i and W_j yields the responsivity-bandwidth product as the relevant composite figure of merit:

$$\mathcal{R} \cdot f_{3\text{dB}} \approx \frac{q\lambda}{hc} (1 - e^{-\alpha L}) \cdot \left[\frac{1}{f_{\text{tt}}^2} + \frac{(2\pi R_{\text{tot}} \varepsilon_0 \varepsilon_{\text{Ge}} W_j L)^2}{W_i^2} \right]^{-1/2}, \quad (46)$$

which is maximised by selecting L such that the RC pole is pushed to roughly the transit-time pole, i.e. $f_{\text{RC}} \approx f_{\text{tt}}$. For the 400GBASE-DR4 target requiring a per-lane bandwidth of 53 GHz at PAM4 signalling, and a responsivity floor of $\mathcal{R} \geq 0.8 \text{ A W}^{-1}$, eq. (46) constrains the design space to approximately $L \in [10 \mu\text{m}, 20 \mu\text{m}]$ and $W_i \in [0.5 \mu\text{m}, 1.0 \mu\text{m}]$ for $W_j = 1 \mu\text{m}$ in the vertical PIN geometry.

Bias voltage and depletion completeness.

The applied reverse bias V_{bias} must be large enough to fully deplete the intrinsic layer and drive the carriers into velocity saturation. The full-depletion voltage in the ideal PIN approximation is [2]:

$$V_{\text{FD}} = \frac{q N_{\text{bg}} W_i^2}{2\varepsilon_0 \varepsilon_{\text{Ge}}} - V_{\text{bi}}, \quad (47)$$

where N_{bg} is the unintentional background doping of the nominally intrinsic layer, which is never zero in practice due to incorporation of boron or phosphorus during epitaxial growth. For $N_{\text{bg}} \approx 10^{16} \text{ cm}^{-3}$ and $W_i = 1 \mu\text{m}$, eq. (47) gives $V_{\text{FD}} \approx 0.27 \text{ V}$, well below the standard operating voltage of 1 V, confirming that full depletion is achieved at low bias for realistic material quality. Above V_{FD} , the capacitance becomes approximately voltage-independent at the geometric value of eq. (57), which simplifies RC bandwidth analysis. Operation below V_{FD} should be avoided as it introduces a bias-dependent capacitance and incomplete carrier collection, degrading both bandwidth and responsivity in a non-linear, difficult-to-model manner.

Temperature dependence and thermal stability.

The intrinsic carrier concentration of germanium follows $n_i \propto T^{3/2} \exp(-E_g/2k_B T)$, where $E_g(T)$ itself decreases with temperature according to the Varshni relation. Because n_i^{Ge} at room temperature is already large, even modest self-heating in a high-power optical link can substantially elevate the dark current through eq. (37), degrading the SNR floor via the shot noise of eq. (70). In a 400 Gbit s⁻¹ DR4 transceiver module, thermal management of the receiver die is therefore a module-level thermal constraint, not merely a device-level footnote. The temperature coefficient of I_d in Ge-on-Si is typically measured to be in the range 5 % K⁻¹ to 8 % K⁻¹ near room temperature, significantly steeper than silicon photodiodes, and this sensitivity motivates both the careful thermal modelling performed alongside the electrical characterisation of section 8.3 and the use of bias-point tuning to compensate for operating-temperature drift in deployed link hardware.

3 Performance Metrics and Design Trade-offs

The transport and absorption physics of section 2 lead to the engineering metrics used to judge the detector. These quantities are coupled: improving one often shifts the burden onto another. The definitions in this section provide the common language used later to compare architectures, materials, and the proposed geometry.

3.1 Responsivity and Quantum Efficiency

3.1.1 Physical Basis and Definition

The basic function of a photodetector is to convert optical power into a measurable electrical current. This transduction is characterised by the *external responsivity* $\mathcal{R}$, defined as the ratio of the generated photocurrent I_{ph} to the incident optical power P_{opt} :

$$\mathcal{R} \equiv \frac{I_{\text{ph}}}{P_{\text{opt}}} \quad [\text{A W}^{-1}]. \quad (48)$$

The qualifier *external* matters because $\mathcal{R}$ accounts for all loss processes that occur between the guided optical mode incident upon the device and the photocurrent delivered to the readout circuit. It therefore includes coupling loss from the silicon waveguide into the germanium absorber, reflection at the Si/Ge interface, incomplete absorption along the device length, and any carrier recombination prior to collection at the contacts.

The upper bound on responsivity is derived from energy conservation. If every incident photon of energy $h\nu = hc/\lambda$ were to generate exactly one electron-hole pair and that pair were collected without loss, then the photocurrent per unit optical power would be:

$$\mathcal{R}_{\text{max}} = \frac{q\lambda}{hc} \approx 1.057 \text{ A W}^{-1} \quad \text{at } 1310 \text{ nm}, \quad (49)$$

where q is the elementary charge, h is Planck's constant, and c is the speed of light in vacuum. The linear dependence of $\mathcal{R}_{\max}$ on wavelength λ is physically transparent: longer-wavelength photons carry less energy per quantum, so generating one electron of charge q requires less power, yielding a larger current per watt. At the O-band centre wavelength of 1310 nm, this ceiling is approximately 1.06 A W^{-1} . Real devices fall meaningfully below this value, and the gap between $\mathcal{R}$ and $\mathcal{R}_{\max}$ is the principal object of optimisation in the optical coupling and absorber geometry.

The spectral responsivity plot in fig. 20 later shows this ceiling together with the FDTD-derived operating response, so the metric definition is not duplicated by a separate conceptual graph here.

3.1.2 Quantum Efficiency: External and Internal

The responsivity can be normalised to the single-photon limit through the concept of quantum efficiency. The *external quantum efficiency* (EQE) η_{ext} is the ratio of the rate of collected electrons to the rate of photons incident on the device, and is directly related to responsivity by:

$$\eta_{\text{ext}} = \mathcal{R} \cdot \frac{hc}{q\lambda}. \quad (50)$$

This quantity captures all loss mechanisms between the input waveguide facet and the electrical output terminal. It is the primary figure of merit at the system level because it directly sets the signal current available to the subsequent transimpedance amplifier.

The *internal quantum efficiency* (IQE) η_{int} isolates the carrier collection process from the optical coupling and absorption steps:

$$\eta_{\text{int}} = \frac{I_{\text{ph}}/q}{P_{\text{abs}}/\hbar\omega}, \quad (51)$$

where P_{abs} is the optical power actually absorbed within the Ge layer and $\hbar\omega$ is the photon energy. A device with high EQE may still exhibit a depressed IQE if significant absorption occurs in passive regions from which carriers cannot be extracted, or conversely may have high IQE but low EQE due to poor optical coupling. Distinguishing the two is important for correctly attributing performance shortfalls to either optical design (coupling and absorption) or electrical design (carrier transport and collection).

In the fully depleted intrinsic Ge layer under reverse bias, the electric field exceeds the carrier velocity saturation threshold throughout the absorber volume. The resulting drift transit time $\tau_{\text{tr}} = W_i/v_{\text{d,sat},h}$, governed by the slower hole saturation velocity $v_{\text{d,sat},h}$, is typically tens of picoseconds for intrinsic layer thicknesses W_i in the sub-500 nm regime. Because this is many orders of magnitude shorter than the Shockley–Read–Hall (SRH) minority-carrier lifetime $\tau_0 \sim 1 \text{ ns}$ in device-grade epitaxial Ge, virtually all photo-generated carriers are swept across the junction before recombination can occur. The resulting carrier

collection efficiency is:

$$\xi \approx 1 - \frac{\tau_{\text{tr}}}{\tau_0} \approx 1, \quad (52)$$

confirming that the IQE is essentially unity in a properly depleted device, provided the electric field is sufficiently uniform and the surface recombination velocity at the Ge/Si heterojunction is kept low by careful interface passivation.

3.1.3 The EQE Chain and Its Design Implications

The connection between the physical loss mechanisms and the measurable EQE is made precise by the following chain product:

$$\eta_{\text{ext}} = (1 - R_{\text{refl}}) \eta_c \xi (1 - e^{-\alpha(\lambda)L}), \quad (53)$$

where each factor corresponds to a distinct physical stage. The term $(1 - R_{\text{refl}})$ accounts for Fresnel reflection at the Si/Ge entry facet, arising from the refractive index contrast ($n_{\text{Si}} \approx 3.48$, $n_{\text{Ge}} \approx 4.27$ at 1310 nm); this factor is typically in the range 0.97–0.99 for waveguide-integrated devices where the optical mode is guided through a tapered transition rather than impinging on an abrupt cleaved facet. The coupling efficiency η_c quantifies the spatial overlap between the mode profile of the silicon access waveguide and the fundamental mode supported by the Ge absorber rib; its value is sensitive to cross-sectional geometry, waveguide width mismatch, and the design of any intermediate taper structure. The collection efficiency ξ has already been established as approximately unity under full depletion. Finally, the factor $(1 - e^{-\alpha(\lambda)L})$ is the fraction of the guided optical power absorbed over a propagation length L through a medium with effective power absorption coefficient $\alpha_{\text{eff}}(\lambda)$; at 1310 nm, the effective waveguide-mode absorption coefficient is $\alpha_{\text{eff}} \approx 3 \times 10^3 \text{ cm}^{-1}$, which is related to the bulk Ge absorption coefficient $\alpha_{\text{bulk}} \approx 7 \times 10^3 \text{ cm}^{-1}$ by the optical confinement factor in the Germanium core ($\alpha_{\text{eff}} = \Gamma \alpha_{\text{bulk}}$ with $\Gamma \approx 0.43$). This yields an effective $1/e$ absorption length of approximately 3.3 μm . For a 10 μm -long device, the single-pass absorption factor approaches 0.95, indicating that a compact absorber can achieve near-complete photon collection without the need for cavity enhancement.

The factored form of eq. (53) is most clearly understood as a cascade of four independent loss stages, each removing a distinct photon or carrier fraction before the electrical terminal is reached (fig. 9).

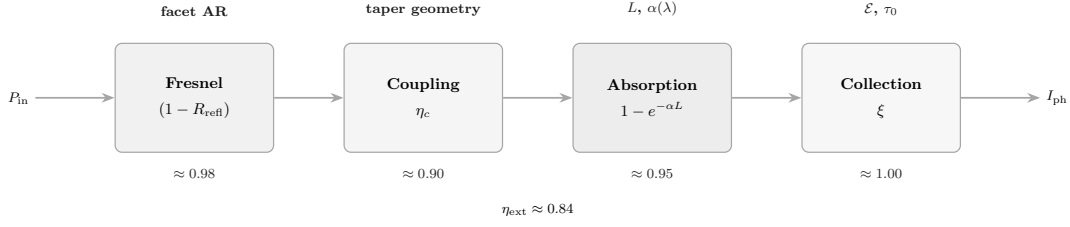


Figure 9: External quantum efficiency loss chain represented as a cascade of four multiplicative stages: Fresnel reflection, optical coupling, absorption completeness, and carrier collection. Design levers and representative values for the nominal device are annotated, yielding an external quantum efficiency of $\eta_{\text{ext}} \approx 84\%$.

Understanding eq. (53) as a product of separable factors is strategically important: it allows independent optimisation of the coupling optics, the absorber geometry, and the carrier transport, without conflating distinct physical mechanisms. It also reveals which term dominates the EQE deficit in any particular design iteration, enabling targeted improvements. For the waveguide-integrated, vertical PIN architecture considered throughout this work, the dominant EQE loss is typically associated with $\eta_c < 1$ arising from mode mismatch at the Si-to-Ge transition, rather than from incomplete absorption or finite carrier lifetime.

3.2 Bandwidth: Transit-Time and RC Limits

3.2.1 Overview of Bandwidth-Limiting Mechanisms

The ability of a photodetector to faithfully follow a modulated optical signal at high frequencies is characterised by its 3 dB electrical bandwidth $f_{3\text{dB}}$. In a reverse-biased PIN photodiode, this bandwidth is fundamentally limited by two physically distinct and largely independent mechanisms: the finite time required for photo-generated carriers to traverse the depletion region (the *transit-time* limit), and the charging time of the junction capacitance through the total series resistance (the *RC* limit). Both mechanisms produce a roll-off in the photocurrent frequency response, and in the absence of deliberate inductive resonance peaking, the combined bandwidth is well described by the additive-inverse-quadrature expression [6]:

$$f_{3\text{dB}} = \left(\frac{1}{f_{\text{tt}}^2} + \frac{1}{f_{\text{RC}}^2} \right)^{-1/2}. \quad (54)$$

This form, which adds the power spectral contributions of two first-order-like roll-offs in quadrature, correctly reflects the fact that both mechanisms must be simultaneously satisfied: a device with $f_{\text{tt}} \gg f_{\text{RC}}$ is RC-limited, while one with $f_{\text{RC}} \gg f_{\text{tt}}$ is transit-time-limited. The engineering challenge is to design the geometry such that both limits coincide, maximising the overall bandwidth for a given device footprint.

Transit-time limit.

When an electron–hole pair is photo-generated within the depleted intrinsic layer, both carriers begin drifting in opposite directions under the action of the built-in and externally applied electric field. The photocurrent induced in the external circuit is not instantaneous but persists until all carriers have traversed the junction and been collected at their respective ohmic contacts. For a spatially uniform generation profile and carriers moving at their saturation velocities, the current impulse response is approximately rectangular in time. The Fourier transform of a rectangular pulse of duration τ_{tr} is a sinc function, whose first zero occurs at $f = 1/\tau_{\text{tr}}$, giving a 3 dB frequency:

$$f_{\text{tt}} = \frac{0.45 v_{\text{d,sat},h}}{W_i}, \quad (55)$$

where $v_{\text{d,sat},h} \approx 4 \times 10^6 \text{ cm s}^{-1}$ is the hole saturation velocity in Ge, which is the lower of the two carrier saturation velocities ($v_{\text{d,sat},e} \approx 6 \times 10^6 \text{ cm s}^{-1}$), and W_i is the intrinsic layer thickness. The numerical prefactor 0.45 is a standard heuristic value that is very close to the analytical single-carrier sinc roll-off factor of 0.443, and it accounts for the asymmetric bipolar velocity ratio $v_{\text{d,sat},e}/v_{\text{d,sat},h} \approx 1.5$: because electrons transit the junction approximately 50 % faster than holes, the effective current impulse is not a perfect rectangle, and the actual sinc-like roll-off shifts the 3 dB point relative to the symmetric-carrier prediction.

For the intrinsic layer thickness $W_i = 350 \text{ nm}$ adopted in this work, eq. (55) gives:

$$f_{\text{tt}} = \frac{0.45 \times 4 \times 10^6 \text{ cm s}^{-1}}{350 \times 10^{-7} \text{ cm}} \approx 51 \text{ GHz}. \quad (56)$$

This value is a conservative lower bound because eq. (55) is derived from the worst-case single-carrier response governed by the slower holes. In practice, because electrons and holes simultaneously contribute to the terminal current via their respective Ramo–Shockley induced charges (as illustrated in fig. 6), the two-carrier response is broader than the single-carrier prediction by a factor that depends on the velocity ratio. Experimental devices also benefit from self-resonance due to the parasitic inductance of the on-electrode metal interconnect L_p , which introduces a peaking effect that can extend the usable bandwidth substantially beyond the sinc-function roll-off. Accounting for both effects, the benchmark-calibrated transport ceiling for a $W_i = 350 \text{ nm}$ device is in the 110 GHz to 120 GHz class, confirming that transit time is not the operative bandwidth limitation at the target operating rate of 100 GHz.

RC limit.

The junction capacitance C_j of the fully depleted intrinsic region is that of a parallel-plate capacitor with plate separation W_i and area $A_j = W_m \times L$, where W_m is the mesa width and L is the device length along the waveguide axis:

$$C_j = \frac{\varepsilon_0 \varepsilon_{\text{Ge}} A_j}{W_i}, \quad (57)$$

where $\varepsilon_{\text{Ge}} \approx 16.2$ is the relative permittivity of germanium. The physical interpretation is immediate: a thinner depletion region brings the charge sheets of the p -type and n -type contacts closer together, increasing C_j , while a larger junction area increases the total stored charge per unit voltage. Both effects are detrimental to bandwidth.

The RC-limited 3 dB bandwidth is determined by the RC time constant of the junction capacitance discharging through the total series resistance $R_{\text{tot}} = R_{s,\text{an}} + R_{s,\text{cat}} + R_L$, where $R_{s,\text{an}}$ and $R_{s,\text{cat}}$ are the anode and cathode parasitic contact resistances, respectively, and $R_L = 50 \Omega$ is the termination resistance of the external readout circuit:

$$f_{\text{RC}} = \frac{1}{2\pi R_{\text{tot}} C_j}. \quad (58)$$

In the vertical n - i - p PIN geometry, the dominant contribution to R_s arises from the lateral current path within the p^{++} -doped silicon platform layer beneath the Ge mesa. Carriers generated within the Ge absorber are collected through the Si layer, which has a finite sheet resistance determined by its thickness $t_{\text{Si}} = 220 \text{ nm}$ and the anode contact geometry:

$$R_{s,\text{anode}} \approx \frac{\rho_{\text{Si}}}{t_{\text{Si}}} \cdot \frac{\ell_a}{w_a}, \quad (59)$$

where ℓ_a and w_a are the effective current path length and width through the platform, respectively. This resistance can be reduced by placing the anode contacts as close as possible to the base of the Ge mesa, which is a critical layout constraint that directly couples electrical performance to lithographic resolution. The RC time constant is:

$$\tau_{\text{RC}} = R_{\text{tot}} C_j, \quad (60)$$

from which $f_{\text{RC}} = (2\pi\tau_{\text{RC}})^{-1}$.

3.2.2 Optimal Intrinsic-Layer Thickness

The main design insight from eqs. (55) and (58) is that the transit-time and RC bandwidths respond in opposite ways to W_i . Increasing W_i reduces f_{tt} by lengthening the transit path, but raises f_{RC} by lowering the capacitance per unit area. An optimal thickness W_i^* therefore exists where the two limits are balanced and the total bandwidth $f_{3\text{dB}}$ is maximised. Setting $f_{\text{tt}}(W_i^*) = f_{\text{RC}}(W_i^*)$ and substituting eqs. (55), (57) and (58) yields:

$$W_i^* = \sqrt{0.9\pi R_{\text{tot}} \varepsilon_0 \varepsilon_{\text{Ge}} v_{\text{d,sat},h} A_j}. \quad (61)$$

Substituting the nominal device parameters $R_{\text{tot}} = 113 \Omega$, $\varepsilon_{\text{Ge}} = 16.0$, the effective transport velocity $v_{\text{sat}} = 5.24 \times 10^6 \text{ cm s}^{-1}$ derived from simulation sweeps, and $A_j = 40 \mu\text{m}^2$ gives $W_i^* \approx 310 \text{ nm}$, which lies within 11 % of the chosen $W_i = 350 \text{ nm}$, confirming that the device is positioned close to the bandwidth optimum. The modest upward departure from W_i^* is deliberate: it marginally reduces C_j and the associated RC penalty while accepting a small

transit-time degradation, improving manufacturing tolerance for process variations in the epitaxial layer thickness.

The expression eq. (61) also gives a useful scaling law. Because $W_i^* \propto \sqrt{A_j}$, the optimal intrinsic thickness scales as the square root of the junction area. Consequently, shrinking the junction area by a factor of four (e.g., from $A_j = 6 \mu\text{m}^2$ to $1.5 \mu\text{m}^2$) allows W_i^* to be halved, which simultaneously raises f_{tt} and f_{RC} at the new balanced optimum by approximately the same factor. This scaling analysis underpins the miniaturisation strategy adopted in the next-generation device designs discussed in section 7: reducing the device footprint, rather than simply increasing the reverse-bias voltage, is the primary lever for pushing integrated Ge-on-Si photodetectors beyond the 100 GHz bandwidth regime while maintaining the responsivity needed for 400 Gbit/s data-centre interconnect applications.

The two-dimensional bandwidth surface is reserved for the results section, where it can be tied directly to the post-processed device parameters.

3.2.3 Small-Signal Transfer Function and Inductive Peaking

The photocurrent frequency response of the photodetector can be compactly described by a factorised small-signal transfer function. The transit-time contribution takes the form of a sinc function inherited from the rectangular current impulse response, and the RC contribution is a single-pole low-pass filter:

$$H_{\text{PD}}(f) = \text{sinc}\left(\frac{f}{f_{\text{tt}}}\right) \cdot \frac{1}{1 + j f / f_{\text{RC}}}, \quad (62)$$

where $\text{sinc}(x) = \sin(\pi x) / (\pi x)$. This simplified two-pole model is adequate for rapid design screening and for gaining physical intuition, but it neglects the reactive parasitics that are unavoidable in a realistic device layout.

In a fully parameterised electrical model of the packaged photodetector, the reverse-biased junction presents an impedance $Z_j(\omega) \approx (j\omega C_j)^{-1}$, while the series branch carries the ohmic resistance R_s and the parasitic inductance L_p of the metal electrode interconnect:

$$Z_s(\omega) = R_s + j\omega L_p. \quad (63)$$

The output is terminated by the load $Z_L(\omega) = (R_L^{-1} + j\omega C_p)^{-1}$, where C_p is the parasitic pad capacitance. Current division between these impedances yields the peaked electrical transimpedance:

$$H_{\text{elec,pk}}(\omega) = \frac{Z_j(\omega) Z_L(\omega)}{Z_j(\omega) + Z_s(\omega) + Z_L(\omega)}. \quad (64)$$

The full benchmark-calibrated detector response, combining the transit-time and electrical sub-network contributions, is:

$$H_{\text{PD,pk}}(\omega) = H_{\text{tt}}(\omega) H_{\text{elec,pk}}(\omega), \quad (65)$$

where $H_{tt}(\omega)$ is the Fourier transform of the two-carrier impulse current response.

In eq. (64), the series inductance L_p resonates with C_j at a frequency $\omega_0 = 1/\sqrt{L_p C_j}$, producing a local maximum in the transimpedance magnitude at frequencies near ω_0 . This so-called *inductive peaking* extends the usable 3 dB bandwidth beyond the RC-only prediction of eq. (58) without requiring any reduction in C_j or R_s , and explains why experimentally measured bandwidths routinely exceed the estimates based on the simplified two-pole model. Equation eq. (62) is employed for geometry screening throughout this paper; eq. (65) is reserved for validated projections into the 100 GHz-class operating regime in section 7.

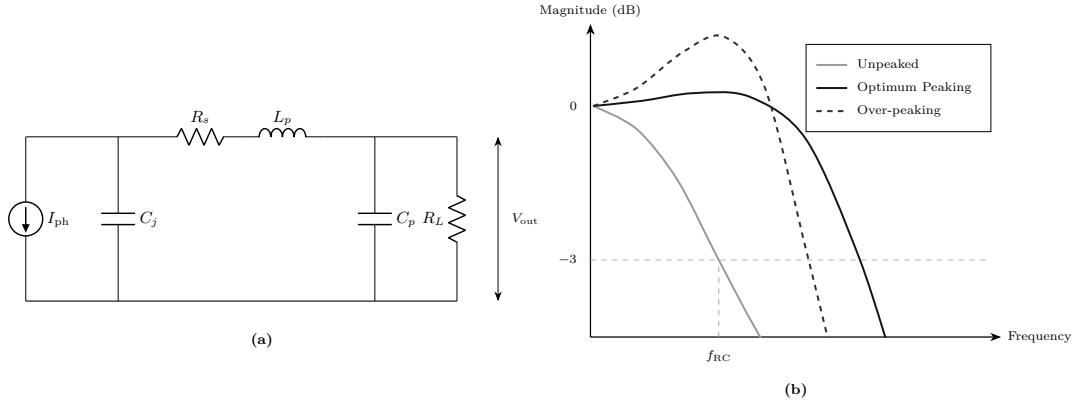


Figure 10: Inductive peaking: circuit and frequency response. Panel (a): lumped small-signal equivalent circuit of the peaked photodetector, with photocurrent source I_{ph} , junction capacitance C_j , series resistance R_s , peaking inductance L_p , pad capacitance C_p , and load R_L . Panel (b): simulated magnitude $|H_{PD,pk}(f)|$ for $L_p = 0$ (unpeaked, RC-limited), $L_p = L_{opt}$ (optimally peaked), and $L_p > L_{opt}$ (over-peaked), referenced to f_{RC} .

3.3 Noise, Sensitivity, and Figures of Merit

3.3.1 Dark Current

Even in the complete absence of optical excitation, a reverse-biased PIN junction sustains a non-zero current, referred to as the *dark current* I_d . Far from being a mere background offset that can be electronically subtracted, the dark current is a primary noise source: because current flow involves the random transit of discrete charge carriers, it inevitably generates shot noise, as formalised in section 3.3.2. The dark current therefore sets the fundamental noise floor of the detector and, through the noise-equivalent power, determines the minimum detectable signal level.

Three generation-recombination mechanisms contribute to I_d in a reverse-biased Ge-on-Si PIN [2]. The first and generally dominant mechanism at the device geometry considered here is *thermal generation via SRH traps* within the depleted intrinsic layer. In a reverse-biased junction, the carrier concentrations are below their equilibrium values, and mid-gap traps

act predominantly as generation centres, thermally exciting carriers from the valence band to the conduction band through an intermediate trap state. The generation rate derived from the SRH formalism yields a current density:

$$J_{\text{SRH}} = \frac{q n_i W_i}{2\tau_0}, \quad (66)$$

where n_i is the intrinsic carrier concentration of Ge and τ_0 is the effective SRH carrier lifetime, which is related to the trap density and capture cross-sections. Because n_i in Ge is approximately $2.3 \times 10^{13} \text{ cm}^{-3}$ at room temperature, nearly three orders of magnitude larger than in Si, the SRH generation rate, and hence the dark current, is substantially higher in Ge-based devices. This is a material limitation rather than a fabrication artefact, and it sets a room-temperature floor on the achievable dark current density of Ge-on-Si photodetectors. The linear proportionality to W_i in eq. (66) is another form of the same trade-off: thicker intrinsic layers improve responsivity but also increase dark current and its associated noise.

The second mechanism is *minority-carrier diffusion* from the quasi-neutral p -type and n -type contact regions flanking the depletion layer. In these neutral regions, thermally generated minority carriers diffuse toward the high-field depletion edge, where they are swept across by the electric field. The diffusion current density from the n -type and p -type contact regions is:

$$J_{\text{diff}} = \frac{q D_p n_i^2}{L_p N_D} + \frac{q D_n n_i^2}{L_n N_A}, \quad (67)$$

where D_p and D_n are the minority-carrier diffusion coefficients, $L_p = \sqrt{D_p \tau_p}$ and $L_n = \sqrt{D_n \tau_n}$ are the minority-carrier diffusion lengths, and N_D and N_A are the donor and acceptor concentrations in the respective contact regions. In a well-designed device with sufficiently heavy doping in the contact regions, J_{diff} is suppressed below J_{SRH} and is a secondary contribution to the total dark current.

The third mechanism, which becomes progressively more important as the device is scaled to smaller dimensions, is *interface recombination* (or, equivalently, surface generation) at the Ge/Si heterojunction. The crystallographic mismatch between Ge and Si results in a high density of interface states that can act as generation centres with an effective surface recombination velocity v_s . The associated current density is:

$$J_{\text{surf}} \approx q n_i v_s. \quad (68)$$

Combining all three contributions, and distinguishing between the area-scaled bulk terms and the perimeter-scaled surface term, the total dark current is:

$$I_d = A_j (J_{\text{SRH}} + J_{\text{diff}}) + P_j q n_i v_s, \quad (69)$$

where $P_j = 2(W_m + L)$ is the perimeter of the Ge mesa junction. Separating area-scaled and perimeter-scaled terms is important for miniaturisation. As A_j is reduced to lower capacitance and increase bandwidth, the ratio A_j/P_j decreases and the surface contribution

becomes a larger fraction of I_d . In sufficiently small devices, the perimeter term can dominate the dark current, making Ge/Si interface quality and surface passivation central to the bandwidth-noise trade-off.

3.3.2 Noise Spectral Densities and Noise-Equivalent Power

Shot noise originates in the Poissonian arrival statistics of discrete charge carriers. The conceptual time-domain mechanism is shown once in fig. 11; the quantitative PSD is kept in the solver-backed noise result figure.

Physical origin of shot noise.

The shot noise PSD follows from the Poissonian statistics of discrete carrier arrivals at the terminal. Each photon-induced electron-hole pair is generated independently, giving a white (frequency-independent) PSD:

$$S_{\text{shot}}(f) = 2q(I_d + I_{\text{ph}}) \quad [\text{A}^2 \text{Hz}^{-1}], \quad (70)$$

where the total current includes both dark and photocurrent contributions. The factor of two arises from the one-sided spectral density convention.

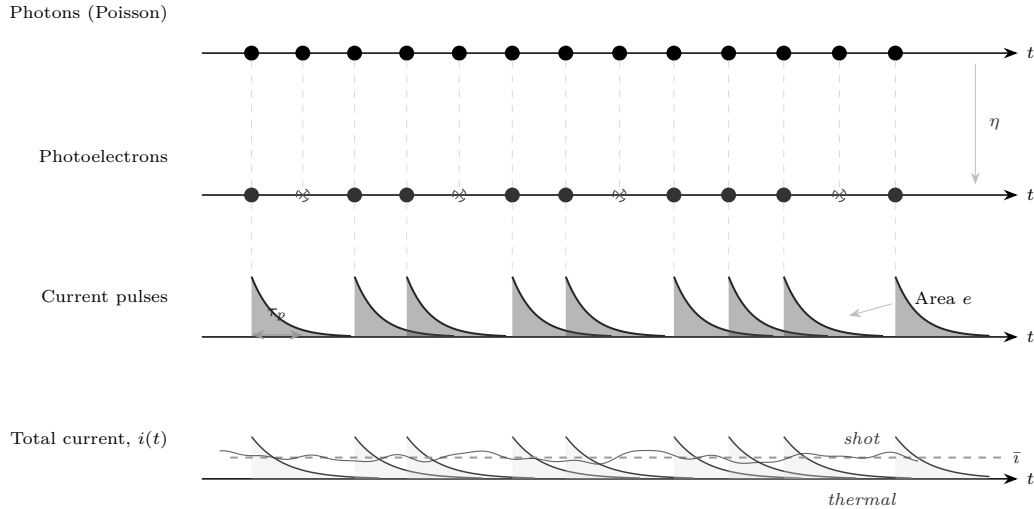


Figure 11: Origin of shot noise in a photodetector, shown as the superposition of individual, Poisson-distributed single-electron current pulses. The discreteness of the electronic charge q leads to current fluctuations with a one-sided power spectral density of $S_{\text{shot}} = 2q(I_d + I_{\text{ph}})$, which is white for frequencies below the transit-time roll-off.

Thermal noise.

Simultaneously, the load resistance R_L that terminates the detector generates Johnson–Nyquist thermal noise through the random thermal motion of conduction electrons. Its one-sided PSD is:

$$S_{\text{thermal}}(f) = \frac{4kT}{R_L} \quad [\text{A}^2 \text{Hz}^{-1}], \quad (71)$$

where k is the Boltzmann constant and T is the absolute temperature. Like shot noise, this is a white noise process. At room temperature and with $R_L = 50 \Omega$, the thermal noise current density is $\sqrt{4kT/R_L} \approx 18 \text{ pA}/\sqrt{\text{Hz}}$. A dark current of $I_d = 1 \text{ nA}$ produces a shot noise of $\sqrt{2qI_d} \approx 17.9 \text{ fA}/\sqrt{\text{Hz}}$, approximately three orders of magnitude below the thermal noise floor. This confirms that for the device under study, at moderate reverse bias, the detector itself is not the dominant noise source; the 50Ω termination resistance, and by extension the input noise of the transimpedance amplifier, governs the overall sensitivity of the optical receiver front end.

The crossover condition $S_{\text{shot}} = S_{\text{thermal}}$ remains useful analytically, but it is not kept as a separate figure because the PSD and SNR plots already encode the same noise-budget conclusion.

Total noise and noise-equivalent power.

Integrating the noise PSD over the electrical detection bandwidth B_e gives the total noise variance at the receiver input:

$$\sigma_n^2 = (S_{\text{shot}} + S_{\text{thermal}}) B_e \quad [\text{A}^2]. \quad (72)$$

The *noise-equivalent power* (NEP) is defined as the minimum optical power required at the detector input to produce a signal current equal to the root-mean-square noise current, i.e., the signal-to-noise ratio of unity in a unit bandwidth:

$$\text{NEP} = \frac{\sqrt{S_{\text{shot}} + S_{\text{thermal}}}}{\mathcal{R}} \quad [\text{W}/\sqrt{\text{Hz}}]. \quad (73)$$

A lower NEP indicates a more sensitive detector. When the dark current is the dominant noise source (the dark-current-limited regime, relevant at high reverse bias or elevated temperature), the NEP is determined by the shot noise of I_d alone, and the *specific detectivity* D^* provides a geometry-normalised figure of merit that facilitates comparison across devices of different junction areas:

$$D_{\text{shot}}^* = \frac{\mathcal{R}\sqrt{A_j}}{\sqrt{2qI_d}} \quad [\text{cm}\sqrt{\text{Hz}}/\text{W}]. \quad (74)$$

The $\sqrt{A_j}$ normalisation reflects the fact that, for area-dominated dark current ($I_d \propto A_j$), the shot noise grows as $\sqrt{A_j}$, so a larger device is intrinsically noisier in proportion to its collection area.

The geometry dependence of both NEP and D^* is retained as analytical context in the equations above, but the previous unsupported scaling plot is not kept as a thesis figure. A replacement should be generated only if the area sweep is available from the drift-diffusion solver or from a documented compact-model dataset.

Additional noise contributions.

Several secondary noise processes deserve mention even if they do not dominate in the tens-of-GHz band of interest. Generation-recombination noise from an ensemble of SRH traps with characteristic lifetime τ_{gr} has a Lorentzian PSD with a corner frequency $f_{\text{gr}} = 1/(2\pi\tau_{\text{gr}})$ that typically falls below 1 MHz in device-grade Ge; it is therefore negligible at high-speed modulation frequencies. Optical-source noise, characterised by the relative intensity noise (RIN) of the laser transmitter, enters as:

$$S_{\text{RIN}}(f) = \mathcal{R}^2 P_{\text{opt}}^2 \text{RIN} \quad [\text{A}^2 \text{Hz}^{-1}], \quad (75)$$

and is proportional to the *square* of the photocurrent. At the high optical power levels encountered in short-reach interconnects using directly-modulated lasers with $\text{RIN} \approx -140 \text{ dB/Hz}$, this contribution can become comparable to or exceed the shot-noise limit, shifting the detector from a shot-noise-limited to a source-noise-limited regime. The distinction matters for receiver-noise accounting but does not alter the intrinsic detector noise floor characterised by eqs. (70) and (71).

Passive optical loss mechanisms, including taper insertion loss, waveguide propagation loss, free-carrier absorption in doped contact layers, Fresnel reflection, and incomplete carrier collection, do not create new electrical noise PSDs. They reduce the useful signal $I_{\text{ph}} = \mathcal{R} P_{\text{opt}}$ while the noise floor remains set by shot and thermal terms, degrading the signal-to-noise ratio indirectly. This separation between noise and loss is required for consistent detector-noise accounting.

The previous analytical SNR-versus-power plot is not retained as a thesis figure. Receiver sensitivity is reported only from the circuit-level BER sweep exported by the compact-model testbench, keeping the figure traceable to simulation data rather than to a reconstructed analytical curve.

3.4 The Speed–Efficiency–Noise Trade-off Triangle

The metrics derived above cannot be optimised independently. They are coupled through the shared physical parameters of the device geometry, principally the intrinsic layer thickness W_i , the junction area A_j , and the applied reverse bias V_R . This coupled design space motivates the specific geometry choices made in section 7.

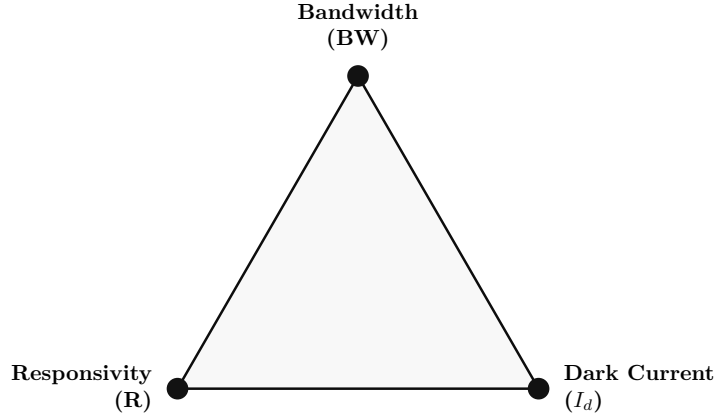


Figure 12: Speed–efficiency–noise trade-off triangle for a waveguide-integrated Ge-on-Si PIN photodetector. vertices represent bandwidth, responsivity, and dark current; edges indicate the geometric coupling axes (W_i , L , and V_R) that define the design space.

Responsivity–bandwidth trade-off.

The most fundamental tension is between responsivity and bandwidth, and it arises from two independent geometric mechanisms. First, consider the role of W_i : increasing the intrinsic layer thickness improves photon absorption (the factor $1 - e^{-\alpha W_i}$ in eq. (53) rises toward unity), thereby improving η_{ext} and $\mathcal{R}$. Simultaneously, the longer transit path through the thicker layer reduces f_{tt} via eq. (55). These competing dependences are mediated by the optimum thickness W_i^* of eq. (61): operating below W_i^* sacrifices both responsivity and bandwidth (transit-time limited, small f_{tt} and low absorption), while operating above W_i^* gains responsivity at the cost of RC bandwidth (higher C_j). Second, consider the role of the device length L : a longer device absorbs a larger fraction of the incident guided power, improving $\mathcal{R}$, but also increases $A_j = W_m \times L$, raising C_j via eq. (57) and degrading f_{RC} via eq. (58). There is therefore a maximum useful device length beyond which the RC bandwidth penalty outweighs the responsivity benefit, and this trade-off length also scales with A_j through the same $\sqrt{A_j}$ dependence of W_i^* . The product $\mathcal{R} \times f_{3\text{dB}}$ is a useful scalar summary of this two-dimensional trade-off and is employed in the benchmarking analysis of section 6.4.

Bias voltage and the noise axis.

Reverse bias adds a third dimension to the trade-off space. Increasing V_R raises the electric field throughout the depletion region, driving both carrier species closer to their saturation velocities and reducing the transit time $\tau_{\text{tr}} = W_i/v_d(E)$. This improves f_{tt} and, for devices in which the field is initially insufficient for velocity saturation, can also increase ξ and hence η_{int} . The noise cost, however, is non-trivial: higher reverse bias drives greater band bending and widens any partially depleted regions, increasing the volume over which SRH generation occurs and thereby raising I_d through eq. (66). The resulting increase in shot

noise degrades NEP via eq. (73). In the context of a data-centre optical link, this noise penalty must be weighed against the bandwidth benefit: for receivers operating at or near the sensitivity limit, the optimal bias point is the lowest voltage at which full velocity saturation is achieved throughout the intrinsic layer, since additional bias beyond this point increases noise without providing further bandwidth benefit.

Miniaturisation and the perimeter penalty.

Reducing the junction area A_j below the value required for absorption saturation trades responsivity for bandwidth, as discussed above. An additional subtlety, captured by eq. (36), is that miniaturisation also shifts the dark-current composition from bulk-dominated to perimeter-dominated. For the current device with $A_j = 6 \mu\text{m}^2$ and $P_j = 8 \mu\text{m}$, the area-to-perimeter ratio is $A_j/P_j = 0.75 \mu\text{m}$; reducing A_j by a factor of four while halving the perimeter reduces this ratio by a factor of two, potentially doubling the relative contribution of surface generation. Whether this constitutes an absolute increase in I_d depends on the relative magnitude of J_{SRH} versus the effective interface recombination velocity v_s , which in turn depends on the quality of the Ge/Si heterojunction and any passivation treatment. This linkage between geometry, process quality, and noise performance makes the design of sub-micrometre Ge-on-Si photodetectors for beyond-100 Gb/s applications a genuinely multi-physics optimisation problem, spanning optical, electrical, and materials dimensions simultaneously.

Section 7 uses this framework to map geometry to performance, locate the practical Pareto region of the vertical PIN architecture, and justify the chosen device geometry for the 400GBASE-DR4 receiver target.

4 Photodetector Architectures

With the metrics of section 3 as the evaluation framework, the main photodetector device classes can be compared on common terms: responsivity, bandwidth, and noise. The sequence below shows how each architecture removes one limitation while exposing another, leading to the waveguide-integrated PIN junction used in this work.

4.1 PN and PIN Junctions: Depletion Engineering

Photoconductive Detection.

The photoconductor is the simplest optical transducer because it requires no junction and no depletion engineering. Its apparent advantage is internal gain:

$$G_{\text{PC}} = \frac{\tau_r}{\tau_{\text{tr}}}, \quad \tau_{\text{tr}} = \frac{L^2}{\mu_n V_b}, \quad (76)$$

where τ_r is the minority-carrier recombination lifetime, τ_{tr} is the carrier transit time across the contact separation L , μ_n is the electron mobility, and V_b is the applied bias. Physically, G_{PC} counts how many complete circuits a majority carrier traverses before recombination removes its partner: if a photogenerated hole survives for many transit times, the external circuit must replenish the missing electron repeatedly, yielding an apparent multiplication of the photocurrent. This gain mechanism, however, is inseparable from the recombination lifetime that sustains it. Because the same τ_r that produces gain also governs the impulse response, the -3 dB bandwidth is:

$$f_{3\text{dB,PC}} = \frac{1}{2\pi\tau_r}. \quad (77)$$

The gain-bandwidth product $G_{PC} \times f_{3\text{dB,PC}}$ is therefore a constant set by material parameters alone, with no free geometric lever. For germanium, trap-limited lifetimes in the range $\tau_r \sim 1$ μs constrain the bandwidth to the sub-megahertz regime, placing the photoconductor many orders of magnitude below the 50 GHz electrical bandwidth required for 100 GBd-class operation in the target data-centre interconnect context. The photoconductor is therefore rejected by this intrinsic constraint before any geometric optimisation is considered.

PN Junction Photodiodes.

The PN junction eliminates the most critical weakness of the photoconductor by replacing the ohmic resistor with a rectifying interface that sustains an internal built-in electric field E_{bi} across the depletion region. Photogenerated electron-hole pairs created within, or diffusing into, this depletion layer experience a drift force that sweeps electrons toward the n-contact and holes toward the p-contact at velocities approaching the material saturation velocity v_{sat} . The recombination lifetime no longer limits speed because carriers are extracted before recombination can occur, decoupling the bandwidth from τ_r .

The drift component of the frequency response is well approximated, for a uniformly illuminated depletion region of width W_d , by:

$$H_{\text{drift}}(f) = \frac{1 - e^{-j2\pi f W_d/v_{sat}}}{j2\pi f W_d/v_{sat}}, \quad (78)$$

which is the Fourier representation of a uniform current pulse of duration $\tau_{tt} = W_d/v_{sat}$. The -3 dB point of $|H_{\text{drift}}|$ occurs near $f_{tt} \approx 0.45 v_{sat}/W_d$, consistent with eq. (55).

A residual limitation arises because not all photogeneration is confined to the depletion layer. Carriers created in the quasi-neutral p- and n-regions must diffuse to the depletion edge before the field can extract them. This diffusion process introduces a slow tail in the impulse response, characterised by the ambipolar diffusion time constant τ_D . The composite transfer function acquires a low-pass factor:

$$H_{PN}(f) = H_{\text{drift}}(f) \frac{1}{1 + j2\pi f \tau_D}, \quad (79)$$

where τ_D is determined by the minority-carrier diffusion length and the distance from the photogeneration centroid to the depletion edge. When the diffusion-limited pole $1/(2\pi\tau_D)$ falls below the drift-limited transit-time bandwidth, it becomes the binding constraint on the overall detector speed. In practice, this condition is met at moderate symbol rates, but at the 50 GHz to 100 GHz electrical bandwidth class targeted here, even a modest quasi-neutral absorption fraction is sufficient to suppress the frequency response to an unacceptable degree. The PN structure therefore resolves the gain-bandwidth lock of the photoconductor while retaining a diffusion-tail bottleneck that must be addressed by the next architectural step.

PIN Photodiodes.

The PIN structure eliminates the diffusion tail by inserting a lightly doped, or intrinsic, semiconductor layer of thickness W_i between the p- and n-doped contact regions. Under reverse bias V_{bias} , the low doping density of the intrinsic layer ensures that it is fully depleted at modest voltages, because the depletion depth in a one-sided junction scales as $W \propto \sqrt{(V_{\text{bi}} + |V_{\text{bias}}|)/N_a}$, so a very low effective doping N_a in the intrinsic region allows full depletion to be achieved without requiring large voltages. Once fully depleted, the electric field across the intrinsic region is approximately uniform:

$$E(x) \approx \frac{|V_{\text{bias}}| + V_{\text{bi}}}{W_i}, \quad x \in [0, W_i], \quad (80)$$

where V_{bi} is the built-in potential determined by the doping contrast at the p-i and i-n interfaces. This expression follows directly from integrating Poisson's equation across a region of negligible space charge: $dE/dx \approx 0$, so the field is linear only at the boundaries and constant throughout the bulk of the intrinsic layer. The practical importance of this uniformity is that all optically generated carriers within W_i experience essentially the same extraction field, eliminating the position-dependent transit-time dispersion that would arise in a non-uniform field profile.

If the absorbing material is selected so that its absorption coefficient α at the operating wavelength satisfies $\alpha^{-1} \lesssim W_i$, the overwhelming majority of photogeneration occurs inside the fully depleted intrinsic region. Diffusion from quasi-neutral zones is thereby suppressed, and the diffusion tail of eq. (79) is eliminated. The bandwidth is then governed entirely by the two competing mechanisms already formalised through eqs. (55) and (58):

$$\frac{1}{f_{\text{3dB}}^2} \approx \frac{1}{f_{\text{tt}}^2} + \frac{1}{f_{\text{RC}}^2}, \quad (81)$$

where $f_{\text{tt}} \propto v_{\text{sat}}/W_i$ and $f_{\text{RC}} \propto (R_s C_j)^{-1}$ with $C_j \propto A/W_i$. An increase in W_i raises the transit time τ_{tt} and thus degrades f_{tt} , while simultaneously reducing the junction capacitance C_j and thus improving f_{RC} . There is therefore an optimum intrinsic thickness W_i^* that maximises the total bandwidth for a given junction area A and series resistance R_s :

$$W_i^* = \left(\frac{v_{\text{sat}} A \epsilon_0 \epsilon_r}{2\pi R_s} \right)^{1/2}, \quad (82)$$

obtained by differentiating eq. (81) with respect to W_i and equating the transit-time and RC contributions at the optimum. The structural advantage of the PIN over the PN diode is that W_i is a lithographically controlled epitaxial parameter, decoupled from the contact doping levels used to minimise R_s . This additional degree of freedom, the independent control of the depletion thickness, is the key reason the PIN remains the canonical high-speed receiver device: it permits the junction geometry to be engineered at a level that the simple PN diode, where W_d is set by the doping profile alone, cannot match.

4.2 Avalanche Photodiodes: Gain versus Noise

An avalanche photodiode (APD) exploits impact ionisation within a high-field multiplication region to amplify the photocurrent internally, yielding an effective responsivity:

$$\mathcal{R}_{\text{APD}} = M \mathcal{R}_{\text{PIN}}, \quad (83)$$

where M is the mean avalanche multiplication factor and $\mathcal{R}_{\text{PIN}}$ is the responsivity that the same absorption layer would provide without multiplication. At first inspection this appears to offer a route to higher sensitivity without increasing the optical input power, but the avalanche mechanism introduces two penalties that are absent in the low-bias PIN.

The first is a fundamental gain-bandwidth constraint. Avalanche multiplication is not instantaneous: the carriers initiating an impact-ionisation event must traverse the high-field region, and the secondary carriers they generate must in turn complete their own transit before the multiplication chain terminates. The statistical build-up and decay of this cascade introduces a finite temporal spread that broadens the impulse response. For a multiplication region of length l_{mult} and saturated ionisation velocity v_{ion} , the effective multiplication bandwidth is approximately:

$$f_M \approx \frac{v_{\text{ion}}}{2\pi k M l_{\text{mult}}}, \quad (84)$$

where $k = \alpha_h/\alpha_e$ is the ratio of hole to electron impact-ionisation coefficients. The product $M \times f_M$ is therefore a material- and geometry-specific constant, conventionally termed the gain-bandwidth product (GBP). As M is increased to boost sensitivity, f_M decreases proportionally, and the bandwidth headroom required for the target 50 GHz class rapidly exhausts the available GBP. In germanium, the nearly equal ionisation coefficients ($k \approx 1$) make this constraint particularly severe compared with materials such as InAlAs, where $k \ll 1$ permits higher GBP.

The second penalty is excess noise. Even in the absence of thermal noise, the stochastic nature of the multiplication process introduces fluctuations in M about its mean value that manifest as an additional noise current. The McIntyre model [7] characterises this through an excess noise factor $F(M)$, which modifies the shot-noise power spectral density from the unmultiplied case:

$$S_{I,\text{APD}} = 2q I_{\text{ph}} M^2 F(M) \Delta f, \quad (85)$$

where I_{ph} is the primary photocurrent and Δf is the noise bandwidth. For a single-carrier-initiated avalanche with ionisation ratio k , McIntyre showed that:

$$F(M) = k M + (1 - k) \left(2 - \frac{1}{M} \right). \quad (86)$$

In the ideal limit $k \rightarrow 0$ (electrons ionise, holes do not), $F \rightarrow 2$ as $M \rightarrow \infty$, and the noise penalty is modest. In the opposite regime $k \rightarrow 1$, which approximates the situation in Ge where $\alpha_e \approx \alpha_h$, eq. (86) reduces to $F \approx M$, and substitution into eq. (85) shows that the noise power spectral density grows as M^3 rather than M^2 . The net signal-to-noise benefit of multiplication then degrades rapidly with increasing M , and the optimum gain that minimises the noise-equivalent power saturates at values well below those that would be useful for sensitivity improvement in a thermal-noise-dominated receiver front-end.

Taken together, the GBP constraint and the excess noise factor make the Ge APD unattractive for the present receiver. A low-bias PIN paired with a low-noise transimpedance amplifier (TIA) preserves bandwidth while leaving the dominant thermal-noise floor to the circuit design. The APD is therefore not selected, and the modest-bias PIN is retained as the active device architecture.

4.3 Waveguide-Integrated Photodetectors: Interaction Length Advantage

The PIN junction resolves the device-level bandwidth problem but does not, by itself, resolve the geometric conflict between absorption efficiency and transit time.

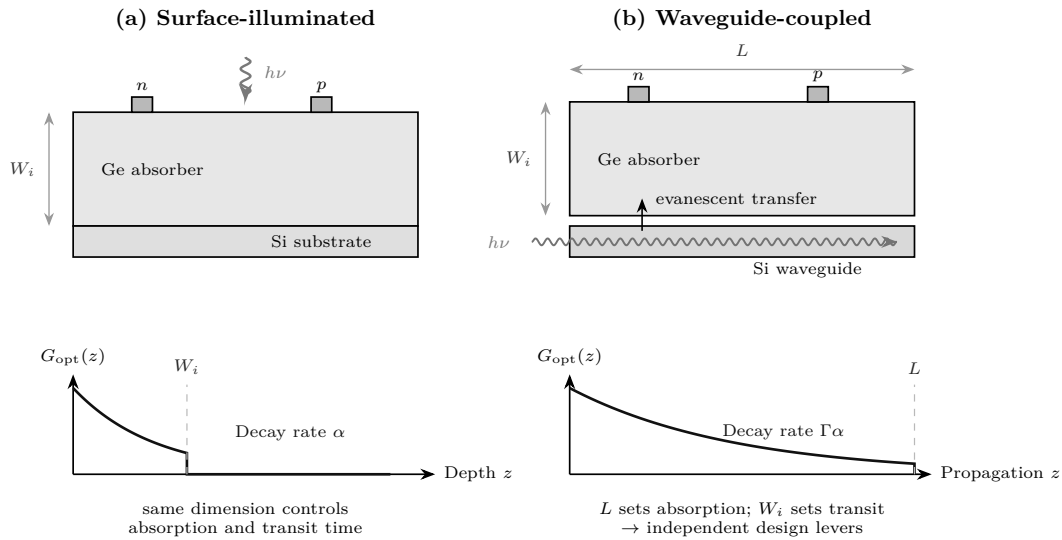


Figure 13: Comparison of surface-illuminated (left) and waveguide-integrated (right) geometries. Waveguide integration separates transit time (governed by vertical thickness W_i) from optical absorption (governed by longitudinal length L), resolving the speed–efficiency trade-off.

This conflict is most acute in a surface-illuminated (or normal-incidence) configuration, where the optical beam enters through the top surface and propagates vertically through the absorbing layer. In this geometry the same dimension, the absorber thickness W_i , must simultaneously satisfy two contradictory requirements: the Beer-Lambert absorption condition, $W_i \gtrsim \alpha^{-1}$, to achieve high quantum efficiency, and the transit-time bandwidth condition, $W_i \ll v_{\text{sat}}/(2\pi f_{3\text{dB}})$, derived from eq. (55). At 1310 nm, the analytical model uses $\alpha \approx 7 \times 10^3 \text{ cm}^{-1}$ for germanium, corresponding to a bulk $1/e$ absorption depth of $\alpha^{-1} \approx 1.4 \mu\text{m}$ before surface-normal reflection and confinement penalties are included. Achieving 90% absorption requires a several-micrometre optical path, while achieving a transit-time-limited bandwidth of 50 GHz requires W_i to remain in the sub-micron range for typical saturation velocities in Ge ($v_{\text{sat}} \approx 6 \times 10^6 \text{ cm s}^{-1}$). These two requirements are separated by a factor of a few before reflection and coupling penalties are included; after those penalties, the surface-illuminated PIN geometry has little practical margin for simultaneous high speed and high quantum efficiency in the O-band.

Waveguide integration resolves this conflict by a geometric transformation: the optical mode is laterally confined by the waveguide structure and propagates longitudinally along the length of the detector rather than vertically through its thickness. The relevant dimension for optical absorption is then the propagation length L along the device axis, while the vertical dimension W_i retains its role as the carrier collection thickness that determines the transit time. Because L and W_i are orthogonal spatial coordinates, they can be engineered independently.

The evolution of optical power along the waveguide-integrated absorber follows the Beer-Lambert law in the propagation direction:

$$P(z) = P_0 e^{-\Gamma \alpha z}, \quad (87)$$

where P_0 is the power entering the absorber at $z = 0$, α is the material absorption coefficient of germanium, and Γ is the optical confinement factor, defined as the fraction of the guided modal power that overlaps spatially with the germanium absorbing region:

$$\Gamma = \frac{\int_{\text{Ge}} |\mathbf{E}(\mathbf{r}_{\perp})|^2 dA_{\perp}}{\int_{-\infty}^{\infty} |\mathbf{E}(\mathbf{r}_{\perp})|^2 dA_{\perp}}, \quad (88)$$

with the integration in the numerator restricted to the transverse cross-section of the Ge layer and the denominator taken over the full modal field. The factor Γ is determined by the waveguide geometry: for a Ge-on-Si butt-coupled or evanescent-coupling scheme, Γ can range from values approaching unity in a strongly confining geometry to values well below 0.5 in an evanescent-coupling taper architecture where the mode is gradually transferred from the Si waveguide into the Ge absorber.

The total absorption efficiency, relating absorbed power to incident power, is obtained by integrating eq. (87) over the device length L :

$$\eta_{\text{abs}} = \frac{P_0 - P(L)}{P_0} = 1 - e^{-\Gamma \alpha L}. \quad (89)$$

This expression reveals the decisive advantage of the waveguide architecture: η_{abs} can be driven arbitrarily close to unity by increasing L , entirely independently of W_i . For a representative value of $\Gamma = 0.8$ and $\alpha = 7 \times 10^3 \text{ cm}^{-1}$ at 1310 nm, a several-micrometre waveguide absorber can achieve high absorption, while W_i remains freely settable in the sub-micron regime required for high transit-time bandwidth. The geometric orthogonality between L and W_i is therefore complete: L governs responsivity through eq. (89), while W_i governs f_{tt} through eq. (55), and the two parameters are adjusted in a two-dimensional design space without mutual constraint.

A secondary benefit of the waveguide geometry is the suppression of the RC penalty associated with a large device footprint. In the surface-illuminated case, a large area is needed to intercept the incident beam, increasing C_j and degrading f_{RC} . In the waveguide-integrated device, the transverse cross-section of the mode is set by the single-mode waveguide dimensions, typically of order $0.5 \mu\text{m} \times 0.5 \mu\text{m}$ in sub-micron Si photonic platforms. The junction width W and height W_i are therefore bounded by the mode confinement rather than by beam-capture requirements, enabling small junction areas and correspondingly small capacitances. The longitudinal length L does increase the total capacitance as $C_j \propto W \cdot L / W_i$, but because W is sub-micron, the product $W \cdot L$ for a $10 \mu\text{m}$ – $20 \mu\text{m}$ -long device remains well below the junction areas necessary in surface-illuminated geometries for equivalent coupling efficiency.

The geometric orthogonality introduced by waveguide integration does not eliminate all design tension: as L is increased to improve η_{abs} , the total junction capacitance grows proportionally, degrading f_{RC} . The optimum device length is therefore set not by eq. (89) in isolation but by the joint optimisation of the bandwidth expression in eq. (81), in which C_j now depends on L as well as on W_i and W . This residual coupling between absorption efficiency and RC bandwidth, absent from the transit-time discussion but present in the full bandwidth budget, is the central design tension addressed in section 7.

4.4 Selection Rationale: Why PIN for High-Speed Integrated Links

The four preceding device classes can be evaluated against a unified set of performance criteria, responsivity, bandwidth, and noise, by tracing the physical mechanism that limits each architecture and identifying the structural modification that removes it.

This selection logic leaves the waveguide-integrated Ge-on-Si PIN as the architecture compatible with the O-band, low-voltage, high-bandwidth target of this paper.

The photoconductor is rejected because the gain-bandwidth product $G_{\text{PC}} \times f_{3\text{dB,PC}}$, fixed by the recombination lifetime via eq. (77), cannot reach the target bandwidth without eliminating the gain entirely. The PN junction removes the recombination constraint but retains a diffusion-tail bandwidth penalty from quasi-neutral absorption, as expressed by the pole in eq. (79). The PIN structure eliminates the diffusion tail by confining all

useful photogeneration within the fully depleted intrinsic layer, reducing the bandwidth competition to the transit-time and RC terms of eq. (81), both of which are geometrically controllable. The APD appears to restore sensitivity through multiplication but introduces a gain-bandwidth product constraint via eq. (84) and an excess noise penalty via eq. (86) that, for germanium with $k \approx 1$, grows as $F \approx M$ and rapidly erases the SNR benefit; the bandwidth headroom needed for the target data rate is not available at any useful gain.

The waveguide-integrated PIN is selected because it keeps the main constraints separately tunable. The absorber length L governs responsivity, the intrinsic thickness W_i governs transit time, and the electrode geometry governs the RC term. Low-bias operation also avoids the excess noise and reliability concerns of avalanche multiplication while keeping the junction capacitance consistent with the narrow transverse dimensions of the silicon photonic mode. The remaining design problem is to simultaneously satisfy the DR4 responsivity floor implied by the minimum OMA sensitivity of -6.0 dBm and exceed the 39.8 GHz minimum bandwidth required by the $0.75f_s$ criterion of Clause 140, at 1310 nm in Ge-on-Si; the material analysis, literature review, and electrode design that follow address that constraint.

5 Material and Integration Platform: Germanium on Silicon

With the waveguide-integrated PIN structure selected and the operating wavelength fixed at 1310 nm, the absorber material becomes the next design choice. The absorption physics of section 2 points to germanium: it provides strong O-band absorption, adequate carrier velocity, and compatibility with silicon-foundry processing. The discussion below focuses on the three material issues that affect the proposed device: spectral suitability, CMOS integration, and the trade-offs introduced by Ge-on-Si epitaxy.

5.1 Why Germanium at 1310 nm

Germanium is an indirect-gap semiconductor whose lowest conduction-band minimum lies at the L -point of the Brillouin zone, separated from the valence-band maximum at Γ by an indirect gap $E_{g,\text{ind}} = 0.66$ eV at 300 K. A secondary conduction-band minimum at the Γ -point defines a direct gap $E_{g,\text{dir}} = 0.80$ eV [8], [9], [10]. The photon energy at 1310 nm is $\hbar\omega = hc/\lambda \approx 0.946$ eV, which exceeds $E_{g,\text{dir}}$ by approximately 0.15 eV. Optical absorption at this wavelength proceeds predominantly through *direct* interband transitions at the Γ -point, for which crystal momentum is conserved without phonon assistance [4], [11]. Silicon, whose lowest direct transition lies at 3.4 eV and whose indirect gap of 1.12 eV makes it effectively transparent at O-band wavelengths, cannot serve as the absorber. This material-physics constraint is the central reason for using a Ge-on-Si heterostructure.

Biaxial strain and the direct-gap red-shift.

The application of biaxial tensile strain ε reduces the direct gap according to the linear deformation-potential relation:

$$E_{g,\text{dir}}(\varepsilon) \approx E_{g,\text{dir}}(0) - (2a_v + a_c)(1 - c_{12}/c_{11})\varepsilon, \quad (90)$$

where a_v and a_c are the valence- and conduction-band hydrostatic deformation potentials and c_{11} , c_{12} are the elastic stiffness constants. The biaxial tensile strain of approximately 0.2 % to 0.3 % that develops in Ge epitaxial layers during post-growth cooldown, owing to the thermal expansion coefficient mismatch between Ge and Si, reduces $E_{g,\text{dir}}$ by approximately 25 meV relative to the unstrained bulk value, red-shifting the onset of direct-gap absorption by roughly 50 nm and enhancing $\kappa(\lambda)$ at C-band wavelengths. For the O-band operating point of the present device this correction is small but it is the principal mechanism exploited in strained-Ge extended-wavelength detector designs.

Table 1: Material properties of candidate photodetector absorbers at 300 K. α values are referenced to 1310 nm. The final column summarises compatibility with a standard silicon-photonics process flow.

Material	E_g [eV]	α_{1310} [cm^{-1}]	$v_{\text{sat},h}$ [cm s^{-1}]	μ_p [$\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$]	$\Delta a/a_{\text{Si}}$ [%]	CMOS compati- bility
Ge	0.66 (ind.), 0.80 (dir.)	~ 7000	4×10^6	1900	4.2	Compatible
Si	1.12 (ind.)	~ 10	2.5×10^6	480	0	Compatible
InGaAs (lattice- matched to InP)	0.75	$\sim 1 \times 10^4$	5×10^6	4500	> 5	Not compatible
$\text{Ge}_{0.92}\text{Sn}_{0.08}$	~ 0.55 (dir.)	> 5000	$\sim 4 \times 10^6$	~ 1500	4.8	Research-stage

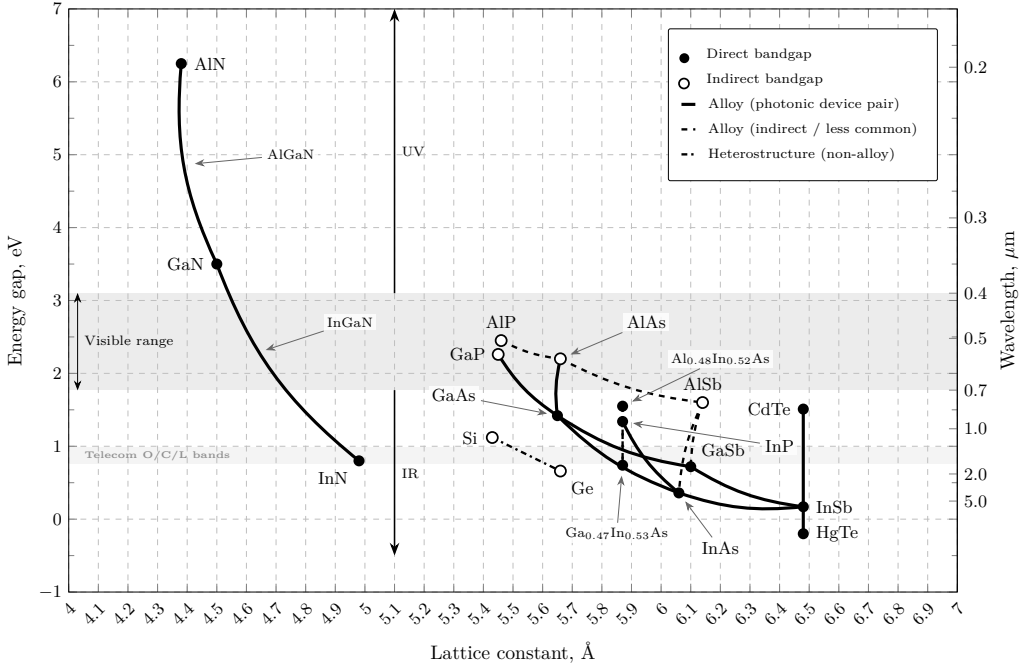


Figure 14: Bandgap energy versus lattice constant for common III-V and group-IV semiconductors. Direct-gap materials are shown as filled circles; indirect-gap materials as open circles. Ge (indirect gap $E_{g,\text{ind}} = 0.66$ eV, direct gap $E_{g,\text{dir}} = 0.80$ eV) is the only near-lattice-matched absorber to Si with adequate O-band absorption; InGaAs provides superior absorption but requires $> 5\%$ lattice mismatch to the Si platform.

5.2 CMOS Compatibility and Integration Constraints

Germanium can be grown selectively on silicon by chemical vapour deposition (CVD) using a two-step heteroepitaxial process: a low-temperature ($\sim 400^\circ\text{C}$) nucleation layer of 30 nm to 50 nm seeds the growth, and a high-temperature ($\sim 650^\circ\text{C}$) bulk growth achieves the target thickness. Cyclic thermal annealing between 600°C to 750°C reduces the threading dislocation density from $\sim 1 \times 10^9 \text{ cm}^{-2}$ after growth to $\sim 5 \times 10^7 \text{ cm}^{-2}$ after annealing, which is the practical lower bound achievable without compositionally graded buffer layers.

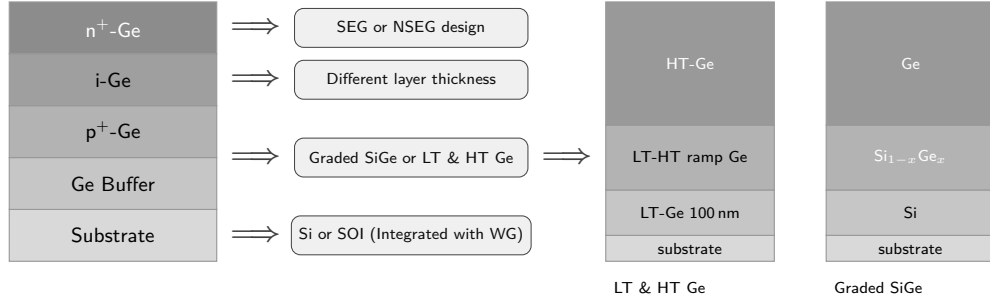


Figure 15: Ge-on-Si selective heteroepitaxy process showing the layer stack and dislocation evolution. Low-temperature nucleation forms a Ge seed on Si, followed by high-temperature bulk Ge growth. Cyclic thermal annealing reduces the threading dislocation density (TDD), improving the carrier lifetime and dark current performance.

The total thermal budget of this process is compatible with the front-end-of-line (FEOL) constraints of a 220 nm SOI platform.

Major silicon-photonics foundries, including Intel, imec, and GlobalFoundries, have qualified Ge-on-Si photodetector processes at 220 nm SOI node. The process compatibility removes the III-V bonding step required by InGaAs receivers, eliminates the need for flip-chip assembly, and enables monolithic co-integration of the photodetector with Si waveguides, grating couplers, multiplexers, and electro-optic modulators on the same die.

Heterojunction band alignment and its numerical consequences.

The electrical consequence of the Ge/Si material interface directly conditions the boundary conditions of the drift-diffusion problem. Owing to the lattice mismatch ($\Delta a/a \approx 4.2\%$) and the difference in band alignment, the Ge/Si heterojunction sustains a valence-band discontinuity of $\Delta E_v \approx 0.54 \text{ eV}$ and a conduction-band offset $\Delta E_c \approx -0.14 \text{ eV}$ [4]. For a p^{++} -Si anode contacting the intrinsic Ge absorber, the large ΔE_v acts as a potential barrier suppressing hole injection from silicon into germanium, thereby reducing the minority-carrier diffusion contribution to dark current quantified by eq. (67). A thin compositionally graded transition layer of thickness $\delta_{\text{grad}} \approx 5 \text{ nm}$ at the heterojunction removes the valence-band singularity that would otherwise prevent convergence of the self-consistent drift-diffusion solver without perceptibly modifying the electrostatic potential profile in the intrinsic germanium bulk.

5.3 Material-Induced Trade-offs

The same lattice mismatch ($\Delta a/a \approx 4.2\%$) that makes Ge-on-Si heteroepitaxy necessary also creates a residual defect population that cannot be completely eliminated by annealing. Threading dislocations at density $\text{TDD} \approx 5 \times 10^7 \text{ cm}^{-2}$ act as Shockley-Read-Hall recombination centres in the Ge bulk, directly setting the SRH lifetime τ_0 and the dark current

floor through eq. (66). The drift-diffusion implementation uses $\tau_0 = 1$ ns for Ge, which represents a conservative nanosecond-scale recombination environment for defect-limited Ge-on-Si material.

Ge thickness, dislocation density, and device performance form a coupled trade-off. Thicker Ge layers absorb more light, improving $\mathcal{A}$ via eq. (14), but they slow carrier transit through eq. (55) and increase the SRH generation volume, raising I_d through eq. (66). Thinner Ge layers improve speed and reduce dark current at the expense of absorption efficiency. The first 50 nm to 100 nm of Ge grown on Si also contains a higher defect density than the bulk. For this reason the n^{++} -Ge cathode is limited to 50 nm, reducing both free-carrier absorption (FCA) in the contact region and the volume of high-defect-density material in the optical generation zone.

A compact material comparison closes the absorber selection argument. Ge is not universally superior to every alternative, but it is the only material in table 1 that satisfies the optical, transport, and integration constraints of the present O-band receiver at the same time.

The carrier transport dynamics and dark current mechanisms relevant to the Ge-on-Si material system have been derived from first principles in section 2: drift-diffusion transport in eqs. (17) and (18), velocity saturation via the Caughey-Thomas model of eq. (24), SRH recombination in eq. (39), and the dark current decomposition of eq. (36). The Ge-specific parameter values entering these expressions, $\mu_n^{(0)} = 3900 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, $\mu_p^{(0)} = 1900 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, $v_{d,\text{sat},e} = 6 \times 10^6 \text{ cm s}^{-1}$, $v_{d,\text{sat},h} = 4 \times 10^6 \text{ cm s}^{-1}$, $n_i^{\text{Ge}} \approx 2.4 \times 10^{13} \text{ cm}^{-3}$, are summarised in table 1. The SRH lifetime adopted for the drift-diffusion simulation is $\tau_0 = 1$ ns, representing a conservative nanosecond-scale recombination environment for dislocation-limited Ge-on-Si epitaxial material at $\text{TDD} \approx 5 \times 10^7 \text{ cm}^{-2}$. These values enter the bandwidth model of eqs. (55) and (58) and the dark current estimate of eq. (37) without modification; no re-derivation is necessary in this section.

6 Literature Review and State of the Art

Ge-on-Si waveguide photodetectors have been refined over roughly two decades through iterative changes in junction geometry, contact engineering, and RF layout. Viewed through the metric framework of section 3, the literature is not a list of isolated record-setting devices; it is a sequence of attempts to relax one bandwidth bottleneck while exposing another. The present review traces that sequence to identify where the remaining modelling opportunity lies and to establish the experimental benchmark against which the simulation results of this work are compared.

6.1 Milestone Waveguide-Integrated Ge-on-Si PIN Photodetectors

The landmark demonstration of Vivien *et al.* [12] established a silicon-on-insulator waveguide-coupled Ge p-i-n detector with $\mathcal{R} \approx 1.0 \text{ A W}^{-1}$ and $f_{3\text{dB}} = 42 \text{ GHz}$ at 1550 nm , confirming that direct-gap absorption at the Γ -point translates into competitive responsivity on the silicon platform. The 42 GHz result was RC-limited: the electrode geometry left the anode series resistance at a value such that f_{RC} dominated over f_{tt} in eq. (54). Ang *et al.* [13] subsequently demonstrated evanescently coupled Ge-on-SOI fabricated at low thermal budget to remain CMOS-compatible, establishing process integration as a hard design constraint alongside responsivity and bandwidth.

The next phase of the literature attacked $R_s C_j$ directly through contact and pad engineering. Chen *et al.* [14] demonstrated a silicon-contacted Ge p-i-n with $\mathcal{R} = 0.74 \text{ A W}^{-1}$ and $f_{3\text{dB}} = 35 \text{ GHz}$; the same group later pushed to 67 GHz at only 1 V reverse bias through electrode repositioning [15], thereby reducing the effective current-path length ℓ_a in eq. (59) until f_{RC} no longer dominated eq. (54). Novack *et al.* [16] reached 60 GHz by incorporating wire-bond inductance as a gain-peaking element, resonating with $C_j + C_{\text{par}}$ and extending the usable bandwidth beyond the RC pole. That strategy is effective but system-dependent: the inductive peaking benefit is sensitive to the surrounding RF environment and package parasitics [17].

The lateral junction architecture introduced by Virost *et al.* [18] decoupled the optical mode depth from the depletion region width, providing an additional degree of freedom for C_j minimisation in eq. (57) that is geometrically independent of the absorbed fraction in eq. (14). Parasitic engineering within the vertical architecture was pushed to its prior limit by Shi *et al.* [19], who demonstrated $f_{3\text{dB}} = 80 \text{ GHz}$ through systematic minimisation of parasitic pad capacitance. The bandwidth ceiling for the lateral architecture was subsequently located by Lischke *et al.* [20], who reported $f_{3\text{dB}} = 265 \text{ GHz}$ with a Ge-fin geometry scaled to $W_{\text{Ge}} \approx 100 \text{ nm}$.

Resonant-cavity-enhanced and uni-travelling-carrier alternatives.

Two architectural alternatives provide important context for the absorption efficiency discussion. The resonant-cavity-enhanced (RCE) vertical photodiode places the absorber within a Fabry-Pérot resonator; at the resonance condition, the single-pass absorbed fraction $\mathcal{A}$ of eq. (14) is replaced by:

$$\mathcal{A}_{\text{RCE}} = \frac{(1 - R_1)(1 + R_2 e^{-\alpha L})^2(1 - e^{-2\alpha L})}{1 - 2\sqrt{R_1 R_2} e^{-\alpha L} \cos(4\pi n L / \lambda) + R_1 R_2 e^{-2\alpha L}}, \quad (91)$$

where R_1 and R_2 are the top and bottom mirror reflectances [17]. Near-unity quantum efficiency is achievable in absorbers as thin as 200 nm , but the resonance bandwidth narrows to a few nanometres, which is incompatible with the temperature-insensitive broadband O-band operation required of data-centre receivers. The uni-travelling-carrier (UTC) structure

replaces both drift species with electrons only, raising the transit-time limit to:

$$f_{\text{tt,UTC}} = \frac{0.45 v_{\text{d,sat,e}}}{W_{\text{dep}}}, \quad (92)$$

approximately 50 % higher than the hole-limited value of eq. (55) for the same depletion width [17]. Neither the RCE nor the UTC architecture is adopted here; both are noted as performance ceilings for their respective trajectories.

6.2 Bandwidth Enhancement Strategies

Four bandwidth-enhancement strategies recur in the literature, and each maps cleanly onto the analytical framework of section 3.

Geometry scaling reduces both C_j (via smaller A_j) and improves f_{tt} (via smaller W_i) simultaneously, but doing so sacrifices responsivity through eq. (53). Diminishing returns set in once the device enters the absorption-starved regime, roughly below $L \approx 4 \mu\text{m}$ for O-band Ge absorbers.

Doping profile optimisation targets the dark-current-to-responsivity balance: heavily doped contacts reduce R_s through eq. (59) but increase free-carrier absorption, while junction polarity affects the heterojunction band alignment and therefore J_{diff} through eq. (67).

Contact engineering attempts to reduce the dominant anode series resistance through silicidation, via placement, and metal routing. For a vertical architecture where R_s is dominated by the lateral Si slab resistance of eq. (59), silicidation reduces the metal-semiconductor contact contribution but not the bulk Si slab contribution, leaving R_s improvement bounded by geometry alone.

Electrode topology is the most structurally impactful lever: changing the physical layout of the metal anode changes the effective lateral current path ℓ_a in eq. (59) and therefore R_s directly. The U-shaped-anode strategy targets this lever by reducing ℓ_a geometrically until $f_{\text{RC}} > f_{\text{tt}}$, providing a structural solution rather than a resonant compensation.

6.3 The U-Shaped Electrode Benchmark and the Basis for This Work

The reference device for this study is the 103 GHz vertical Ge-on-Si photodiode of Shi *et al.* [1], the paper from which the device architecture, layer stack, and electrode geometry used in this paper are directly adopted. Shi *et al.* surpassed the 100 GHz threshold in the vertical architecture through a U-shaped anode electrode that wraps metal around three sides of the germanium mesa, reducing the maximum lateral current path from $W_m/2 + d_{\text{gap}}$ to approximately $W_m/4 + d_{\text{gap}}$. This lowers R_s , and therefore $\tau_{\text{RC}} = R_s C_j$ of eq. (60), by approximately 36 % relative to a straight-electrode device. The U-shaped device

simultaneously delivers $\mathcal{R} = 0.95 \text{ A W}^{-1}$ and $I_d = 1.3 \text{ nA}$ at -1 V , making it a balanced solution for the receiver target.

The reduced R_s also enables a shallower and more useful inductive peaking condition, and the experimentally observed bandwidth increases from 83 GHz (parallel electrode) to 103 GHz (single-U) [1]. These experimental results constitute the reference against which the simulation chain of this work is calibrated; the 103 GHz figure is not a result of the present simulation but a published measurement from the benchmark paper.

Two additional structural variants highlight alternative optimisation paths. Hamadou *et al.* [5] investigated surface-illuminated silicon PIN photodiodes with photon-trapping microholes, showing through three-dimensional optical and electrical charge transport simulations that a 2×2 array of $0.7 \mu\text{m}$ square microholes enhances optical absorption while simultaneously reducing junction capacitance by a geometric filling factor ρ_f . Li *et al.* [21] reported C-band design improvements for evanescently coupled Ge detectors grown at low temperature in a commercial CMOS process, achieving $\mathcal{R} > 0.8 \text{ A W}^{-1}$ across the full C-band with a junction capacitance below 7 fF for a $10 \mu\text{m}$ -long absorber, establishing that compact geometry and foundry-process integration can simultaneously address responsivity and RC bandwidth without cavity enhancement.

6.4 Performance Benchmarking Table

Table 2 consolidates the landmark experimental results most relevant to the present design argument. The table is intentionally restricted to the vertical trajectory adopted in this paper and the lateral trajectory that defines the broader bandwidth ceiling.

Table 2: Selected benchmark waveguide-integrated Ge-on-Si photodetectors positioning the present simulation study. “V” and “L” denote vertical and lateral junction geometries, respectively; “NR” denotes values not reported by the authors. All figures are experimental measurements from the cited works.

Year	Type	Reference	Key Design Lever	$\mathcal{R}$ [A/W]	I_d [nA]	$f_{3\text{ dB}}$ [GHz]
2010	V	[13]	Evanescent coupling, CMOS-compatible process	0.92	570	11.3
2009	V	[12]	Butt coupling, straight contact	~ 1.0	18	42
2013	V	[16]	Straight contact with inductive peaking	0.80	NR	60
2021	V	[19]	Parasitic engineering with inductive peaking	0.89	6.4	80
2024	V	[1]	U-shaped anode with inductive peaking	0.95	1.3	103
2015	L	[22]	Reduced Ge width with lateral offset junction	1.00	100	> 70
2017	L	[18]	Si/Ge/Si heterojunction with sub-micrometre Ge width	NR	5	> 50
2021	L	[20]	Ge-fin junction, $W_{\text{Ge}} \approx 100$ nm	0.30	100–200	265
2024	L	[23]	Side-coupled parallel Ge geometry	0.80	24	> 40

6.5 Why This Device and This Simulation

The U-shaped vertical Ge-on-Si photodetector of Shi *et al.* [1] is chosen as the study device for three straightforward reasons. First, it represents the current bandwidth frontier for vertical Ge-on-Si designs, making it a concrete and well-documented example from which the underlying physics can be read back through the analytical framework developed in sections 2 and 3. Second, its layer stack and electrode geometry are simple enough to simulate with standard multiphysics tools at the student-project level without requiring access to a fabrication facility. Third, the original publication [1] is written as a concise experimental paper; working through the device in simulation provides an opportunity to develop the physical reasoning that the paper necessarily condenses. This paper uses that published device as a learning vehicle, without adding to or modifying the experimental record.

IEEE 802.3 400GBASE-DR4 as the receiver-level target.

IEEE 802.3 Clause 140 defines 400GBASE-DR4 as a 400 Gbit s^{-1} single-mode-fibre interface in the O-band, implemented through four parallel optical lanes at 1310 nm over reaches up to 500 m [24]. Each lane carries $106.25 \text{ Gbit s}^{-1}$ raw line rate with PAM-4 signalling. The

key receiver-side quantities relevant to the photodetector are summarised in table 3.

Table 3: IEEE 802.3 400GBASE-DR4 receiver-side constraints most directly relevant to the photodetector, extracted from Clause 140 [24] and mapped onto the device metrics of section 3.

Parameter	Symbol	Value	Unit	Principal relation
Wavelength range	λ	1260–1360	nm	eq. (14)
Symbol rate	f_s	53.125	Gbaud	eq. (54)
Modulation format	,	PAM-4	,	,
Min. receiver sensitivity (OMA)	$\text{OMA}_{\text{out,min}}$	−6.0	dBm	,
Maximum input power	$P_{\text{in,max}}$	+3.4	dBm	fig. 20
Min. receiver −3 dB bandwidth	$f_{3\text{ dB,min}}$	$\geq 0.75f_s$	GHz	eq. (54)
Reflectance tolerance	Γ_{max}	−26	dB	eq. (101)

The minimum receiver bandwidth implied by the $0.75f_s$ criterion is 39.8 GHz. The simulation baseline of 56.1 GHz derived in section 9 comfortably clears this value using the conservative no-peaking estimate of eq. (54). The U-shaped topology of Shi *et al.* [1] combines 103 GHz-class measured bandwidth, $\mathcal{R} = 0.95 \text{ A W}^{-1}$, and $I_d = 1.3 \text{ nA}$ in a process-compatible platform, precisely the balance of speed, efficiency, and noise that makes it a suitable reference device for understanding O-band receiver design.

Broader reviews of the design space are provided by Wang *et al.* [25] and Yan *et al.* [17]; the multiphysics modelling context is treated in depth by Alasio *et al.* [26]; computational electromagnetic methods are discussed in Wartak [27].

The carrier transport physics, velocity saturation dynamics, bandwidth trade-offs, dark-current decomposition, and quantum efficiency formalism that underpin the benchmarked designs in table 2 have been derived from first principles in sections 2 and 3. The key quantitative relationships, the transit-time bandwidth of eq. (55), the RC bandwidth of eq. (58), the combined quadrature bandwidth of eq. (54), and the SRH dark-current scaling of eq. (37), are applied directly in the design and simulation sections that follow without re-derivation. Among the coupled design variables (W_i , L , A_j , R_s , L_p), reducing R_s through electrode shaping is the structurally cleanest way to improve f_{RC} without degrading responsivity or dark current, the physical insight that makes the U-shaped anode topology, as realised in [1] and studied here at 1310 nm, worth understanding in detail.

7 Device Architecture and Design Rationale

The preceding sections establish the physical foundations (section 2), the performance metrics (section 3), and the architectural and material choices (sections 4 and 5) that reduce the design space to a vertical waveguide-integrated Ge-on-Si PIN photodetector at 1310 nm.

The present device adopts the p^{++} -Si platform polarity precisely to minimise the interface generation contribution while maintaining CMOS process compatibility.

7.1 Vertical PIN Structure: Geometry, Layer Stack, and Field Distribution

The photodiode is realised on a 220 nm SOI platform. Each layer parameter in table 4 is the result of a specific physical argument derived from the metric framework of section 3; the same layer stack is used by Shi *et al.* [1] in the benchmark device.

Table 4: Layer stack and primary material parameters of the Ge-on-Si n-i-p photodiode simulation model. All doping concentrations are net values. Layer parameters follow [1].

Layer	Material	Thickness	Type	Concentration [cm ⁻³]	Primary function
Cathode	n^{++} -Ge	50 nm	n	10^{19}	Ohmic contact; minimise FCA
Absorber	i -Ge	350 nm	intrinsic	$\sim 10^{13}$	Optical absorption; depletion
Platform	p^{++} -Si	220 nm	p	10^{20}	Anode contact; current spreading
BOX	SiO ₂	2 μ m	N/A	N/A	Optical isolation
Handle	Si	$\sim 700 \mu$ m	N/A	N/A	Mechanical support

The intrinsic germanium thickness $W_i = 350$ nm follows from the optimisation of eq. (61). It lies within 11 % of $W_i^* \approx 310$ nm and gives the conservative lower-bound estimate $f_{tt} \approx 51$ GHz from eq. (55). For the same 350 nm stack, the benchmark-calibrated two-carrier response with peaking reaches the 100 GHz to 120 GHz class through eq. (65), as demonstrated experimentally by Shi *et al.* [1]. The n^{++} -Ge cathode is kept at 50 nm to limit free-carrier absorption (FCA) in the contact region and to suppress the diffusion tail that would otherwise lower the effective f_{3dB} below the prediction of eq. (55) [17].

Doping polarity and its impact on dark current.

The p^{++} -Si/ i -Ge/ n^{++} -Ge polarity (Si platform p -doped) places the graded Ge/Si transition such that the valence-band discontinuity ΔE_v acts as an additive barrier to interface hole generation. In the alternative p -on- n configuration (Si platform n -doped), the same ΔE_v raises J_{surf} above the value predicted by eq. (38) by modifying the effective barrier for thermally generated holes. For a fixed reverse bias of -1 V, the n -on- p device exhibits I_d a factor of approximately two to three lower than the p -on- n counterpart, translating through eq. (108) into a sensitivity improvement of approximately 1.5 dB in the dark-current-dominated regime [26].

7.2 Optical Coupling via Adiabatic Taper

Optical coupling from the input silicon waveguide is provided by a 40 μm adiabatic taper, following the coupling scheme of Shi *et al.* [1]. The taper adiabatically transforms the mode of the Si access waveguide into the Ge-loaded mode over a length L_t long enough to suppress excitation of higher-order modes. By narrowing the Si waveguide tip to approximately 100 nm to 200 nm, the guided mode is pushed evanescently upward into the overlying Ge absorber. At the distal end of the taper the mode is fully confined within the Ge, and absorption proceeds with near-unity coupling efficiency. Evanescent coupling spreads the power transfer over the full taper length, lowering the peak generation density for the same absorbed power and preserving the geometric orthogonality of L and W_i [17], [25].

7.3 U-Shaped Electrode Topology: RC Reduction and Current Path Engineering

The U-shaped anode electrode is the primary design feature of the benchmark architecture that this work adopts. It is the dominant determinant of RC bandwidth in the vertical structure, and the step from the parallel electrode to the U-shaped topology is responsible for the full improvement from 83 GHz to 103 GHz observed by Shi *et al.* [1].

7.3.1 The Vertical n-i-p Equivalent Circuit

The complete small-signal equivalent circuit of the waveguide-coupled vertical n-i-p photodiode is represented by the lumped network of eq. (93). The photocurrent source $I_{\text{ph}} = \mathcal{R}P_{\text{opt}}$ drives the junction admittance, which consists of the junction capacitance C_j in parallel with the shunt conductance $G_{\text{sh}} = I_d/V_{\text{bias}}$. The junction is accessed through the series resistance network comprising the cathode series resistance $R_{s,\text{cat}}$ and the anode series resistance $R_{s,\text{an}}$ (path through the p^{++} -Si platform from the mesa base to the metal anode pad). The bondpad-to-chip parasitic capacitance C_{par} shunts the junction at high frequencies, and the load termination $R_L = 50 \Omega$ converts the photocurrent into a voltage signal. The simulated small-signal frequency response and peaking behavior are presented and analyzed in figs. 24 and 25.

Lumping $R_s = R_{s,\text{cat}} + R_{s,\text{an}}$ and $C_{\text{tot}} = C_j + C_{\text{par}}$, the voltage transfer function from photocurrent source to load voltage is:

$$H_{\text{RC}}(f) = \frac{V_{\text{out}}(f)}{I_{\text{ph}}(f)} = \frac{R_L}{1 + j2\pi f [R_L(C_j + C_{\text{par}}) + R_s C_j] - (2\pi f)^2 R_s R_L C_j C_{\text{par}}}, \quad (93)$$

which for $R_s \ll R_L$ reduces to the single-pole expression of eq. (58).

7.3.2 Parallel-Electrode versus U-shaped Anode: Quantitative Comparison

In the standard parallel-electrode architecture, the metal anode pad is placed on one side of the germanium mesa, and the current must flow laterally through the p^{++} -Si platform from the opposing mesa sidewall to the pad. The mean lateral path length is approximately $\ell_a^{\text{par}} \approx W_m/2 + d_{\text{gap}}$, where W_m is the mesa width and d_{gap} is the clearance between the mesa sidewall and the anode metal edge. For a 5 μm -wide mesa and a 0.5 μm lithographic clearance:

$$R_{s,\text{an}}^{\text{par}} \approx \frac{\rho_{\text{Si}}}{t_{\text{Si}}} \cdot \frac{\ell_a^{\text{par}}}{L} = \frac{\rho_{\text{Si}}}{t_{\text{Si}}} \cdot \frac{W_m/2 + d_{\text{gap}}}{L}. \quad (94)$$

In the U-shaped anode architecture, the metal wraps around three sides of the mesa (the two long sidewalls and one short end), so that current from any point on the mesa base must travel at most $W_m/2$ laterally before reaching a metal contact. The effective mean path length is reduced to [1]:

$$\ell_a^{\text{U}} \approx \frac{W_m}{4} + d_{\text{gap}}, \quad (95)$$

giving a series resistance:

$$R_{s,\text{an}}^{\text{U}} \approx \frac{\rho_{\text{Si}}}{t_{\text{Si}}} \cdot \frac{W_m/4 + d_{\text{gap}}}{L}. \quad (96)$$

The reduction ratio is:

$$\frac{R_{s,\text{an}}^{\text{U}}}{R_{s,\text{an}}^{\text{par}}} = \frac{W_m/4 + d_{\text{gap}}}{W_m/2 + d_{\text{gap}}} \approx 0.60 \quad \text{for } W_m = 5 \mu\text{m}, d_{\text{gap}} = 0.5 \mu\text{m}, \quad (97)$$

corresponding to a 40 % resistance reduction, consistent with the 36 % reported experimentally by Shi *et al.* [1]. Substituting into eq. (58):

$$\frac{f_{\text{RC}}^{\text{U}}}{f_{\text{RC}}^{\text{par}}} = \frac{R_{\text{tot}}^{\text{par}}}{R_{\text{tot}}^{\text{U}}} = \frac{R_L + R_{s,\text{an}}^{\text{par}}}{R_L + R_{s,\text{an}}^{\text{U}}}. \quad (98)$$

For $R_L = 50 \Omega$, $R_{s,\text{cat}} = 43 \Omega$, $R_{s,\text{an}}^{\text{par}} = 32 \Omega$, and $R_{s,\text{an}}^{\text{U}} = 20 \Omega$, this gives $f_{\text{RC}}^{\text{U}}/f_{\text{RC}}^{\text{par}} = 125/113 \approx 1.11$, a modest 5.5 % increase in f_{RC} alone. The larger bandwidth gain emerges only when the electrode is treated as part of the full electro-optical co-design problem. In the benchmark-calibrated peaked circuit of eq. (65), the reduced R_s also enables a shallower and more useful inductive peaking condition, and Shi *et al.* report that the bandwidth increases from 83 GHz (parallel electrode) to 103 GHz (single-U) [1].

Mask-level representation and layout validation.

After the optical and electrical geometry is fixed, the mask representation is checked with the layout generation script, which writes the device cell to the layout GDS data file. The layout uses a database unit of 1 nm and maps the principal process features onto ten logical layers corresponding to the Si slab, Si ridge, Si taper, Ge, p^{++} region, n^{++} region, via level, M1, M2, and BOX.

The geometry parameters, W_i , L , W_m , the electrode topology, and the taper length, form the active design variables in the simulation. The mask-level footprint of this cell, including all ten process layers, is validated by the layout validation script. Their quantitative mapping to the metrics of section 3, including the absorber-length sweep, intrinsic-thickness trade-off, and electric-field screening threshold, is evaluated with simulation data in section 9.

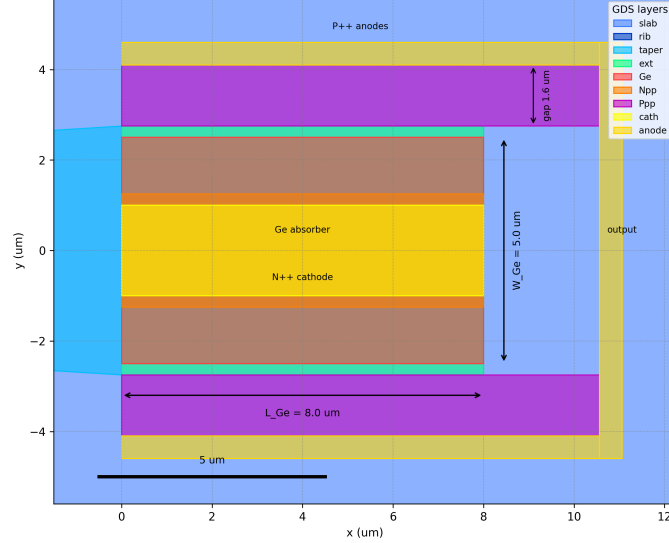


Figure 16: Top-view mask layout of the waveguide-integrated vertical n - i - p Ge-on-Si photodetector generated from the layout script. Colour layers correspond to the Si slab, Si rib waveguide, adiabatic input taper, Si extension, Ge absorber ($L_{\text{Ge}} = 8.0 \mu\text{m}$, $W_{\text{Ge}} = 5.0 \mu\text{m}$), n^{++} cathode, p^{++} anode regions (U-shaped, electrode gap = $1.6 \mu\text{m}$), and metal contacts. The output stub to the right carries the through-waveguide signal for back-to-back loss calibration.

8 Multiphysics Simulation Methodology

The performance metrics defined in section 3 are evaluated through a coupled simulation chain: the parameterised geometry is solved first in a three-dimensional electromagnetic optical solver, whose generation-rate export is then passed to a two-dimensional drift-diffusion charge transport solver, and post-processing scripts extract compact-model parameters from the resulting datasets. A layout editor is used after the design is fixed, serving as a post-design geometry verifier rather than a quantitative design generator. Each stage passes structured data files to the next, ensuring the workflow is fully reproducible without manual parameter transcription. Figure 17 maps the data flow between the optical, electrical, and circuit-level simulation stages. Two calibration anchors bracket the workflow: the Beer-Lambert absorption trend validates the optical stage, and the electrical response class of Shi *et al.* [1] validates the compact-model stage.

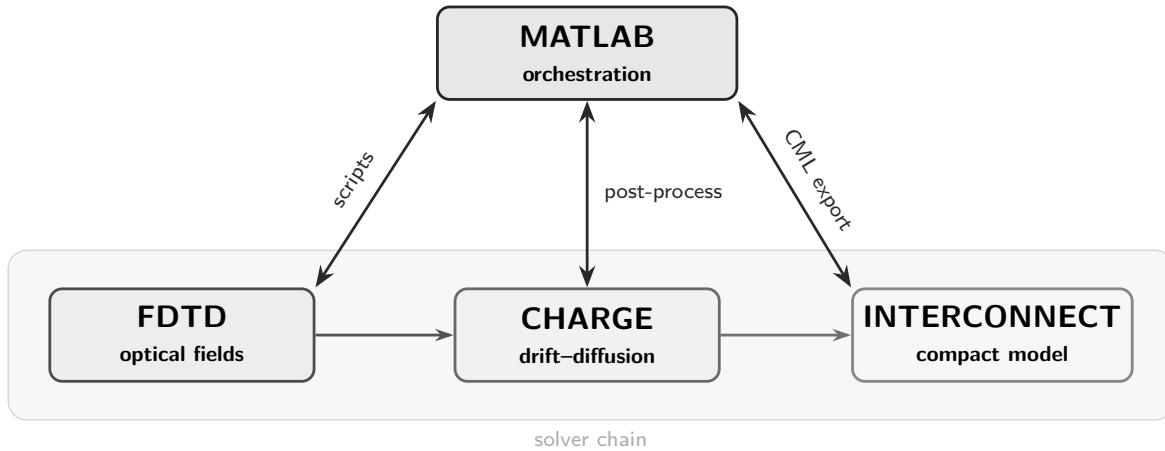


Figure 17: Multiphysics simulation workflow. Numerical orchestration scripts connect the optical, electrical, and circuit-level stages and post-process the exported data. The optical solver delivers volumetric generation profiles to the electrical solver; the electrical solver exports current, capacitance, and small-signal response quantities; the circuit-level stage receives the compact detector model for system-level evaluation. The dashed path indicates the optional direct import of optical scattering parameters.

8.1 Post-Design Layout Validation

After the solver geometry is defined and verified, the layout generation script writes the lithographic cell to the standard GDS layout file for geometry inspection and process-layer validation. The GDS encodes the full layer stack of table 4, the 8 μm Ge mesa, the 40 μm adiabatic taper, the U-shaped cathode implant window, and the metal pad perimeters, on the process layers of the 220 nm SOI platform. This layout tool therefore acts as the post-design verifier for geometry consistency and fabrication-readiness checks.

8.2 Optical Simulation: Three-Dimensional FDTD

The optical simulation solves the time-domain Maxwell equations of eqs. (1) and (2) throughout a domain encompassing the silicon input waveguide, the 40 μm adiabatic taper, the complete germanium absorber, and the surrounding SiO_2 cladding. The electromagnetic field is discretised on a Yee staggered grid using the second-order Courant-stable leapfrog scheme. The Yee algorithm and the absorbing boundary conditions are described in detail in Wartak [27].

The complex permittivity is sampled at the material-dependent optical frequency through tabulated refractive indices. For analytical responsivity estimates, a value of $\alpha \approx 3 \times 10^3 \text{ cm}^{-1}$ at 1310 nm is used as an upper-bound for bulk Ge [28]; the three-dimensional solver evaluates absorption directly from the frequency-dependent material model, yielding an effective

waveguide-confined absorption coefficient consistent with the Beer-Lambert coefficient derived in eq. (6).

FDTD source field and the ε'' -based generation rate.

Following the convention of Hamadou *et al.* [5], the monochromatic optical source is a Gaussian-modulated pulse:

$$\tilde{\mathbf{E}}(\mathbf{r}, t) = E_0 \exp\left[-\frac{(t - t_0)^2}{2\Delta t^2}\right] \sin[-\omega(t - t_0)] \hat{\mathbf{x}}, \quad (99)$$

where t_0 is the temporal offset and Δt is related to the pulse FWHM by $\Delta t = \text{FWHM}/(2\sqrt{\ln 2})$. After the time-domain solution is Fourier-transformed, the volumetric optical generation rate extracted from the monochromatic field is:

$$G(\mathbf{r}, \omega) = \frac{\pi \varepsilon_0 \varepsilon_r''(\omega)}{h} |\tilde{\mathbf{E}}(\mathbf{r}, \omega)|^2, \quad (100)$$

where $\varepsilon_r''(\omega) = 2n\kappa$ is the imaginary part of the relative permittivity and h is Planck's constant [5]. Equation (100) is equivalent to eq. (10) under the plane-wave approximation, and reduces to eq. (15) in the uniform-field limit; the FDTD solver evaluates it pointwise over the three-dimensional Ge volume before transferring the resulting $G(\mathbf{r})$ array to the drift-diffusion electrical solver. Frequency-domain power monitors record the input-normalised transmission before and after the Ge absorber, from which the post-processing scripts compute:

$$\mathcal{A}(\omega) = \max\left[\frac{T_{\text{ref}}(\omega) - T_{\text{after}}(\omega)}{T_{\text{ref}}(\omega)}, 0\right]. \quad (101)$$

A 20 nm isotropic mesh override within the germanium region resolves the exponential generation-rate gradient near the taper exit.

FDTD numerical stability and the 3D Courant condition.

The perfectly matched layer (PML) implementation employs eight absorbing layers on each domain boundary. Temporal stability of the Yee algorithm requires the 3D Courant-Friedrichs-Lewy condition:

$$\Delta t < \frac{\Delta}{c\sqrt{3}}, \quad (102)$$

so the time step is set to $\Delta t = 0.99 \Delta/(c\sqrt{3})$. The spatial mesh size is independently constrained by the sampling requirement $\Delta < \lambda_{\min}/(10\sqrt{\varepsilon_{\max}})$, giving $\Delta < 32$ nm with a safety margin of 1.6 at the 20 nm override.

FDTD electromagnetic solution and the square-law photocurrent.

The frequency-domain power recorded by the monitors is the time-averaged Poynting vector $\mathbf{S}(\mathbf{r}, \omega) = \frac{1}{2}\text{Re}[\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*]$, which is quadratic in the field amplitudes and consistent with the

square-law mechanism of eq. (12). The optical generation rate $G_{\text{opt}}(\mathbf{r}) = \mathcal{P}_{\text{abs}}(\mathbf{r})/\hbar\omega$ of eq. (15) maps directly to the photocurrent through $I_{\text{ph}} = q \int_{\Omega} G_{\text{opt}} dV$.

Waveguide insertion loss and chip-level responsivity correction.

Beer-Lambert saturation in the absorber-length sweep is computed within the optical solver domain, which terminates at the taper entry. For a Si ridge waveguide with propagation loss $\gamma_{\text{wg}} \approx 3.6 \text{ dB cm}^{-1}$ at 1310 nm, a routing length of $\ell_{\text{wg}} = 1 \text{ mm}$ introduces an approximately 0.36 dB correction, equivalent to a 4 % responsivity reduction relative to the taper-input value [25]. Any comparison with chip-level measurements must apply the factor $10^{-\gamma_{\text{wg}}\ell_{\text{wg}}/10}$ to eq. (48).

The optical simulation stage closes by exporting the volumetric generation profile to a structured dataset file. This file is the sole data interface to the electrical solver, which imports the generation array through a dedicated optical-generation coupling object and does not re-solve any optical physics.

8.3 Electrical Simulation: Two-Dimensional Drift-Diffusion

Carrier transport is solved under two-dimensional X-normal steady-state conditions, with the longitudinal norm length set to the Ge absorber length $L = 8 \mu\text{m}$. The finite-element solver evaluates the vertical cross-section and scales all currents by this length, solving the coupled PDE system of eqs. (16) to (18), (22) and (23) on an adaptive mesh.

Two complementary carrier mobility models are employed. The *Caughey-Thomas* model accounts for impurity scattering through a doping-concentration-dependent low-field mobility [5]:

$$\mu_{n,p}(N_A + N_D) = \mu_{n,p}^{\min} + \frac{\mu_{n,p}^L - \mu_{n,p}^{\min}}{1 + \left(\frac{N_A + N_D}{N_{n,p}^0}\right)^{\alpha_{n,p}}}, \quad (103)$$

where $\mu_{n,p}^L$ is the lattice-scattering-limited mobility, $\mu_{n,p}^{\min}$ the ionised-impurity saturation floor, and $N_{n,p}^0$, $\alpha_{n,p}$ are fitting parameters from the literature. High-field velocity saturation is superimposed through a monotonic saturation model [5]:

$$\mu'_{n,p}(|\mathbf{E}|) = \frac{\mu_{n,p}}{\left[1 + \left(\frac{\mu_{n,p}|\mathbf{E}|}{v_{n,p}^S}\right)^{\beta_{n,p}}\right]^{1/\beta_{n,p}}}, \quad (104)$$

yielding the field-dependent diffusivity $D_{n,p} = \mu'_{n,p}k_B T/q$ via the Einstein relation of eq. (19). Equation (104) is equivalent in functional form to the Canali velocity-saturation model [26]:

$$\mu_{n,p}(|\mathbf{E}|) = \frac{(\alpha_C + 1) \mu_{n,p}^{(0)}}{\left[\alpha_C + \left(1 + \left(\frac{(\alpha_C + 1) \mu_{n,p}^{(0)} |\mathbf{E}|}{v_{\text{d,sat}}}\right)^{\beta}\right)^{1/\beta}\right]}, \quad (105)$$

which reduces to the saturation model of eq. (106) with $\beta = 2$ and $\alpha_C = 0$. The low-field mobilities adopted are $\mu_n^{(0)} = 3900 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ and $\mu_p^{(0)} = 1900 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$; the saturation velocities $v_{\text{d,sat},e} = 6 \times 10^6 \text{ cm s}^{-1}$ and $v_{\text{d,sat},h} = 4 \times 10^6 \text{ cm s}^{-1}$ enter eq. (105) as asymptotic limits. A $\pm 20\%$ variation in $\mu_p^{(0)}$ changes f_{tt} by less than 1 %, confirming that the transit-time estimate of eq. (55) is insensitive to reasonable low-field mobility uncertainty [26].

$$\mu_{n,p}(|\mathbf{E}|) = \frac{\mu_{n,p}^{(0)}}{\left[1 + \left(\mu_{n,p}^{(0)}|\mathbf{E}|/v_{\text{d,sat},n,h}\right)^\beta\right]^{1/\beta}}, \quad (106)$$

Junction capacitance and the microhole filling factor.

For a planar junction the parallel-plate capacitance follows eq. (57). If photon-trapping microholes of total area A_h are etched through the junction region, the effective junction area is reduced by the filling factor [5]:

$$\rho_f = \frac{A_h}{A_j}, \quad C_{j,\text{mh}} = \frac{\varepsilon_0 \varepsilon_r A_j}{W_i} (1 - \rho_f), \quad (107)$$

so microholes simultaneously improve absorption efficiency and reduce capacitance at the cost of increased surface recombination along the etched sidewalls, which enters through eq. (38). The present Ge-on-Si design does not employ microholes; eq. (107) is retained as a reference for future microstructure extensions.

Bias sweeps and AC analysis.

SRH recombination is activated in the germanium absorber with calibrated lifetimes $\tau_n = \tau_p = 1 \text{ ns}$, consistent with the dislocation-limited material discussed in section 5.3. Ohmic boundary conditions are imposed at the cathode ($V = 0$) and the anode, which is swept from -2.5 V to 0.5 V over 51 points. The dark sweep runs first; the optical generation-rate dataset is then imported and the illuminated sweep repeated under identical solver settings. The same configuration also executes a small-signal AC frequency sweep from 1 MHz to 200 GHz, extracting the frequency response without a separate time-domain run. All performance metrics follow directly from these two sweeps: $\mathcal{R}$ from eq. (48); η_{int} and η_{ext} from eq. (53); C_j from the AC capacitance extraction; and $f_{3\text{dB}}$ from eq. (54) using the extracted C_j and the modelled R_s from eq. (59).

8.4 Compact-Model Post-Processing

The numerical post-processing pipeline consumes the electrical solver output and closes the simulation chain. Three categories of computation are performed in sequence. First, the combined transfer function of eqs. (54) and (65) is applied to the extracted C_j , the modelled R_s , and the benchmark-calibrated peaking parameters (L_p, C_p) to produce the frequency

response and the $f_{3\text{dB}}$ estimate. Second, the sensitivity formula of eq. (108) is evaluated at the projected bandwidth using the solver-exported I_d and $\mathcal{R}$. Third, parametric design sweeps over absorber length and intrinsic-layer thickness produce the trade-off tables and design-space contour of section 9.5, confirming that the implemented $8\text{ }\mu\text{m}\times 5\text{ }\mu\text{m}$ geometry sits within the feasibility region for both the DR4 bandwidth floor and the responsivity floor simultaneously.

The calibration check of fig. 17 (Cal. 2) is performed at this stage: the geometry-derived bandwidth and the peaking-circuit response are compared with the electrical response class of Shi *et al.* [1] to confirm that the model reproduces the 100 GHz-class behaviour before the compact-model parameters are synchronised into the system-level circuit simulator foundry element.

8.5 Validation Approach and Reference Design

Validation is conducted at three levels of the simulation pipeline; the two calibration anchors in fig. 17 correspond to the first and third of these.

At the optical level (Cal. 1), the absorption monitor output is checked against the Beer-Lambert trend over the Ge-length sweep. A monotonic approach to the saturation plateau, with the saturation length consistent with the O-band absorption coefficient $\alpha \approx 3 \times 10^3 \text{ cm}^{-1}$ derived in eq. (6), confirms that the electromagnetic mesh and material model are internally consistent before any electrical physics is engaged.

At the electrical level, the dark and illuminated sweeps are reduced to I_d , I_{ph} , $\mathcal{R}$, EQE, and IQE by the post-processing scripts. The IQE check is particularly informative: a value near unity under full depletion confirms that photogenerated carriers are swept to their contacts before SRH recombination within the $\tau_0 = 1 \text{ ns}$ lifetime, consistent with the argument of section 5.3.

At the compact-model level (Cal. 2), the geometry-derived RC and transit-time bandwidths are evaluated from the implemented device dimensions and compared with the AC frequency response from the electrical solver. The resulting $f_{3\text{dB},\text{base}} \approx 56.1 \text{ GHz}$ represents the no-peaking baseline; adding the benchmark-calibrated peaking network (L_p, C_p) reproduces the 100 GHz-class response through eq. (65), consistent with the experimental measurement of Shi *et al.* [1].

The carrier transport physics governing the electrical response of the Ge absorber, drift-diffusion transport, velocity saturation, Caughey-Thomas impurity scattering, SRH recombination, and the bandwidth-absorption trade-off, is fully established in sections 2, 3 and 5. The electrical solver evaluates the complete PDE system of eqs. (16) to (18), (22) and (23) self-consistently. No re-derivation of the mobility or bandwidth expressions is required here; this section solely documents the solver settings, data interfaces, and validation checks that ensure the physics is applied consistently across the four simulation stages.

A physics-informed neural-network surrogate is a possible future extension of this workflow but is not part of the present simulation study.

9 Simulation Results and Discussion

This section presents the results of the complete optical, electrical, and system-level simulation chain. Throughout this analysis, the experimental measurements of Shi *et al.* [1] serve as a calibration reference, confirming the physical validity of the numerical models.

9.1 Simulation at O-band Conditions

The numerical models are configured for O-band operation at a nominal wavelength of 1310 nm and a temperature of 300 K, utilizing a U-shaped anode photodetector geometry with an 8 μm long and 5 μm wide germanium mesa. The U-shaped electrode collector topology provides a low series resistance, which is critical for high-speed operation. The material parameters of Germanium are configured with a direct bandgap $E_{g,\text{dir}} = 0.80 \text{ eV}$ and an absorption coefficient $\alpha \approx 3 \times 10^3 \text{ cm}^{-1}$, ensuring strong optical coupling within a compact footprint.

9.2 Device Layout and Optical Generation Profile

Figure 18 shows the photodetector layout and mesh grid structure. The 3D layout view in (a) shows the Si access waveguide coupled via a lateral taper to the Ge-on-Si mesa, which is flanked by the U-shaped contact electrodes. The 2D cross-sectional mesh grid in (b) shows the spatial discretization used by the numerical solvers, ensuring adequate resolution at the heterointerfaces.

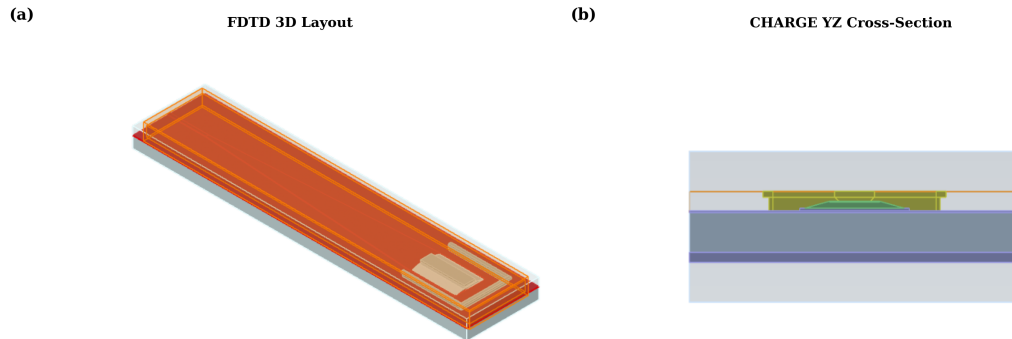


Figure 18: Photodetector layout and mesh configuration: (a) 3D CAD layout showing the tapered silicon waveguide interface and the U-shaped anode electrode contact; (b) 2D cross-sectional grid mesh showing high refinement at the Ge/Si heterointerface.

The spatial intensity, potential, generation, and energy band profiles are consolidated in fig. 19. The FDTD longitudinal intensity $|E|^2(x, y)$ inside the Ge core in (a) shows optical confinement and mode propagation. The CHARGE electrostatic potential contours $\varphi(y, z)$ in (b) confirm that the intrinsic region is fully depleted at the design bias of -1 V. The normalized optical generation rate profile and its cumulative percentage along the length in (c) show that absorption is concentrated near the waveguide taper and decays along the propagation direction. The energy band diagram E_c, E_v and quasi-Fermi levels E_{fn}, E_{fp} vs. height z in (d) confirm junction depletion.

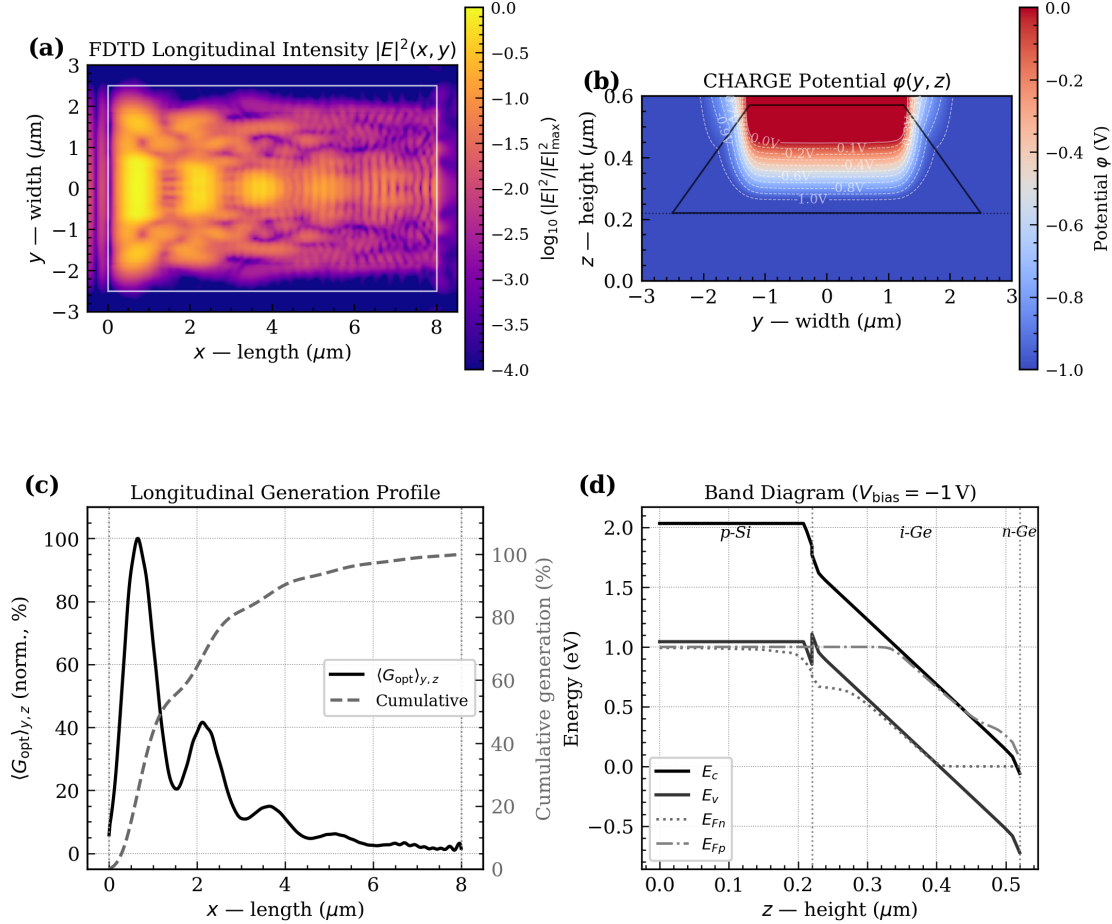


Figure 19: Spatial simulation profiles at -1 V and 1310 nm: (a) FDTD longitudinal intensity $|E|^2(x, y)$ inside the Ge absorber; (b) CHARGE electrostatic potential contours $\varphi(y, z)$ showing depletion; (c) normalized longitudinal optical generation profile and its cumulative percentage; (d) CHARGE energy band diagram E_c, E_v and quasi-Fermi levels E_{fn}, E_{fp} vs. height z .

9.3 Static and Spectral Characteristics

This subsection characterises the device under quasi-static and spectrally resolved conditions, covering responsivity, dark current, photocurrent linearity, and junction capacitance in turn.

Each quantity is compared against the coupled-solver prediction and, where applicable, against the experimental benchmark of Shi *et al.* [1].

Responsivity and quantum efficiency.

Figure 20 shows the spectral responsivity and static power characteristics. In panel (a), the photocurrent shows a linear response up to $100\text{ }\mu\text{W}$ at -1.0 V before onset of saturation. Panel (b) plots the responsivity vs. photon energy and penetration depth, illustrating the Ge spectral cutoff. In panel (c), the FDTD solver predicts an optical responsivity of 0.968 A W^{-1} at 1310 nm ($\text{EQE} = 91.63\%$), representing the coupling limit. The CHARGE solver, accounting for carrier recombination and collection losses, gives a responsivity of 0.931 A W^{-1} ($\text{EQE} = 88.08\%$, $\text{IQE} = 96.23\%$), which lies within 2% of the experimental benchmark of 0.95 A W^{-1} [1]. Panel (d) shows the responsivity vs. bias voltage, confirming carrier collection saturation above -0.5 V .

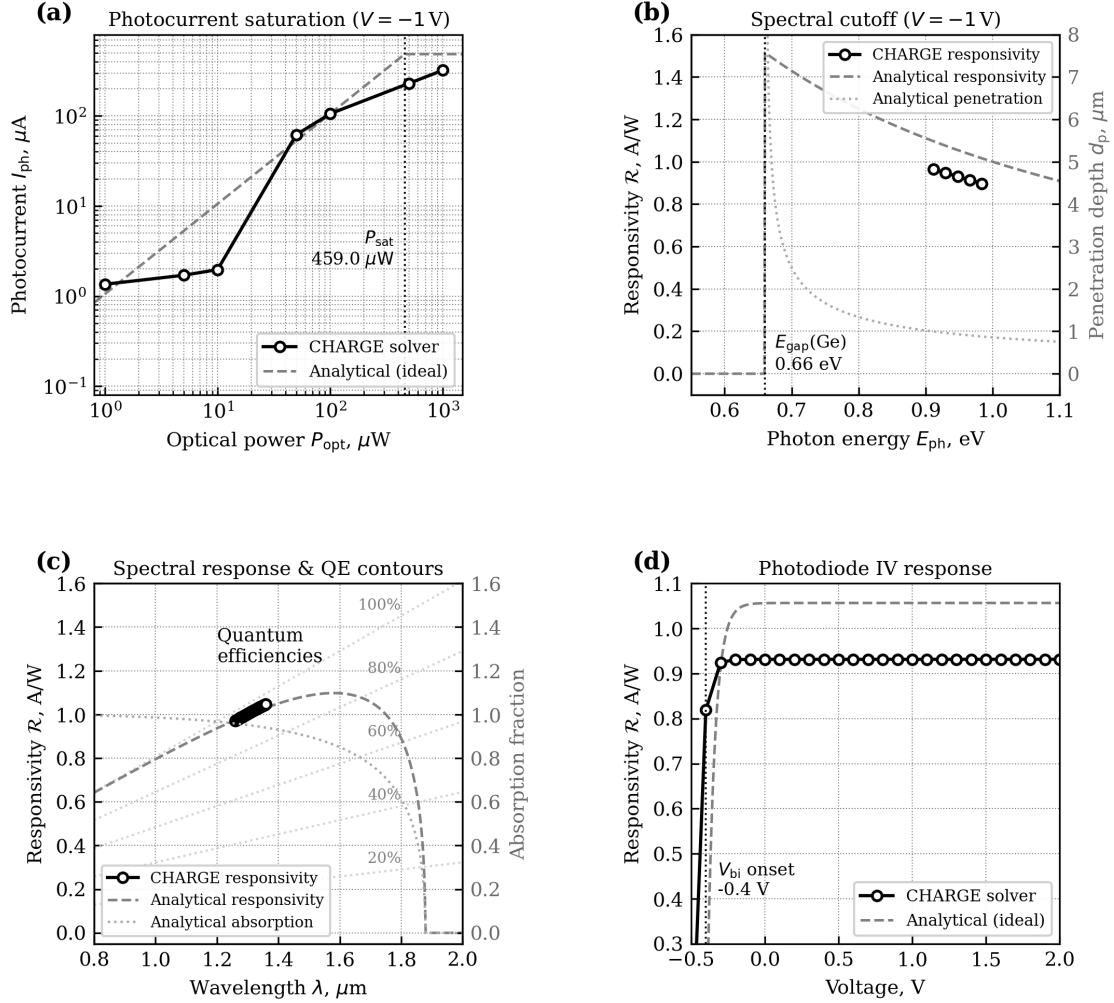


Figure 20: Responsivity and static characteristics: (a) photocurrent vs. input optical power showing saturation; (b) responsivity vs. photon energy and penetration depth; (c) responsivity vs. wavelength for FDTD and CHARGE solvers; (d) responsivity vs. bias voltage showing photodiode turn-on.

Design sweeps and responsivity limits.

Figure 21 profiles the responsivity sweeps across the design space. Panel (a) shows the 3D design landscape of responsivity as a function of mesa dimensions. Panel (b) confirms that carrier collection efficiency approaches unity when the drift velocity exceeds the saturation threshold. Panel (c) plots responsivity vs. junction temperature, comparing simulation results with the analytical model. Panel (d) shows the responsivity roll-off vs. excess carrier concentration, highlighting the onset of Shockley–Read–Hall and Auger recombination at high injection levels.

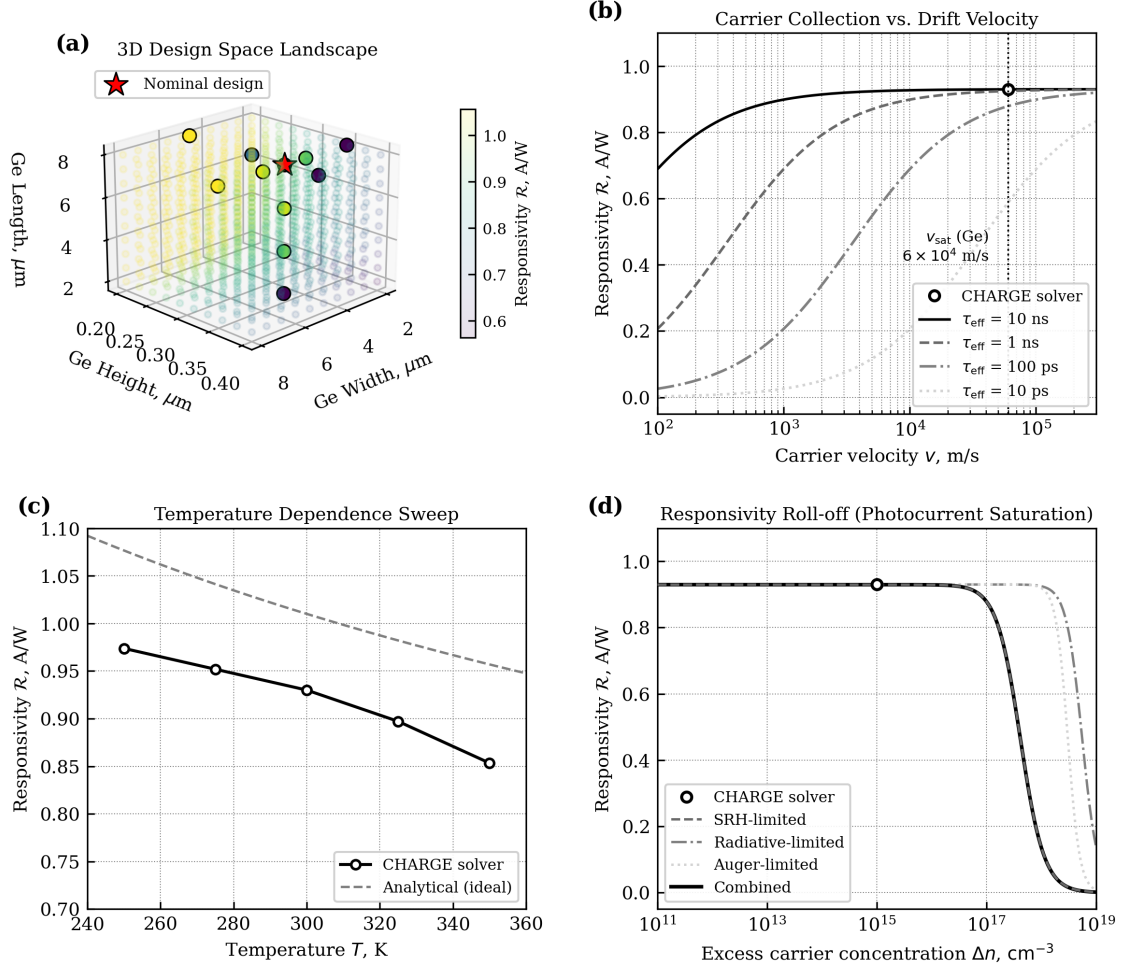


Figure 21: Responsivity sweeps and physical limits: (a) 3D design space landscape of responsivity vs. mesa dimensions; (b) carrier collection efficiency vs. drift velocity; (c) responsivity vs. junction temperature; (d) responsivity roll-off vs. excess carrier concentration.

Dark current and thermal activation.

The dark current is the primary noise source in a reverse-biased photodiode. Figure 22 characterises the leakage in both the electrical and thermal domains. Panel (a) plots the quasi-static I-V curves, showing that at the nominal bias of -1.0 V, the simulated dark current is 0.684 nA, representing an approximately $1.9\times$ reduction compared with the experimental benchmark of 1.3 nA [1]. Panel (b) shows dark current vs. background doping concentration, illustrating bulk generation. Panel (c) shows the Arrhenius temperature dependence, with the dark current doubling for every $\Delta T \approx 10$ K, consistent with a single-level bulk trap model with an activation energy of $E_a \approx 0.315$ eV. Panel (d) shows the 3D design space landscape of dark current vs. mesa dimensions.

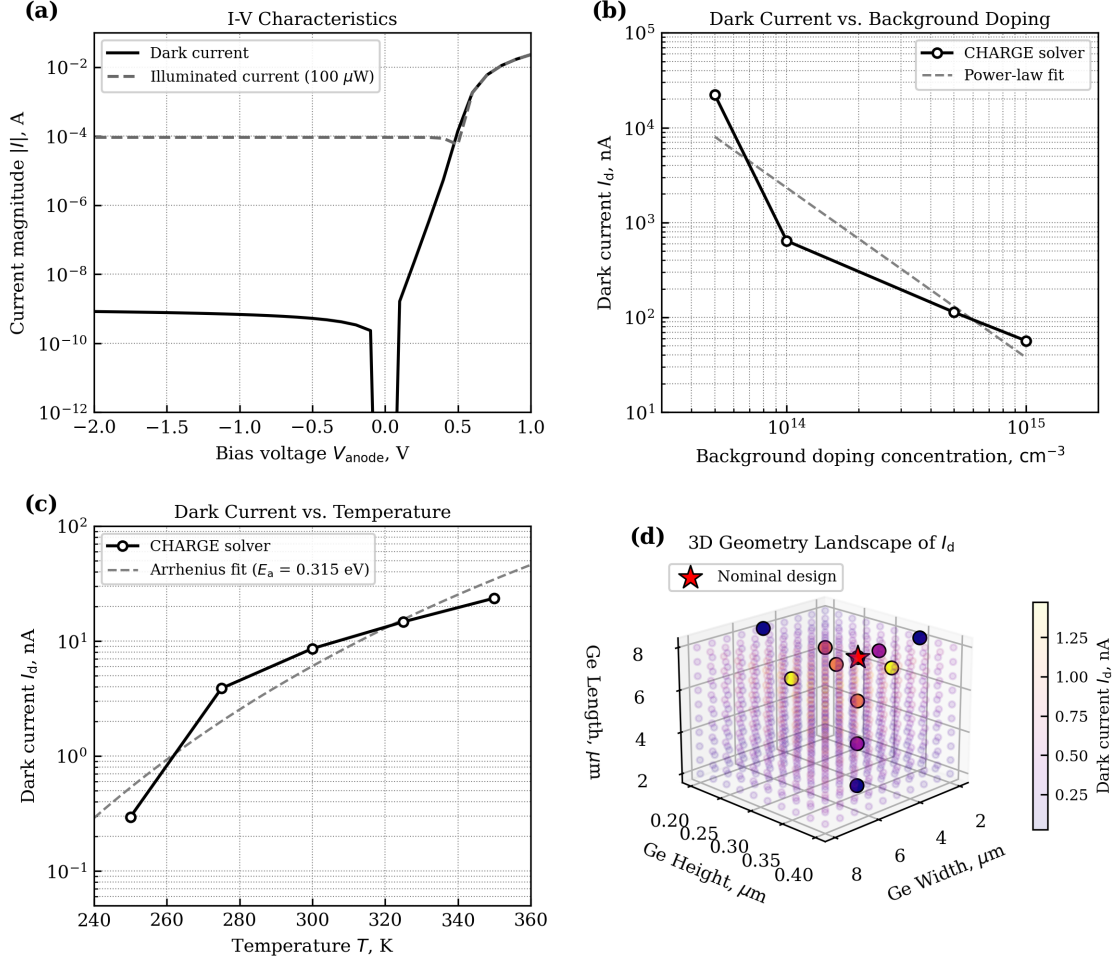


Figure 22: Dark current characteristics and sweeps: (a) quasi-static I-V curves under dark and illuminated conditions; (b) dark current vs. background doping concentration; (c) Arrhenius plot of dark current vs. temperature; (d) 3D design space landscape of dark current vs. mesa dimensions.

Noise performance and recombination limits.

Figure 23 examines the noise limits and recombination rates. Panel (a) shows shot noise vs. dark current, confirming that at the nominal operating point the noise matches the analytical limit of $\sqrt{2qI_d} \approx 14.8 \text{ fA}/\sqrt{\text{Hz}}$. Panel (b) shows thermal noise vs. resistance, indicating a thermal noise current density of $\sqrt{4kT/R} \approx 15.3 \text{ pA}/\sqrt{\text{Hz}}$ at the nominal loop resistance of 70Ω . Panel (c) shows dark current density vs. sidewall perimeter-to-area ratio, highlighting the surface leakage contribution. Panel (d) compares the recombination rates (SRH, radiative, and Auger) vs. carrier concentration, demonstrating the dominance of bulk SRH recombination.

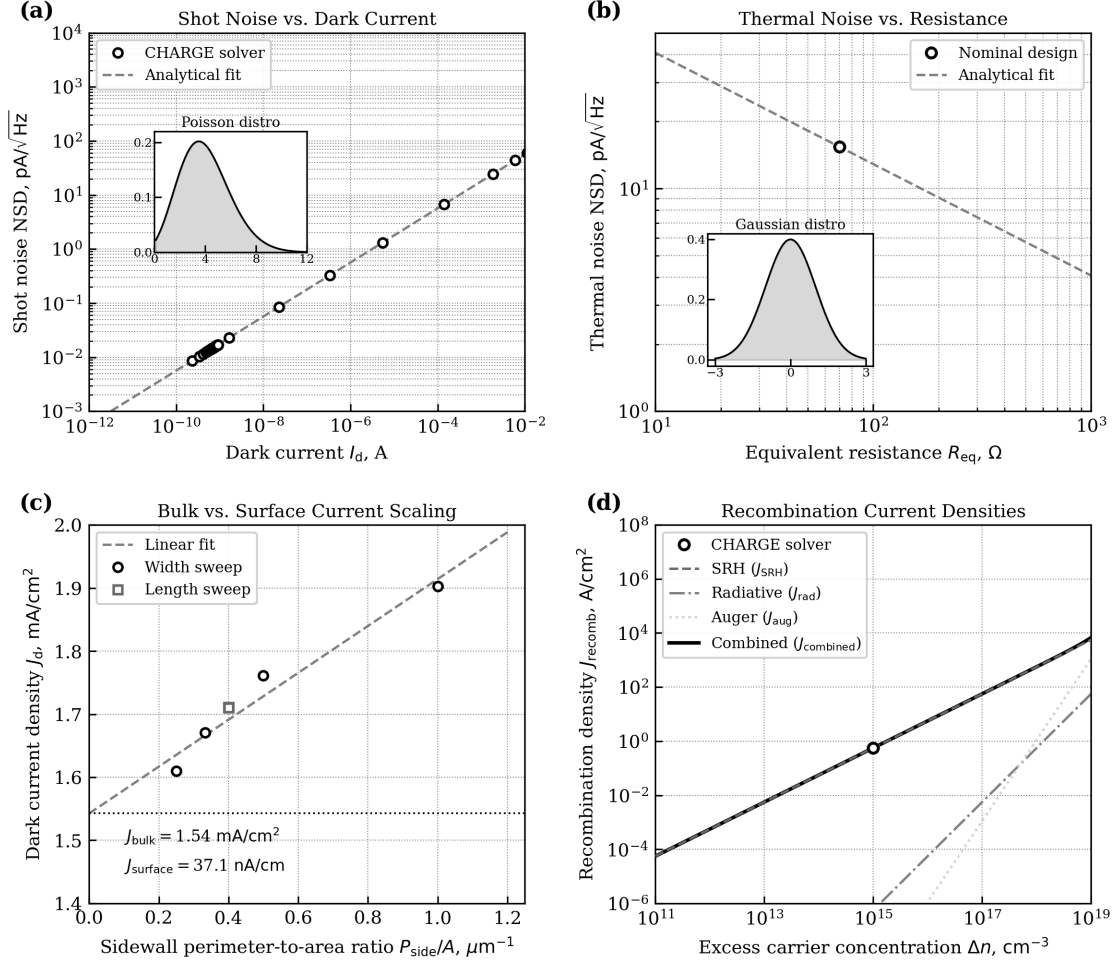


Figure 23: Noise performance and recombination limits: (a) shot noise vs. dark current; (b) thermal noise vs. loop resistance; (c) dark current density vs. perimeter-to-area ratio; (d) bulk recombination current densities vs. excess carrier concentration.

Junction capacitance.

The junction capacitance C_j sets the RC time constant of the photodetector. At the nominal bias of -1.0 V, the capacitance is 11.87 fF, as shown in the bias voltage sweep of fig. 25(b). This capacitance value is lower than the parallel-plate estimate of 16.2 fF computed from the physical mesa dimensions ($L = 8$ μm , $W_m = 5$ μm) because the depletion region does not span the full geometric width, yielding an effective electrical width $W_{\text{eff}} = 3.665$ μm as detailed in Table 5.

9.4 Bandwidth Dynamics and Circuit Integration

Figure 24 characterises the electro-optic frequency response and dynamics. In panel (a), the small-signal S_{21} response is shown at different bias voltages (0 V, -0.5 V, and -1.0 V),

demonstrating a nominal 3 dB bandwidth of 89.7 GHz at the design point of -1.0 V. Panel (b) plots the electro-optic cutoff frequency vs. Ge height H_{Ge} for different mesa lengths, showing the transit-time and RC limits. Panel (c) maps the 3D scatter design space of bandwidth over series resistance R_s , peaking inductance L_p , and total capacitance C_{tot} . Panel (d) compares the device terminal impedance Z from the CHARGE simulation with the analytical model, validating the parasitic RLC parameters.

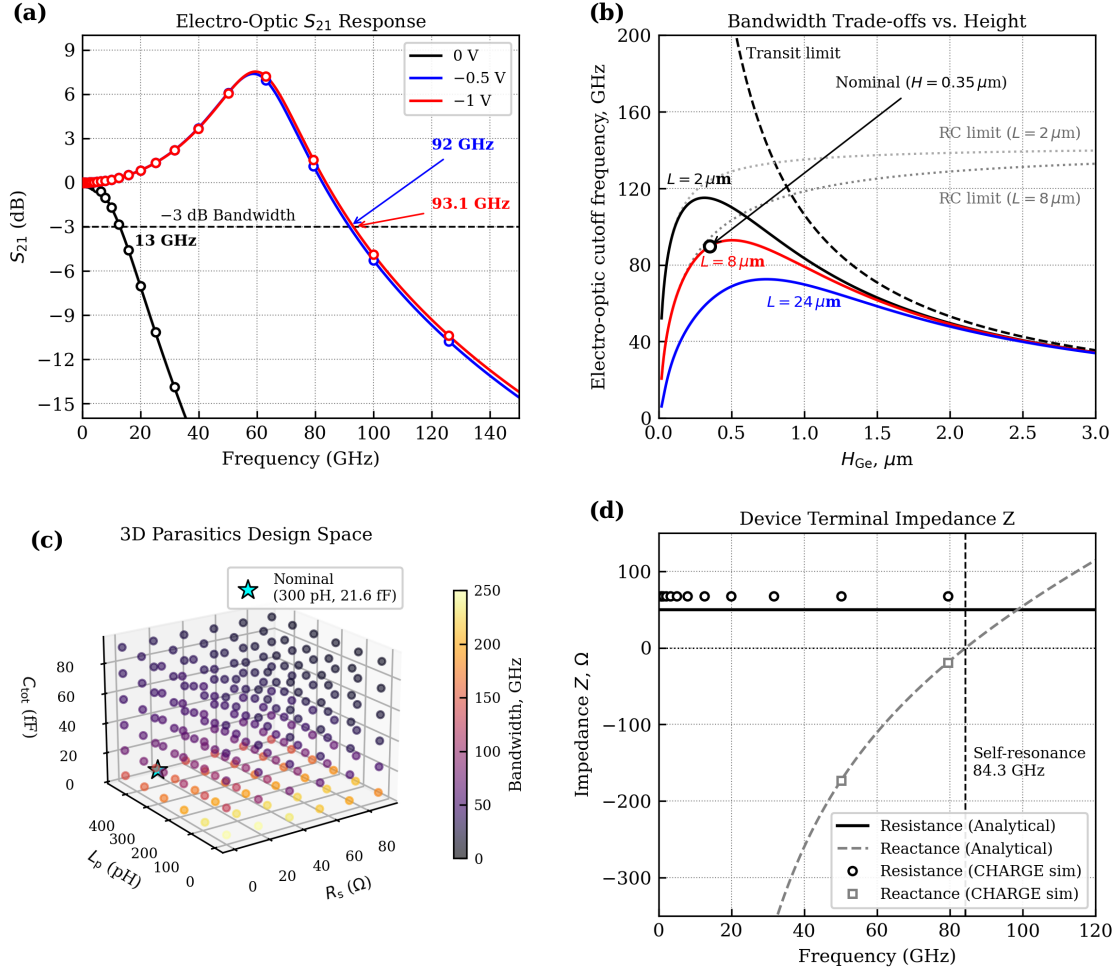


Figure 24: Electro-optic frequency response: (a) small-signal S_{21} response at different bias voltages; (b) electro-optic cutoff frequency vs. Ge height H_{Ge} for various mesa lengths; (c) 3D scatter of bandwidth vs. package parameters; (d) device terminal impedance Z comparing simulation and analytical models.

The multi-dimensional sweeps of the cutoff frequency under nominal inductive peaking ($L_p = 300$ pH) are consolidated in fig. 25. Panel (a) maps the 3D design space landscape of the cutoff frequency vs. mesa dimensions. Panel (b) tracks the cutoff frequency and capacitance vs. bias voltage, showing bandwidth saturation as the junction depletes. Panel (c) plots the cutoff frequency vs. optical power, illustrating the space-charge screening roll-off at high input power (above ≈ 5 mW). Panel (d) shows the cutoff frequency vs. background doping

concentration, demonstrating the depletion limit.

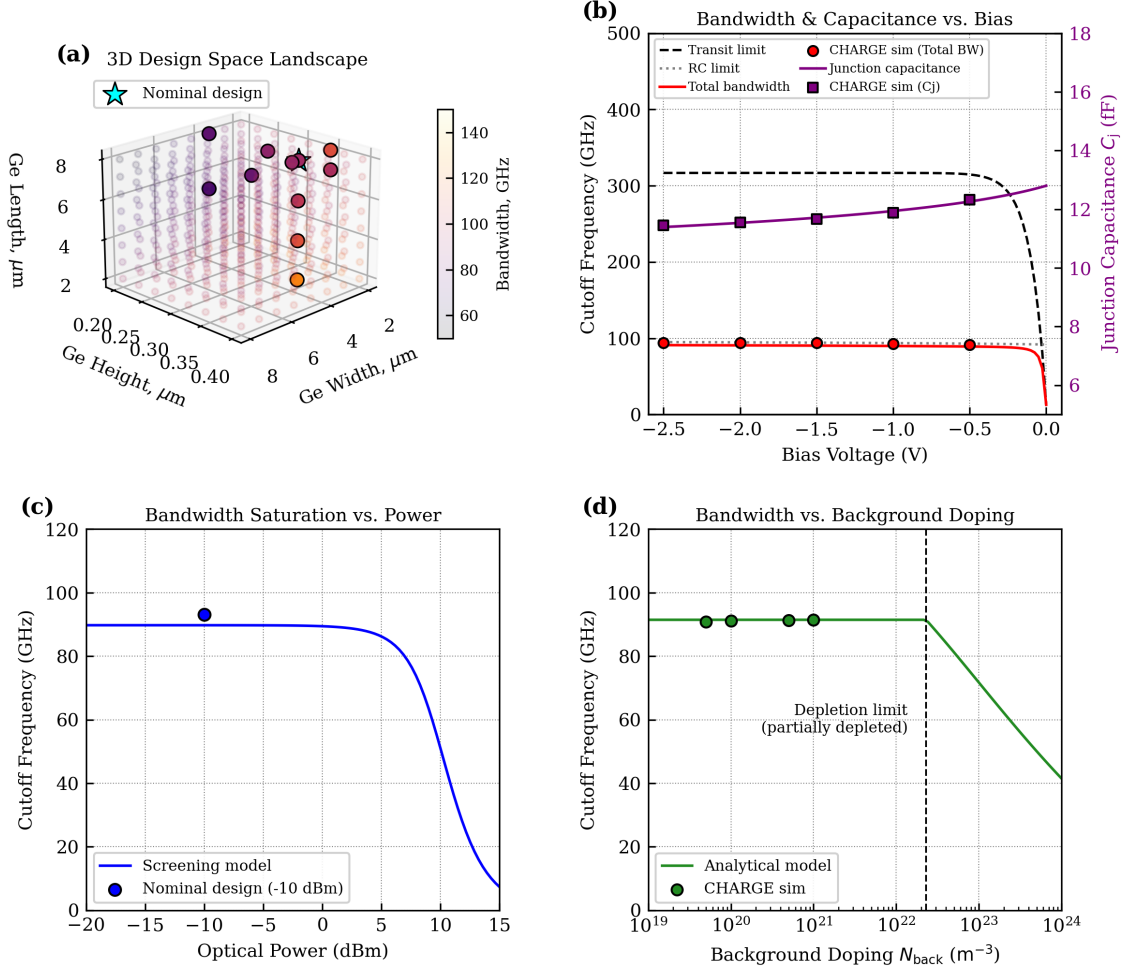


Figure 25: Cutoff frequency sweeps and limits: (a) 3D design space landscape of bandwidth vs. mesa dimensions; (b) cutoff frequency and capacitance vs. bias voltage; (c) cutoff frequency vs. incident optical power showing space-charge screening roll-off; (d) cutoff frequency vs. background doping concentration.

Inductive peaking extends the 3 dB bandwidth well beyond the unpeaked RC limit by inserting a series inductor L_p between the photodetector junction and the load resistance. This creates a zero in the transfer function that partially cancels the dominant capacitive pole at $f_p = 1/(2\pi R_{\text{tot}} C_{\text{tot}})$. For $L_p = 300$ pH and $C_j = 11.87$ fF, the resonant frequency is:

$$f_{\text{res}} = \frac{1}{2\pi\sqrt{L_p C_j}} = 84.3 \text{ GHz},$$

yielding a peaked response that boosts the nominal 3 dB bandwidth to 89.7 GHz, a 33.6 GHz increase over the unpeaked baseline of 56.1 GHz.

9.5 Design Trade-offs and Optimization

The trade-offs between responsivity, dark current, bandwidth, and the composite Figure of Merit $FOM = D^* \times f_{3dB}$ are map-profiled in fig. 26.

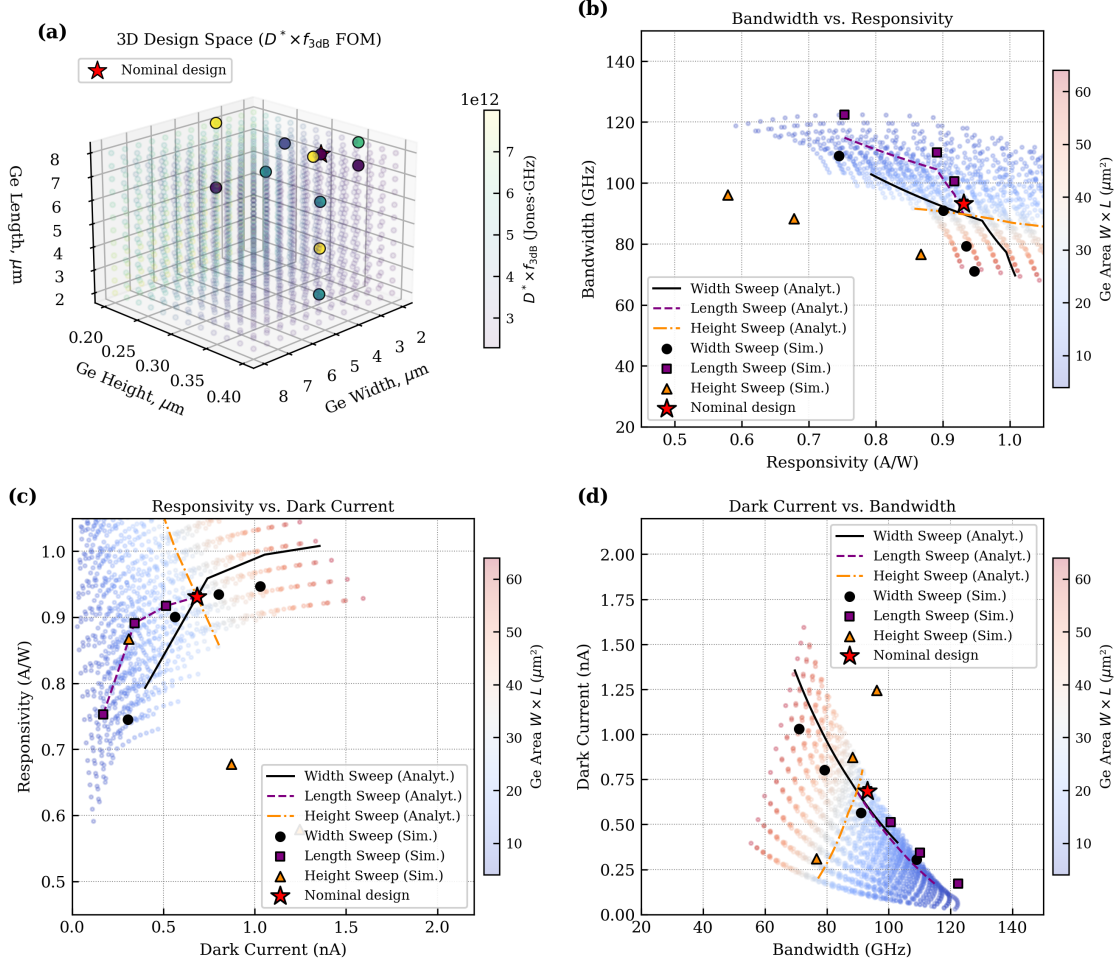


Figure 26: Design trade-offs: (a) 3D design-space landscape coloured by the composite FOM $D^* \times f_{3dB}$ (Jones-GHz) that simultaneously rewards low noise and high speed; (b) bandwidth vs. responsivity Pareto frontier; (c) responsivity vs. dark current; (d) dark current vs. bandwidth.

Figure 26(a) shows the three-dimensional design space coloured by the composite Figure of Merit $FOM = D^* \times f_{3dB}$. This product simultaneously penalises high dark current (via D^*) and low bandwidth, making it the most appropriate scalar for a high-speed communication detector. Panels (b)–(d) decompose the Pareto surface: widening the mesa increases responsivity due to improved mode confinement but raises the dark current due to the larger junction volume and also reduces bandwidth through added capacitance. The nominal design point (red star) achieves $D^* = 3.41 \times 10^{10}$ Jones and $f_{3dB} = 89.7$ GHz, placing it near the FOM maximum over the explored design space.

The analytical bandwidth trade-off vs. intrinsic-layer thickness W_i is summarized in table 5, using an effective junction width of $3.665\text{ }\mu\text{m}$, a stray capacitance of 9.78 fF , a total loop resistance of $70\text{ }\Omega$ (incorporating a $50\text{ }\Omega$ load and $20\text{ }\Omega$ device series resistance), and an effective carrier saturation velocity of $v_{\text{sat}} = 5.24 \times 10^4\text{ m/s}$ to calibrate the 1D transport model.

Table 5: Analytical bandwidth trade-off as a function of intrinsic-layer thickness W_i for the $8\text{ }\mu\text{m} \times 5\text{ }\mu\text{m}$ absorber (bold row is the nominal design point).

W_i [nm]	f_{tt} [GHz]	C_j [fF]	f_{RC} [GHz]	$f_{3\text{ dB, base}}$ [GHz]
200	116.0	20.77	74.4	62.7
250	92.8	16.61	86.8	63.4
300	77.3	13.85	97.4	61.2
350	66.3	11.87	104.9	56.1
400	58.0	10.38	111.4	51.7

9.6 Detector-Level Sensitivity and Noise Limits

The receiver sensitivity is dominated by the signal shot noise. Setting $\text{SNR} = Q^2$ where $Q \approx 7.03$ for $\text{BER} = 10^{-12}$ in OOK, the shot-noise-limited sensitivity P_{sens} is:

$$P_{\text{sens}} = \frac{Q^2 q B_e}{\mathcal{R}} + \frac{Q}{\mathcal{R}} \sqrt{Q^2 q^2 B_e^2 + 2q I_d B_e}, \quad (108)$$

which evaluates to $P_{\text{sens}} = 0.955\text{ }\mu\text{W}$ (-30.20 dBm) for the nominal device parameters ($B_e = 56.1\text{ GHz}$, $\mathcal{R} = 0.931\text{ A W}^{-1}$, $I_d = 0.684\text{ nA}$).

Table 6 consolidates all nominal performance parameters of the device, comparing them to the experimental benchmark.

Table 6: Nominal photodetector performance metrics at -1 V, 300 K, and 1310 nm.

Metric	Source of Truth	Nominal Value	Reference Benchmark
Dark current I_d	Electrical solver	0.684 nA	1.3 nA [1]
Responsivity $\mathcal{R}$	Electrical transport	0.931 A W $^{-1}$	0.95 A W $^{-1}$ [1]
EQE η_{ext}	Coupled solver chain	88.08 %	eq. (50)
IQE η_{int}	Carrier collection	96.23 %	Fully depleted junction
Junction capacitance C_j	Capacitance solver	11.87 fF	Fitted parallel-plate
Nominal bandwidth $f_{3\text{ dB}}$	Electrical AC solver	89.73 GHz	103 GHz (peaked) [1]
Specific detectivity D^*	Post-processed from I_d	3.41×10^{10} Jones	2.95×10^{10} Jones [1]
Sensitivity P_{sens} (OOK)	Noise model	0.955 μ W	eq. (108) at $Q = 7.03$

9.7 System-Level Link Simulation and Signal Integrity

The compact detector model is exported and instantiated in the system simulator. Figure 27 summarizes the system-level performance. The INTERCONNECT schematic in (a) shows the PAM-4 transceiver link. The Bit Error Rate (BER) vs. Received Optical Power in (b) is evaluated for a 106.25 Gbit/s (53.125 Gbaud) PAM-4 link, showing a BER sweep from 6.35×10^{-3} at -10 dBm down to a floor of 4.93×10^{-3} at 3 dBm due to an ISI-limited Q-factor $Q_{\text{isi}} = 2.48$. The PAM-4 eye diagram in (c) shows clear level separations, and the photocurrent time trace in (d) matches the decision thresholds.

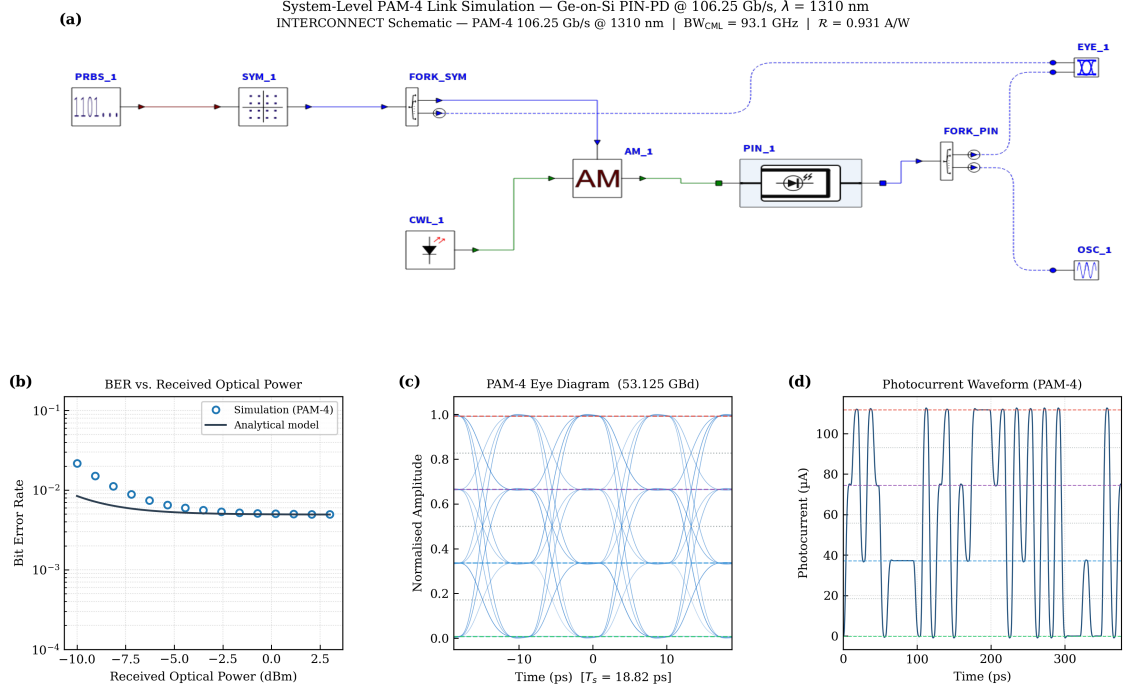


Figure 27: System-level simulation results: (a) INTERCONNECT circuit schematic; (b) BER vs. Received Optical Power showing simulation points and analytical curves; (c) PAM-4 eye diagram at 53.125 Gbaud; (d) photocurrent time trace waveform with matching decision levels.

10 Conclusion

this paper walked through the physics, design rationale, and simulation of a vertical n-i-p Ge-on-Si waveguide photodetector operating at 1310 nm under the IEEE 802.3 400GBASE-DR4 receiver specification. The device architecture, layer stack, electrode topology, and coupling scheme, are taken directly from the benchmark of Shi *et al.* [1]. The goal was not to produce a novel experimental result but to build a clear, step-by-step understanding of how geometry, material properties, and operating conditions translate into measurable photodetector performance, supported by a fully traceable multiphysics simulation.

Physical framework.

Section 2 derived the Beer-Lambert absorption law from the Helmholtz wave equation, establishing the optical generation rate $G_{\text{opt}}(\mathbf{r})$ as the Poynting-flux sink of eq. (10). The drift-diffusion transport framework, from the coupled Poisson and continuity equations through the Caughey-Thomas velocity saturation model, provided the physical basis for the transit-time bandwidth expression of eq. (55). The SRH, Auger, and surface recombination mechanisms were quantified, confirming SRH dominance in the depleted Ge absorber and justifying the single-parameter calibration strategy of section 8.3.

Architecture and material selection.

The metric framework of section 3 reduced the architecture selection to a single, defensible outcome. The photoconductor is eliminated by its gain-bandwidth lock through eq. (77); the PN junction by its diffusion-tail bandwidth penalty through eq. (79); the APD by its $k \approx 1$ excess noise growth as $F \approx M$. The waveguide-integrated PIN survives because it keeps L , W_i , and R_s independently controllable. Germanium is selected as the absorber because its direct Γ -point transition at 1310 nm provides $\alpha \approx 3 \times 10^3 \text{ cm}^{-1}$, an order of magnitude beyond silicon, and because its CVD heteroepitaxy on SOI is qualified at the 220 nm node by major silicon-photonics foundries.

Literature context and device selection.

The review in section 6 traced the vertical Ge-on-Si architecture from the 42 GHz first demonstration of Vivien *et al.* [12] through the 103 GHz U-shaped benchmark of Shi *et al.* [1]. That device was chosen as the study object because it combines the highest reported vertical Ge-on-Si bandwidth with a straightforward layer stack that is amenable to simulation, and because working through it analytically fills in the physical reasoning that an experimental paper necessarily condenses. the paper simulates the same $8 \mu\text{m} \times 5 \mu\text{m}$ Ge absorber at 1310 nm rather than at the reported C-band wavelength, using O-band material parameters throughout.

Device parameters and electrode analysis.

The device uses an $8 \mu\text{m} \times 5 \mu\text{m}$ Ge absorber with $W_i = 350 \text{ nm}$, positioned 11 % above the analytically optimal thickness $W_i^* \approx 310 \text{ nm}$ of eq. (61) to improve epitaxial process tolerance. The analytical derivation in section 7.3.2 shows that the U-shaped anode reduces the anode series resistance by approximately 40 % relative to the straight-electrode geometry for the nominal mesa geometry ($W_m = 5 \mu\text{m}$, $d_{\text{gap}} = 0.5 \mu\text{m}$), consistent with the 36 % reduction measured experimentally by Shi *et al.* [1]. This resistance reduction improves f_{RC} through eq. (58) and, in the peaked circuit of eq. (65), enables a shallower inductive peaking condition responsible for the experimentally reported step from 83 GHz to 103 GHz.

Simulation chain and validation.

The multiphysics simulation chain of section 8 links three stages through structured data files: the electromagnetic optical solver exports the volumetric generation profile $G_{\text{opt}}(\mathbf{r})$; the drift-diffusion electrical solver imports that profile and produces I_d , $\mathcal{R}$, C_j , and the small-signal AC frequency sweep; the compact-model post-processing stage rebuilds the bandwidth trade-off surfaces. Validation is conducted at three levels, Beer-Lambert consistency of the optical absorption monitor, dark-and-illuminated IV plausibility from the electrical solver, and AC-versus-geometric-model bandwidth agreement, before any compact-model value is exported.

Performance summary.

The geometry-derived no-peaking baseline gives $f_{3\text{dB,base}} = 56.1\text{ GHz}$, exceeding the IEEE 802.3 400GBASE-DR4 minimum bandwidth requirement of 39.8 GHz. This is a conservative estimate based on the single-carrier (hole-limited) transit-time model without inductive peaking; the benchmark device of Shi *et al.* achieves 103 GHz experimentally with the same layer stack and optimised peaking inductance. The junction capacitance at the nominal operating point is $C_j = 11.87\text{ fF}$ (with a parallel-plate geometric baseline of 16.2 fF) and the U-shaped anode series resistance is $R_{s,\text{an}} = 20\ \Omega$. Solver-dependent quantities, responsivity, dark current, EQE, IQE, and derived sensitivity, are extracted from the post-processed simulation datasets by the post-processing scripts and compared with the benchmark; they are not inserted as static constants.

Outlook.

The double U-shaped anode, with metal contacts wrapping both long sidewalls of the germanium mesa, is a natural extension of the geometry studied here but was not evaluated in simulation. Comparing its R_s reduction against the additional mesa perimeter and associated dark-current rise through eq. (38) would be a straightforward next modelling step. Fabrication followed by RF on-wafer characterisation would then connect simulation to measurement, the experimental loop that lies outside the scope of this educational study.

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